

The Intrinsic Valuation of Biodiversity Loss^{*}

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Abstract

We explore the costs of the loss of animal life in a consequentialist total utilitarian framework that incorporates the intrinsic value of living beings through their sentience. Moral philosophy and neuroscience define sentience as the capacity for valenced experience. Observable performance on cognitive tests represents a conservative lower bound on sentience under plausible neuroscientific assumptions. We estimate the relationship between cognition and neuron counts to cardinalize sentience. We collect population and neuron data for a wide range of animal species, and aggregate them into a single measure. Across a wide range of parameters, the aggregate sentience losses of animals in the past half century outstrip the sentience gains to humans from population growth 100-fold. Breaking even in welfare terms requires assigning a correspondingly small welfare weight to animals, at which each square kilometer of the Amazon still holds more than a million dollars' worth of biodiversity. Including the benefits of economic growth to humans does not alter these conclusions.

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1. INTRODUCTION

The rise of humanity has transformed the lives of billions. Steady improvement since the Industrial Revolution in frontier economies, and more recently in the developing world, has brought a total transformation of what it means to live and thrive on this planet. By any reasonable metric, a given human almost anywhere in the world lives a life better than their grandparents', and one unimaginable to their distant ancestors. This rising material prosperity has sustained continuous increases in the human population.

At the same time, the spread of humanity has been accompanied by dramatic declines in wild population counts in the natural world. Many species have been lost entirely through extinction, and many more of those that remain have suffered large population declines. These trends have occurred through a combination of habitat loss, competition for resources, and pollution, all direct consequences of economic growth.

This paper attempts to account for what has been lost during humanity's rise. To do so, we consider a consequentialist total utilitarian view, according to which there exists a welfare function aggregated across all relevant living beings. Moral philosophy argues that living beings that exhibit sentience may reasonably be included in the welfare function, where sentience is defined as the capacity for valenced experience. While sentience is difficult to measure, observable performance on cognitive tests represents a conservative lower bound on sentience under plausible neuroscientific assumptions. We estimate the relationship between cognition and neuron counts to cardinalize this lower bound on sentience. We collect population and neuron data for a wide range of animal species, and aggregate them into a single valuation measure. Across a wide range of parameters, the aggregate sentience losses of animals in the past half century outstrip the sentience gains to humans from population growth 100-fold. Converting sentience to welfare and accounting for economic growth makes little difference to this comparison. Less conservative choices can imply relative losses in excess of 10^6 . Equivalently, breaking even in welfare requires placing a relative welfare weight on animal sentience between 3×10^{-7} and 1×10^{-3} . Even under this small weight, the Amazon still holds more than a million dollars' worth of biodiversity per square kilometer.

We clarify two aspects of our approach before describing it in more detail. First, we employ a capacious definition of biodiversity loss, which takes into account changes in overall wildlife populations ([IPBES 2019](#)). We focus primarily on changes in quantities,

and do not attempt to value species variety *per se*. Second, our approach is distinct from the instrumental valuation of wildlife such as ecosystem services. Our goal is to quantify the intrinsic value of lost wildlife. We build on recent progress in moral philosophy and neuroscience to define which living beings to include in the welfare function, and what weight they might receive.

We start our analysis by defining the scope of the consequentialist total utilitarian approach. Only individuals with moral status enter the social welfare function. A broad consensus in moral philosophy, including utilitarianism, is that a living being has moral status if and only if it exhibits some degree of sentience. Sentience is defined as the capacity for valenced experience. Experience in this context means that there is “something it is like” for the living being to be in a given state (Nagel 1974), and can equivalently be defined as phenomenal consciousness. Valence is the property of experience being positive or negative for the living being. These properties encompass pain and pleasure, but also fear, anger, and their analogs in animals.

Sentience is a product of the brain: the majority of philosophers and neuroscientists hold that valenced experience is ultimately a physical phenomenon realized in the brain. In particular, neuroscientific evidence points to two brain regions that determine sentience: the midbrain (an evolutionarily more ancient brain region) and the pallium (an evolutionarily more recent brain region that also determines cognition). These brain regions are differentially developed across species. As a result, different species are likely to experience different degrees of sentience. For instance, vertebrates are generally accepted as exhibiting an ability for valenced experience. Given their distinct neurological architecture, insects and some families of mollusks are plausible candidates, but reasonable people may disagree about their moral status. The definition of sentience excludes plants, bacteria, and animals such as jellyfish, as they have either no neurological systems or extremely rudimentary ones, preventing them from being conscious.

Equipped with the notion of sentience that delineates which species might reasonably be included in the welfare function, we must then assign a cardinal utility to individuals of each species, sometimes called a welfare range in the context of animal welfare. Thus, we need a cardinal measure of sentience. Such a measure of sentience is still beyond existing neuroscientific measurement. However, neuroscientific evidence points to a plausible lower bound on the sentience of other animals relative to humans using measures of cognition.

Valence relies on two brain regions, the evolutionarily older midbrain and the evolutionarily more recent pallium, which controls cognition. Since the midbrain is more similar across species than the pallium, the ability for valenced experience is likely more evenly spread than cognition across animals. In addition, neuroscientific functional arguments suggest that cognition scales more steeply in overall neuronal capacity than sentience. Therefore, variation in cognition across species is likely greater than variation in sentience, allowing us to construct a lower bound on the sentience of other animals relative to humans.

Direct measures of cognition are available for many species, by observing their ability to enumerate or solve puzzles to retrieve food. Leveraging again that cognition is tightly linked to neuronal capacity, we estimate the relationship between cognitive performance and total and pallial neuron counts for species for which both measures are available. We then predict cognition for all other species using their neuron counts. We obtain a cardinal measure of variation in cognition across all species with available neuron data, and thus a lower bound on the sentience of other animals relative to humans.

This lower bound is cardinal under the important assumption that the cognition scale that we select represents cognition without the need for an additional transformation. To partly address this challenge, we consider three different scales to cardinalize cognition, all relative to humans. The first scale is the most transparent: the test success rate itself. The second scale is a logistic transformation of the test success rate, which corresponds to a microfounded choice model ([Bradley and Terry 1952](#); [Rasch 1960](#); [McFadden 1974](#); [Yellott 1977](#)). The third scale is a logarithmic transformation of the failure rate, which corresponds to an underlying model with multiple trials ([Rasch 1960](#); [Prentice and Gloeckler 1978](#); [McCullagh and Nelder 1989](#)). We choose the most conservative specification for our baseline results, and systematically display how departing from this baseline changes our results.

Given a cardinalization of sentience, we draw a distinction between sentience and welfare. The social welfare function takes sentience as an input, but requires decision-makers to take a stance on the welfare weight of animals relative to humans. Thus, we first focus on sentience only, and report welfare as a function of the relative welfare weight in a second step.

Of course, one can reasonably disagree with many components of this calculation. While not devoid of challenges, this exercise has the advantage of being amenable to

aggregation and quantitative analysis, and makes underlying assumptions transparent. We thus view it as a useful first step in the quantification of the intrinsic value of biodiversity loss, but certainly not the last. To assess the role of each step, we report a wide range of sensitivity analyses.

Which species are the most sentient? Unsurprisingly, great apes are among the most sentient animals. Their sentience scores are often above 0.5 times that of humans. Sentience is correlated with body mass, but only imperfectly. In particular, many small animals are highly sentient. Mice have a sentience score of 0.02 relative to humans, while cows have a sentience score of only 0.3, despite being 30,000 times heavier. Similarly, crows have a sentience score of 0.1, salmon of 0.005, and crickets of 0.002.

We then combine our measures of sentience with changes in wildlife population counts by species. We collect data on population trends at the species, order, and group levels for a wide range of animals. Population loss has been significant across taxa. It ranges from around 45% for birds and mammals to 90% for herptiles (reptiles and amphibians) since 1970. These substantial population declines compound with large initial populations: mammals were roughly 40 times more numerous than humans in 1970, fish were a million times more numerous, and insects were several billion times more numerous. The disappearance of this large number of sentient animals represents a potentially significant reduction in aggregate sentience.

We find substantial losses for non-human life. We first focus on aggregate sentience, which has the advantage of not requiring us to take a stance on the welfare weight of sentient animals relative to humans. In that case, animal sentience losses outweigh population gains to humans by 2 to 6 orders of magnitude. That is, animal sentience losses are 100 to 10^6 times larger than the sentience gains accruing to humans from population growth, or in fact similarly large relative to the sentience of the current human population. The lower end of this range excludes insects and the upper end includes them, both at our baseline estimates. We then vary the parameters scaling the relationship between neurons and cognition around these estimates.

First, when we average across tests and cardinalizations instead of using the most conservative scaling of cognitive tests, animal sentience losses excluding insects are more than 50,000 times larger than the sentience gains to humans.

Second, we consider piecewise scalings that further downweight all animals relative to humans by leveraging cognitive tests that are specific to primates and humans only and thus may better capture humans' singular ability for cognition and, potentially,

sentience. At our most conservative estimate of cognitive scaling within animals, the losses to fish alone are larger than human gains. At our most conservative estimate of cognitive scaling between animals and humans, the losses to animals exceed human gains by multiple orders of magnitude.

Including the substantial increase in the populations of farmed animals does not change these conclusions. Farmed animal sentience is of the same order of magnitude as human population sentience. Although farmed animals represent approximately 40% of current vertebrate biomass on Earth, biomass does not scale one-for-one with sentience. The large declines from initially large populations of smaller but still sentient wild animals are quantitatively much more significant. Adjusting farmed animal sentience for their low quality of life would further reduce their contribution to sentience increases.

Converting sentience into welfare requires us to take a stance on the welfare weight of animals relative to humans. The strict utilitarian social welfare function weighs all sentience equally and assigns a welfare weight of 1 to animals, so that aggregate welfare losses are equal to aggregate sentience losses. We also consider a social welfare function that discounts animal welfare beyond their degree of sentience. When the welfare weight of non-human animal life is zero, other animals do not matter and one recovers a standard welfare function. Varying the welfare weight between 0 and 1 traces out the range of possible valuations of biodiversity loss. When the animal welfare weight is between 3×10^{-7} and 1×10^{-3} , the loss of animal life exactly equals the value of human gains depending on the specification.

Quality of life, a species' realized welfare relative to its welfare range, also contributes to aggregate welfare. Our baseline exercise assumes that humans and all other animals had the same quality of life in the state of nature. Incorporating the benefits of economic growth for humans reduces animal welfare losses relative to human gains somewhat, but animal welfare losses remain more than 100 times larger than gains to humans. This conclusion emerges because welfare gains from population increases outweigh economic gains at standard values of statistical life (VSL) for humans, and standard utility functions compress even sizable consumption gains. In addition, assuming lower quality of life for animals compared to humans in the state of nature is isomorphic to assigning a smaller welfare weight to them. Given the large aggregate animal sentience losses that we find, one would need to assume that animals are much closer to indifference between life and death to offset these losses.

We convert these values into a willingness to pay for animal life. We start from a human VSL of USD 14 million. With a welfare weight of one and adjusting for life expectancy, a great ape's life is valued at USD 0.64 million, a crow's life at USD 40,000 and a salmon's life at USD 910. These values exceed stated preferences for biodiversity protection by several orders of magnitude ([Loomis and White 1996](#)). Next, we consider a welfare weight of 1×10^{-3} that approximately equalizes animal losses and human gains (in the conservative case excluding invertebrates, before accounting for economic growth). In that case, a great ape's life is valued at 640 dollars, a crow's life at 40 dollars and a salmon's life at 0.91 dollars. Animal losses and human gains break even despite these smaller values because of the substantial population declines over the last five decades.

These results reveal a large discrepancy between a sentience-based valuation of animal welfare and the valuation implied by human behavior and willingness to pay. Through the lens of our framework, this contrast suggests that humans assign welfare weights to other animals that are substantially below one. We do not argue for a particular value of the animal welfare weight. Rather, our objective is to highlight the importance of this question given the substantial losses in animal sentience over the past half century. In particular, seemingly small departures from a zero animal welfare weight can lead to large aggregate welfare costs from biodiversity loss.

Breaking even in welfare space, or even further downweighting animals, does not preclude large aggregate dollar-equivalent losses. We define human gains to include the contribution of population gains, which thus exceed annual Gross Domestic Product (GDP) at conventional VSLs. At an animal welfare weight of 1×10^{-3} that approximately equalizes animal losses excluding insects and human gains, the welfare cost from biodiversity loss represents 89% of global GDP in 2025.

We finally study how these costs are distributed across the planet. We quantify the value of today's biodiversity across the globe to propose statistics that can be used in cost-benefit analysis of local conservation policies. We combine detailed range maps that cover the current spatial distribution of species with our measures of sentience at the pixel level. The Amazon basin, the Congo basin, South-East Asia and Australia host the largest concentrations of sentience. With an animal welfare weight of one, each square kilometer in the Amazon basin alone is worth USD 220 million. When we adjust it for break-even animal welfare weights, including the gains from economic growth, we obtain a value of USD 1.1 million, still several times the economic value of

deforestation embedded in local land prices.

Related Literature. A large literature in economics and ecology has attempted to place a value on biodiversity through the instrumental value of ecosystem services: the direct benefits and services that animal and plant life provide to humanity simply by existing, without being priced by the market, such as pollination or pharmacological discovery ([Costanza et al. 1997](#); [Simpson et al. 1996](#); [Rausser and Small 2000](#); [Daily et al. 2000](#); [Barbier 2007](#); [Bateman et al. 2013](#); [Frank and Sudarshan 2024](#); [Taylor and Druckenmiller 2022](#); [Frank 2024](#); [Giglio et al. 2024, 2025](#)).

A second strand of literature studies the existence value of biodiversity or species: individuals may value the continued existence of biological variety even when they do not expect to use or directly experience it, for instance for aesthetic or informational reasons ([Krutilla 1967](#); [Weitzman 1998](#); [Metrick and Weitzman 1998](#)).¹

Our work is conceptually distinct from both of these strands. Our approach explores the implications of considering other sentient life as valuable in and of itself as a direct bearer of welfare, in the same way that a human life has intrinsic value regardless of any service to other humans. This approach is grounded in a long tradition in moral philosophy reviewed in [Singer \(1975\)](#) and [Birch \(2024\)](#).²

A growing literature in welfare economics builds on this tradition and treats non-human animals as direct bearers of welfare ([Carlier and Treich 2020](#); [Fleurbaey and Van der Linden 2021](#)).³ [Blackorby and Donaldson \(1992\)](#) and [Espinosa and Treich \(2021, 2025\)](#) consider both human and animal utilities in the social welfare function in the context of animal research and farming for meat.⁴ [Budolfson et al. \(2024\)](#) propose using a total utilitarian framework adjusted for quality of life in order to evaluate policies around meat consumption and biodiversity protection, but do not estimate the welfare score of animals. [Johansson-Stenman \(2018\)](#) uses survey data from Sweden to show that there is likely widespread public support for assigning at least some moral weight to the welfare of animals.

Closest to our paper, [Budolfson and Spears \(2019\)](#) propose using neuron counts of

¹Contingent valuations of stated willingness to pay for species protection, such as those in [Loomis and White \(1996\)](#), may be interpreted as the direct value humans place on a species' existence as an additional argument of their utility.

²The intrinsic valuation of biodiversity is compatible with the existence valuation to the extent humans choose to assign existence value equal to sentience.

³[Adhami et al. \(2023\)](#) consider a related total utilitarian perspective counting human lives only to re-evaluate the gains from economic growth.

⁴The latter explicitly allow for the possibility that captive animals have negative welfare, in particular due to poor captive living conditions. We assume that, in the state of nature, wild animal lives are, on average, not characterized by net negative welfare.

broad taxonomic groups to assign utility potentials to animals in a social welfare function. [Espinosa \(2024\)](#) evaluates animal protection policies in animal research, farming, and hunting for select species with a similar framework. These papers do not estimate the mapping between neurons and utility potentials and instead explore various possible parametrizations. Our approach builds on these ideas, with two important distinctions. First, we anchor the mapping between neuron counts and sentience using observed performance on cognitive tests—while also conducting extensive sensitivity analyses around our main estimate. Second, we collect and harmonize population and neuron count data for thousands of species and taxonomic groups to quantify the costs of animal population declines. The resulting measure of the intrinsic value of biodiversity loss is, to our knowledge, new to the literature.

In contemporaneous work, [Eden et al. \(2026\)](#) develop a related approach to ours. They also assess the consequences of changes in animal populations using neuron counts to cardinalize animal welfare. Our approaches differ in two dimensions. First, we use observed cognitive tests to anchor how sentience relates to neuron counts in addition to exploring possible parametrizations of this scaling. Second, they assume that many wild animals have negative welfare in their state of nature, implying that wild animal population declines improve aggregate welfare. We view these two approaches as complementary: they reflect how various defensible assumptions can lead to a wide range of conclusions, and highlight the need for further work in this area.

2. CONCEPTUAL FRAMEWORK

2.1 Welfare Function

We define total welfare of animals excluding humans at date t as the sum of the welfare of all animal species living at date t :

$$(1) \quad \mathcal{W}_{At} = \sum_{s \in \mathcal{S}} \lambda_s \times u_s(x_{st}) \times N_{st},$$

where $\mathcal{S}$ is the set of species excluding humans, λ_s is a weight capturing the relative moral status of species s , and $u_s(x_{st})$ is the instantaneous utility of a representative individual of species s at date t , which we allow to depend on a generic vector x_{st} . N_{st} is the number of individuals of species s living at date t . The welfare of humans $s = 0$

at date t is:

$$(2) \quad \mathcal{W}_{0t} = u_0(x_{0t}) \times N_{0t},$$

where the moral status of humans is normalized to 1.

Welfare functions of the form in equations (1) and (2) have been proposed in many studies of animal welfare (Budolfson and Spears 2019; Espinosa 2024; Budolfson et al. 2024). In this section, we review conceptual definitions of each component of the welfare function in (1) and (2) that are general enough to apply, at least in principle, to many species. We postpone our discussion of the potential comparability between human and animal welfare to Section 2.4. We propose empirical definitions of these components in Section 3.

2.2 Moral Status and Sentience

Only entities with moral status, that is, those with intrinsic moral worth, enter the social welfare function. A broad consensus in moral philosophy, and in particular in utilitarianism, argues that *sentience* is both necessary and sufficient for moral status $\lambda_s > 0$ (Singer 1975, 1979; Hare 1981; DeGrazia 1996; Birch 2024).

The concept of sentience is the natural generalization of the human ability to experience positive or negative feelings and emotions. Formally, sentience is the capacity for valenced experience. In this context, experience means phenomenal consciousness, that is, “there is something it is like for the subject to be in that state” (Nagel 1974). Experience excludes unconscious processing, such as reflexes. Valence is the property of “feeling bad or feeling good to the subject.” Pain and pleasure are two obvious examples of valenced experiences.

This position, known as sentientism, follows naturally from utilitarianism, the moral foundation of modern welfare economics. Indeed, in the consequentialist-utilitarian framework, valence is the relevant scale for moral evaluation. The objective is to maximize the good, where the good is understood as positively valenced experience and the absence of negatively valenced experience across all individuals. Which species these individuals belong to is incidental to utilitarianism. Including non-human sentient beings in the social welfare function amounts to a consistency requirement.⁵

⁵Sentientism also emerges from other influential moral theories such as Korsgaard’s neo-Kantianism and Nussbaum’s capabilities approach, despite starting from fundamentally different premises. We discuss classical and preference utilitarianism, as well as these alternative theories, and whether they are

Of course, reasonable individuals can disagree about the implication that species other than humans have moral status. Nonetheless, assigning at least some moral weight to the welfare of non-human living beings strikes us as a widely held view: the existence of animal welfare laws, the widespread care people demonstrate for pets, and the numerous protections for endangered species all point in this direction.

Sentience and cognition. One may reasonably ask whether cognition should also determine moral status. Cognition is not typically regarded as directly relevant for moral status in most contemporary ethical theories, including sentientism.⁶ Cognition may nevertheless be indirectly relevant for welfare: it may play a mediating role by amplifying or dampening affective intensity, or expanding the variety and structural richness of valenced experience. We return to the relationship between cognition and sentience in Section 3.

2.3 Utility and Sentience

Having defined who may enter the social welfare function, we define what constitutes welfare. We adopt the mental-state theory of welfare, which posits that welfare consists of consciously experienced valenced states (Crisp 2006; Feldman 2002, 2004). Under this view, the (human) utility function represents specific valenced experiences: a positively valenced experience represents higher utility, and a negatively valenced experience represents lower utility.

The mental-state theory of welfare corresponds to the standard consequentialist-utilitarian framework.⁷ It is a generalization of classical hedonism, as it relies on all valenced affective states (fear, anxiety, social reward, playfulness, etc.), as opposed to only pain and pleasure.⁸

While it was originally developed in the context of human welfare, the mental-state theory extends naturally to other animals because it relies on sentience rather than specifically human traits, and is thus consistent with sentientism (Singer 1975; DeGrazia 1996; Birch 2024). To the extent that many species exhibit sentience, they also have a

consistent or inconsistent with sentientism, in more detail in Appendices A.1 and A.2.

⁶Only certain deontological traditions such as orthodox Kantian ethics tie moral status directly to rational agency. However, this view has been shown to lead to important paradoxes.

⁷In particular, the value of an action depends solely on its consequences for individual welfare. That is, there are no other morally relevant properties such as rights, freedom, or equality, except insofar as they affect welfare.

⁸The mental-state view is also fully compatible with a preference-satisfaction view of welfare, in its evidential reading: satisfaction of preferences is evidence of welfare, and welfare is valenced experience. We provide a more detailed discussion in Appendix A.1.

utility function u_s that represents the strength of their valenced experiences.

Utility, moral status, and welfare weights. Without further restrictions, degrees of moral status λ_s are isomorphic to degrees of sentience embedded in the utility function u_s . Therefore, our baseline analysis starts by normalizing $\lambda_s = \mathbb{1}\{\sup u_s > 0\} \in \{0, 1\}$ to be an indicator variable encoding whether a living being is sentient at all. Moral status then indicates whether a species has any sentience at all, and the utility function measures how much sentience this species has.⁹

We additionally assess the consequences of assigning sentient animals a welfare weight strictly below 1, further discounting their weight in the social welfare function relative to their measured sentience. To do so, we consider cases in which the utility function u_s is fixed to our cardinal measure of sentience, and animals other than humans have $\lambda_s \in (0, 1)$, where λ_s now plays the role of the relative welfare weight that this species receives relative to humans holding sentience fixed.

Welfare ranges. The utility function defines the capacity for welfare, sometimes called the welfare range, as the total range between its maximum and minimum utility:

$$(3) \quad \bar{u}_s = \sup_{x_{st} \in \mathcal{X}_s} u_s(x_{st}) - \inf_{x_{st} \in \mathcal{X}_s} u_s(x_{st}),$$

where the range spans a set of possible vectors $\mathcal{X}_s$ that might include food quantity and quality, safety, comfort, novel experiences, and so on.¹⁰ We further normalize $\inf_{x_{st} \in \mathcal{X}_s} u_s(x_{st}) = 0$ for all s . This normalization imposes that all species achieve weakly positive utility. We return to the possibility of negative utility in Section 5.5.¹¹

We can then rewrite the utility function as $u_s(x_{st}) = \bar{u}_s \times \tilde{u}_s(x_{st})$, where $\tilde{u}_{st} \equiv \tilde{u}_s(x_{st})$ captures the difference between *capacity* for welfare and *realized* welfare. The component $\tilde{u}_{st}$ captures the quality of life with respect to potential.

⁹This normalization corresponds to the so-called unitarian view. Alternatively, the hierarchical view loads differences in degrees of sentience into differences in moral status as opposed to differences in the utility function. For our purposes, these two views are isomorphic.

¹⁰We assume that either the utility function is bounded or the set $\mathcal{X}_s$ is compact.

¹¹Our approach accommodates $\inf_{x_{st} \in \mathcal{X}_s} u_s(x_{st}) < 0$ as long as the average welfare range experienced by the average member of each species over the set of possible x_{st} , $\mathbb{E}_s[u_s(x_{st})]$, is weakly positive. In that case, differences in experienced welfare relative to potential are isomorphic to rescaling the species' welfare weight.

2.4 Species Welfare and Total Utilitarianism

Having defined sentience, the utility function and moral status, we consider the welfare of a single species, defined as:

$$(4) \quad \mathcal{W}_{st} = N_{st} u_s(x_{st}).$$

The formulation in equation (4) continues to build on the standard consequentialist-utilitarian framework, whereby we can rank states of the world by the sum or average of individual welfare. This property requires inter-individual cardinal comparability of welfare and additive aggregation across individuals.

The formulation in equation (4) adopts total rather than average utilitarianism. That is, the social welfare of a species is the sum of the welfare of all individuals of that species, not the average. Total utilitarianism is the natural setup for our investigation. Asking about the welfare effects of the loss of animal populations is only relevant if the welfare criterion is not invariant to the scale of populations. In addition, a total utilitarian social welfare function with variable population emerges from seemingly desirable axioms, although it also features well-known limitations (Kuruc et al. 2022).¹²

Aggregating species welfare (4) into total animal welfare (1) involves one additional key property: welfare is comparable across animal species in addition to being comparable across individuals. The mental-state theory and sentientism offer a natural justification for this property: sentience is, in principle, comparable across individuals and species.¹³

Aggregating species welfare into total animal welfare (1) without more structure requires us to use the welfare function with care. For instance, eradicating a species might seem like an isolated event from the perspective of the welfare function. In practice, however, it may trigger changes in other species' populations as ecosystems reorganize. Thus, the welfare function (1) can be used for ex-post accounting, once population changes have been observed. Using it for counterfactuals requires more caution: it necessitates using population changes that include ecosystem-wide reorganizations.

¹²The first axiom is 'same number Pareto': the welfare ordering respects Pareto improvements among fixed populations. The second axiom is 'non-anti-egalitarianism': society does not prefer inequality. The third axiom is 'mere addition': holding other individuals' utilities constant, adding a person who values living does not reduce social welfare.

¹³Of course, instantaneous utility differs from lifetime utility. Because humans tend to live longer than most animals, the lifetime utilities of humans and other animals differ more than their welfare capacities. However, our welfare function does not require us to adjust for life expectancy by species because it is defined as flow utility for each living individual.

One may aggregate animal welfare (1) and human welfare (2) into total welfare:

$$(5) \quad \mathcal{W}_t(\boldsymbol{\lambda}) = \mathcal{W}_{0t} + \mathcal{W}_{At}(\boldsymbol{\lambda}),$$

which further requires that welfare is comparable across animals and humans. This property has the same content as requiring that welfare is comparable across animal species.

Crucially, comparability of welfare does not imply equal weights on sentience: our formulation allows for smaller welfare weights assigned to animals $\lambda_s < 1$. Changing the vector $\boldsymbol{\lambda} = \{\lambda_s\}_{s \geq 1} \in [0, 1]^{|S|}$ traces out the Pareto frontier defined by the welfare function (5).

To isolate the role of sentience from the role of welfare weights, we consider two comparisons. First, we compare aggregate animal sentience, defined as $\mathcal{W}_{At}(\mathbf{1})$, to aggregate human sentience $\mathcal{W}_{0t}$, holding quality of life fixed. This first comparison has the advantage of abstracting from the welfare weights assigned to animals relative to humans, about which one may reasonably disagree. Of course, it only compares units of sentience and does not translate into a common welfare metric. Second, we map out how aggregate animal welfare $\mathcal{W}_{At}(\boldsymbol{\lambda})$ compares to aggregate human welfare $\mathcal{W}_{0t}$ as a function of animal welfare weights $\boldsymbol{\lambda}$ and economic growth embedded in $\tilde{u}_{0t}$. This second comparison requires us to take a stance on animal welfare weights $\boldsymbol{\lambda}$.

Having established the conditions under which the welfare function (5) emerges, we now turn to the measurement of each of its components.

3. MEASURING SENTIENCE

3.1 Measurement Challenges

Our exercise requires the measurement of two objects for a large number of species: moral status λ_s , strictly positive if species s is sentient, and $\bar{u}_s$, a cardinal measure of the range of valenced experiences of species s . These objects are inherently difficult to measure: conscious experience gives subjects a first-person point of view that cannot be fully apprehended from outside but only from the inside. This property constitutes the ‘hard problem’ of consciousness (Chalmers 1995).

Our measurement approach relies heavily on the established link between sentience and the brain. The majority of contemporary philosophers and neuroscientists hold the

view that consciousness and valenced experience are ultimately physical phenomena, realized in physical processes occurring in the brain. Mental states, such as thoughts, feelings and subjective experiences, arise from neural activity, and a complete physical account of the brain is also a complete account of mental states (Papineau 2002; Kim 2005).¹⁴ Therefore, variation in neural architecture is a principled starting point for understanding interspecies variation in sentience and welfare ranges.

Empirical frameworks for measuring sentience have emerged as the question of animal sentience has migrated from scientific and bioethical inquiry toward explicit legal recognition over the past three decades. The first systematic criteria emerged from the 1991 report of the Institute of Medical Ethics working party on the use of animals in biomedical research (Smith and Boyd, eds 1991). The European Union committed to regard animals as “sentient beings” in Article 13 of the Lisbon Treaty, in force since 2009, building on an earlier protocol to the 1997 Treaty of Amsterdam. The United Kingdom enshrined sentience in domestic law through the Animal Welfare (Sentience) Act 2022. The legal landscape is actively evolving, and we attempt to build on the most up-to-date conceptual framework. In particular, we borrow heavily from the framework in Birch (2024), which builds on the review that informed that Act (Birch et al. 2021).

3.2 Behavioral Markers of Sentience

The literature defines markers of sentience. These markers, while individually neither necessary nor sufficient to determine sentience, are to be understood as “symptoms” of sentience, each raising the probability that an individual is sentient. They are useful not only to determine sentience, but also to identify which brain regions are responsible for valenced experience.

The starting point is humans, who can report a valenced experience at the same time as displaying a particular marker. These markers are then generalized to other animals by analogy. The literature has converged on three types of markers: behavioral markers such as avoiding a particular location, physiological markers such as an autonomic stress response, and pharmacological markers such as reaction to an analgesic. Following Birch (2024), these criteria are:

¹⁴This view is called materialism. The main alternative is dualism, which holds that to explain consciousness, we need to posit at least some fundamentally non-physical properties. There is, however, no scientific basis for this claim. It is beyond the scope of this paper to settle this debate, and we accept the materialist position. Appendix B.1 provides a brief account of the debate and of the arguments that have made materialism the contemporary mainstream.

1. **Motivational trade-offs:** The animal engages in dynamic decision-making, weighing the adverse impacts of noxious stimuli against the value of rewards.
2. **Flexible self-protection:** The animal exhibits flexible self-protective behaviors, including wound tending and guarding.
3. **Associative learning:** The animal demonstrates associative learning by forming connections between noxious stimuli and neutral cues.
4. **Analgesia preference:** The animal expresses a preference for analgesics or anesthetics when injured.

These markers are useful to define the scope of sentience. However, comparable and cardinal measures of these markers are not available for a broad range of species. Thus, for the purpose of measuring welfare ranges in a systematic way across species, we turn to the brain regions that are related to behavioral markers of sentience.

3.3 Neurological Markers of Sentience

The neural basis of sentience. Two broad categories of brain structure have been identified as most likely responsible for valenced experience and thus for the markers listed above. The first are evolutionarily ancient subcortical structures, in particular the periaqueductal gray and the mesolimbic dopamine system. We will refer to these structures as the ‘midbrain’ for brevity. The second are evolutionarily more recent integrative brain regions: the cortex in mammals, and functional analogs in other taxa. We will refer to these structures as the ‘pallium’ for brevity.

The midbrain is generally understood to govern anoetic consciousness: the most basic notion of phenomenal consciousness that does not involve self-referential content. The pallium contributes to noetic and autonoetic consciousness, which are more complex forms of phenomenal consciousness. They respectively involve the recognition that one is experiencing and the embedding of that experience in time. While specific theories disagree about whether anoetic, noetic or autonoetic consciousness is necessary to achieve sentience, all sides agree that sentience emerges from the architecture of the brain.¹⁵

¹⁵Some theories hold anoetic consciousness to be sufficient for valenced experience, with pallial mechanisms further elaborating this experience (Panksepp 1998; Merker 2007; Damasio and Carvalho 2013). Competing theories hold that subcortical activity is unconscious and needs cortical re-representation to become an experience of any kind. These theories include, among others, the Global Neuronal Workspace theory (Dehaene et al. 2017), higher-order theory (LeDoux and Brown 2017; Rosenthal 2005), and recurrent processing theory (Lamme 2006). All locate consciousness in different brain mechanisms but share the requirement of cortical involvement. We provide more details in Appendix B.2.

For the determination of sentience based on brain architecture, [Birch \(2024\)](#) proposes four neurological markers. As for the behavioral markers discussed in Section 3.2, none of these markers is strictly necessary or sufficient for sentience. Rather, each criterion that is satisfied raises the probability that the animal is sentient. These criteria are:

5. **Nociceptors:** The animal possesses receptors located in neurons that respond specifically to noxious stimuli.
6. **Integrative brain regions:** The animal possesses brain regions capable of integrating information from various sensory sources.
7. **Integrated nociception:** Neural pathways within the animal link nociceptors to integrative brain regions.
8. **Analgesia:** Behavioral responses to noxious stimuli are modulated by chemical compounds affecting the nervous system.

The evolutionary basis of sentience. The neural architecture responsible for sentience is the product of evolution. These evolutionary roots explain why sentience is likely to be shared by many species. The leading theories of sentience treat valence as an evolved adaptation that serves to motivate fitness-improving actions and discourage fitness-reducing ones ([Ginsburg and Jablonka 2019](#); [Godfrey-Smith 2017](#)). Under this view, valenced experience plays three roles: it represents fitness-relevant information, it serves as a common ‘currency’ in the brain for decision-making, and it facilitates learning by tagging outcomes as good or bad. Therefore, animals with shared evolutionary history are likely to have similar valenced responses, with similar scopes for welfare experience supported by similar nervous systems ([Browning 2019](#)).

3.4 Sentience Candidates

Because sentience can rarely be established with complete certainty, we adopt [Birch \(2024\)](#)’s definition of a sentience candidate, and a candidate for moral status: “A system *S* is a sentience candidate if there is an evidence base that (a) implies a realistic possibility of sentience in *S* that it would be irresponsible to ignore when making policy decisions that will affect *S*, and (b) is rich enough to allow the identification of welfare risks and the design and assessment of precautions.”

Applying the markers 1–8 to the available evidence, [Birch \(2024\)](#) classifies as sentience candidates all vertebrates (mammals, birds, fish, reptiles, and amphibians), adult coleoid cephalopods in the phylum Mollusca, decapods in the suborder Pleocyemata

of the phylum Arthropoda, and adult insects in the phylum Arthropoda, though he notes that the evidence is sparser for insects.

By contrast, decapods of the Dendrobranchiata suborder, including penaeid shrimps, together with insect larvae, spiders, gastropods, and nematode worms are instead “investigation priorities”: they fall short of sentience candidacy, but further evidence could plausibly elevate them to it. Plants and unicellular organisms are not sentient.

3.5 Welfare Ranges and Cognition

With an empirical definition of sentience linked to brain architecture, we are now ready to construct a cardinal measure of sentience for each species. While the moral status of different species has received considerable attention, the existing literature does not provide the cardinalization of welfare ranges that our exercise requires.

To construct a cardinal measure of welfare ranges $\bar{u}_s$ for a large number of species, we start from the idea that species with similar brains will have similar welfare ranges, and hence variation in neural architecture is a principled starting point for understanding interspecies variation in welfare ranges. Following Section 3.3, we take $\bar{u}_s$ to be a function of brain characteristics, which are observed for many species. We posit a general relationship that allows for the midbrain and the pallium to be relevant for sentience and thus for welfare ranges:

$$(6) \quad \bar{u}_s = (z_{m,s}^{\alpha_u} z_{p,s}^{1-\alpha_u})^{\kappa_u},$$

where $z_{m,s}$ is the number of midbrain neurons of species s and $z_{p,s}$ is the number of pallial (cortical) neurons of species s . α_u captures the relative contributions of midbrain and pallial neurons to welfare ranges. κ_u captures the overall scaling in neural architecture.

The problem then reduces to estimating the elasticity of welfare range with respect to brain characteristics. The central question for interspecies comparison is the curvature of this relationship: Do welfare ranges exhibit decreasing, constant, or increasing returns to brain characteristics?

Measuring $\bar{u}_s$ directly to fit this function is not feasible. Existing tests for sentience typically use experimental protocols that are not designed to be comparable across species. Sentience indices typically aggregate binary tests by relying on “scorecard approaches” that take values in a bounded range by construction, which would artificially

compress welfare ranges.

We circumvent this issue by relying on measures of cognition across species. Standardized cognitive tests yield continuous performance measures that are comparable across species. We can thus fit the relationship between test performance and brain characteristics for the species for which such tests exist, and extrapolate it to the remaining species. Cognition is not sentience, as detailed in Section 3.3. However, we claim that using cognition is conservative: the neuroscientific evidence suggests that there is wider dispersion in cognition than in sentience across species, and hence relying on cognition is more likely to understate than to overstate the welfare range of non-human animals. That is, measures of observed cognition will allow us to provide a lower bound on the welfare ranges of non-human animals relative to those of humans.

To formalize the content of this claim, we write cognition as a function of midbrain and pallial neurons, similarly to welfare ranges:

$$(7) \quad c_s = (z_{m,s}^{\alpha_c} z_{p,s}^{1-\alpha_c})^{\kappa_c},$$

where the parameters α_c, κ_c may differ from their counterparts for welfare ranges.

Total neuron count is $z_{tot,s} = z_{m,s} + z_{p,s} + z_{o,s}$, where $z_{o,s}$ is the number of other neurons (e.g., in the cerebellum). For expositional simplicity, and as we show later empirically in Appendix Figure D.1, we posit that neuron counts in each brain region are log-linear in total neuron count:

$$z_{p,s} = z_{tot,s}^{\gamma_p}, \quad z_{m,s} = z_{tot,s}^{\gamma_m}.$$

3.6 Cognition as a Lower Bound on Welfare Ranges

We are now ready to state the main assumption that lets us construct a cardinal lower bound on sentience.

Assumption 1 (Sentience and Cognition).

1. Pallial neurons grow faster with total neuron counts than midbrain neurons: $\gamma_p \geq \gamma_m$.
2. Cognition depends relatively more on the pallium than sentience: $\alpha_u \geq \alpha_c$.
3. Cognition scales more strongly with neuron counts than sentience: $\kappa_u \leq \kappa_c$.

Below, we present the main arguments supporting these assumptions in a concise and nontechnical form. A more detailed discussion of the scientific evidence underlying

these assumptions is provided in Appendix B.4.

The first part of Assumption 1 is that the pallium grows relatively faster with brain size than the midbrain. This assumption reflects the empirical evidence (e.g., Kverková et al. 2022). Indeed, through evolution, increases in brain size are accompanied by disproportionate expansion of the pallium, while the midbrain remains comparatively conserved, causing pallial neuron counts to scale more steeply with total neuron counts.

The second part of Assumption 1 requires that sentience weighs the midbrain more heavily than the pallium, relative to cognition. Neuroscientific evidence indicates that cognition (working memory, abstract reasoning, and planning) occurs primarily in the cortex. By contrast, sentience relies heavily on the midbrain for all anoetic, noetic and auto-noetic theories of sentience (Panksepp 1998; LeDoux and Brown 2017). In addition, sentience is thought to predate flexible cognition evolutionarily, likely implying that it relies more on older brain structures (Panksepp 1998; Damasio 1999; Damasio and Carvalho 2013).

The third part of Assumption 1 encodes the idea that cognition depends weakly more on overall neuron counts than sentience. Neuroscientific evidence suggests that cognition is strongly linked to total neuronal capacity (Herculano-Houzel 2017; Dicke and Roth 2016). Cognitive activity is associated with high-dimensional neural signals, whose complexity rises with neuron count without obvious anatomical limits (Rigotti et al. 2013; Stringer et al. 2019). By contrast, valenced experience is associated with low-dimensional neural signals (Levy and Glimcher 2012). These signals are in turn associated with neurochemical reactions that eventually saturate (Koob and Volkow 2010). These properties are consistent with the role of valence as a common scale for decisions.

With Assumption 1 in hand, we state our main result.

Proposition 1 (Cognition as a Lower Bound on Sentience). *Suppose that humans $s = 0$ have the largest cognition. Under Assumption 1, for all species $s \in \mathcal{S}$:*

$$\frac{d \log \bar{u}_s}{d \log z_{tot,s}} \leq \frac{d \log c_s}{d \log z_{tot,s}} \quad \text{and so} \quad \frac{c_s}{c_0} \leq \frac{\bar{u}_s}{\bar{u}_0}.$$

Proof. $\frac{d \log \bar{u}_s}{d \log z_{tot,s}} = \kappa_u(\alpha_u \gamma_m + (1 - \alpha_u) \gamma_p) \leq \kappa_c(\alpha_u \gamma_m + (1 - \alpha_u) \gamma_p) \leq \kappa_c(\alpha_c \gamma_m + (1 - \alpha_c) \gamma_p) = \frac{d \log c_s}{d \log z_{tot,s}}$, where the first inequality uses Assumption 1.3 and the second inequality uses Assumptions 1.1 and 1.2. The second part of the Proposition follows from integrating the first part between $z_{tot,s}$ and $z_{tot,0}$. $\square$

Proposition 1 formalizes the idea that cognition is a lower bound on sentience. It is conservative in that several of the inequalities involved in the proof of Proposition 1 are likely to be strict.

Of course, Proposition 1 is useful only if we have access to a cardinal measure of cognition. If we observe cognition for all species, there is no further requirement. If we observe cognition for a subset of species, we also need to observe neuron counts for this same subset in order to extrapolate to all species. We thus require:

Assumption 2 (Cognition is Observable). *Either one of the following conditions holds:*

1. Relative cognition $\frac{c_s}{c_0}$ is observable for all species $s \in \mathcal{S}$.
2. Relative cognition $\frac{c_s}{c_0}$ and total neuron counts $z_{tot,s}$ are observable for a subset of species $s \in \bar{\mathcal{S}} \subseteq \mathcal{S}$.

Assumption 2 emphasizes that choosing a measure of cognition sets the scale of the lower bound. Given any measure of cognition, a transformation thereof leads to different values of the lower bound. Thus, choosing a measure of cognition is an important choice and is not innocuous. For this reason, we explore how our results change when we vary the normalization of cognition metrics.

In practice, we observe cognition measures only for a subset of species. We rely on Assumption 2.2 and leverage the relationship between cognition and total neuron counts. Therefore, we estimate relationships of the form:¹⁶

$$(8) \quad \log c_s = \log a + \beta \log z_{tot,s} + \epsilon_s, \quad s \in \bar{\mathcal{S}},$$

and use the estimated coefficient β to extrapolate to all species. Combining an estimate of β from equation (8) with Proposition 1, we obtain our lower bound:

$$(9) \quad \frac{\bar{u}_s}{\bar{u}_0} \geq \left(\frac{z_{tot,s}}{z_{tot,0}} \right)^\beta, \quad s \in \mathcal{S}.$$

¹⁶When the relationships of midbrain and pallial neurons to total neurons have perfect predictive power with exactly the same exponents across all species, we can equivalently consider the elasticity of cognition with respect to total or pallial neurons and use the predicted values. In practice, the R-squared values of these relationships are not exactly one, and the scaling exponents do vary slightly across taxa. In particular, γ_p is somewhat larger for primates than for other mammals due to a larger number of pallial neurons relative to total neurons for this order. In that case, it may be more conservative to estimate the elasticity of cognition with respect to pallial neurons, which are the best predictor of cognitive performance across species (MacLean et al. 2014; Herculano-Houzel 2017). We use both strategies in practice and obtain similar results in Appendix Figure F.2.

3.7 Cardinalizing Cognition

Which measures of cognition should one use? Our main data consist of success probabilities p_{js} on a wide range of tests j at the species level s . Consider the following examples of cardinalization. One may define cognition as the success probability, or its logarithm, on a particular test k , $\log c_s = \log p_{ks}$. One may also aggregate across all tests and use instead $\log c_s = \mathbb{E}[\log p_{js}|s]$.

These examples highlight two challenges. The first challenge is that any nonlinear transformation of success probabilities can in principle be used to define cognition. This first challenge cannot be fully resolved with our data alone because we do not observe the fundamental measure of sentience. Thus, any measure of cognition necessarily involves researcher discretion. The second challenge is how to aggregate across the broad range of tests at our disposal.

We address both challenges by considering a broad range of possible cardinalizations, adopting the most conservative assumptions within the scope of the data, and exploring sensitivity beyond the scope of the data.

Specifically, to address the first challenge, we start by considering three cardinalizations that map observed success probabilities p_{js} into test-specific cognition c_{js} .

Cardinalization 1: identity. The first map is the identity:

$$(10) \quad p_{js} = c_{js}, \text{ or, equivalently, } \log c_{js} = \log p_{js}.$$

The identity map has the advantage of simplicity and transparency. However, it does not correspond to an underlying model of test success at the individual level. In particular, $p_{js} = c_{js}$ does not need to remain bounded below one for all possible values of cognition. Of course, choosing the identity rather than an isoelastic power function is a choice.

Cardinalization 2: logistic. The second transformation we consider is the logistic function:

$$(11) \quad p_{js} = \frac{c_{js}}{1 + c_{js}}, \text{ or, equivalently, } \log c_{js} = \log \frac{p_{js}}{1 - p_{js}}.$$

This transformation corresponds to an underlying model in which each individual of species s chooses between two options: a behavior A that leads to success at test j , and a behavior B that leads to failure. Succeeding at test j leads to a reward $R > 0$, while

failing at it delivers 0. The individual receives a signal $\log c_{js} + \varepsilon_{js}$ that behavior A leads to success, and a signal ε'_{js} that behavior B leads to success. Higher cognition leads to a more accurate set of signals. When $\varepsilon_{js}, \varepsilon'_{js}$ are independent and Gumbel distributed with scale parameter 1, choosing the behavior with the higher signal leads to success with the probability given in equation (11). Of course, setting the scale parameter of the signal draws to 1 is a choice. Without this normalization, an isoelastic transformation of cognition is also admissible.

Cardinalization 3: Poisson trials. The third transformation we consider is the Poisson trials function:

$$(12) \quad p_{js} = 1 - e^{-c_{js}}, \text{ or, equivalently, } \log c_{js} = \log(-\log(1 - p_{js})).$$

This transformation corresponds to an underlying model in which each individual of species s successfully attempts the test at Poisson rate $c_{js} > 0$. There is a unit amount of time to succeed at the test. The number of successful attempts is Poisson distributed with mean c_{js} , so that at least one attempt succeeds with the probability given in equation (12). Once more, any isoelastic transformation of cognition c_{js} would also be admissible, so that using the particular normalization in (12) is a choice.

In our baseline results, we use the most conservative cardinalization, which implies the largest slope β between log cognition and log neuron counts. Selecting the cardinalization with the largest value of the slope β ensures that cognition, and thus sentience, falls most rapidly as neuron counts decline.

Nevertheless, in all three cardinalizations, an isoelastic transformation of cognition is admissible. Therefore, we systematically report sensitivity with respect to our estimate of the slope β between cognition and neurons. This sensitivity analysis is isomorphic to applying an isoelastic transformation to cognition, and thus relaxing the underlying normalization. To understand why, suppose that we apply the transformation $c'_s = c_s^\nu$, where c_s is observed cognition as measured from Cardinalizations 1, 2 or 3, and c'_s is true cognition. Equation (8) becomes: $\log c'_s = \nu \log a + \nu \beta \log z_{tot,s} + \nu \varepsilon_s$. Therefore, varying β around its estimate is isomorphic to varying ν around 1.

With a test- and species-specific measure of cognition c_{js} in hand, we are ready to address the second challenge: aggregating across multiple tests. We start by estimating the test-specific relationship between neuron counts and cognition as in equation (8) and obtain test-specific slopes β_j .

Our baseline and most conservative cardinalization selects the maximum of these

test-specific slopes as the relationship between neuron counts and overall cognition: $\beta^{\max} = \max_j \beta_j$, which we then use in the lower bound given in equation (9). Selecting the largest value of the slopes β_j ensures that cognition, and thus sentience, falls most rapidly as neuron counts decline.

Our second approach is to estimate a pooled regression across multiple tests j and species s , of the form:

$$(13) \quad \log c_{js} = \log a_j + \beta^{\text{pooled}} \log z_{\text{tot},s} + \epsilon_{js}.$$

We then use the pooled estimate β^{pooled} in the lower bound given in equation (9).

We consider three estimators for equations (8) and (13): Ordinary Least Squares (OLS), Poisson Pseudo-Maximum Likelihood (PPML) and Generalized Linear Model (GLM). OLS is the most transparent, but drops test-species pairs where the success probability is zero in the data. PPML drops test-species pairs where the success probability is one in the data. GLM does not drop either, and generally produces conservative estimates of scalings β . For both these reasons, we use GLM as our baseline estimator, and report our results under OLS and PPML in the Appendix.

Equipped with an approach to construct an empirical bound on welfare ranges, we discuss in Section 4 the animal population data, the cognition data, and the neuron count data that we use to estimate all the components of the welfare cost of biodiversity loss.

4. DATA AND MEASURES OF SENTIENCE

4.1 Population Data

We begin by collecting and harmonizing data on population-level trends for wild animals at different levels of aggregation. We restrict our analysis to vertebrates (Chordata), mollusks, and arthropods, as these correspond to the criteria for sentience candidates and moral status as discussed in Section 3.4. We construct a hierarchical population dataset at the species, order, and group level. We combine estimates of current population levels with estimates of relative population change over time to estimate the absolute change in population size.

Current population levels. For mammals and birds, we obtain species-level population counts from Greenspoon et al. (2025) and Wikipedia’s compilations of species

counts after verifying its sources. We obtain a total count for mammals by aggregating species-level counts. For birds, we rely on the total count in [Bar-On et al. \(2018\)](#). For fish, we build our own counts: mesopelagic and non-mesopelagic fish from the biomass estimates of [Bar-On et al. \(2018\)](#), and freshwater fish separately, as detailed in Appendix C.4. For arthropods, we obtain a count from [Bar-On et al. \(2018\)](#) and a separate count for insects from [Smithsonian Institution \(1996\)](#). Given the lack of data for herptiles, we develop our own estimate, detailed in Appendix C.5. We provide validations of these estimates in Appendix C.3.

Cephalopods (belonging to mollusks) and decapod crustaceans (belonging to arthropods) belong to our list of sentience candidates. However, no source allows us to enumerate their populations within the broader arthropod and mollusk taxa. Therefore, they are dropped from our analysis.

Population changes. Our main data source is the data underlying the Living Planet Index (LPI). Maintained by the Zoological Society of London and WWF, these data contain time-series population data for 29,559 populations across 4,844 vertebrate species, from 1970 to the present. The Living Planet Index is the most widely used global measure of change in the abundance of wild vertebrates. It is one of the indicators the Parties to the Convention on Biological Diversity adopted in 2022 to monitor the Kunming-Montreal Global Biodiversity Framework, including its goal on the abundance of wild species ([Convention on Biological Diversity 2022](#)). It informs intergovernmental assessments such as the IPBES global assessment ([IPBES 2019](#)), and its method and database underlie many official national and regional indicators.

We replicate the LPI methodology to aggregate these population time series for each major vertebrate group (mammals, birds, fish, herptiles), as well as at the order level for well-covered orders (supported by more than 100 population series). The LPI has a number of known caveats, discussed in Appendix C.1. To validate our estimates of population changes at the group and order level, we conduct extensive cross-checks of the LPI-estimated changes with other sources (see Appendix Table C.3). This analysis shows that LPI-estimated changes have the same order of magnitude as alternative sources, and Appendix Table C.2 shows that they remain similar when trimming outlying population series.

The LPI does not cover invertebrates. We obtain an estimate of the insect population change from [Dirzo et al. \(2014\)](#). We complement the LPI with species-level population changes for some mammals, obtained from [Greenspoon et al. \(2025\)](#). Finally, for farmed

TABLE 1: Current Populations of Major Animal Taxa

TAXON	CURRENT POP.	HISTORICAL POP.	DECLINE (%)
Birds	3.00×10^{11}	$5.49 (3.82\text{--}7.73) \times 10^{11}$	45 (21–61)
Fish	8.07×10^{15}	8.66×10^{15}	7 (6–7)
Herptiles	1.45×10^{12}	$1.45 (0.53\text{--}4.16) \times 10^{13}$	90 (73–97)
Insects	1.00×10^{19}	1.82×10^{19}	45
Mammals	7.84×10^{10}	$14.5 (8.6\text{--}26.0) \times 10^{10}$	46 (9–70)

Notes: Current population refers to 2020. Historical population refers to 1970. Confidence bounds for the LPI estimates use the bootstrap procedure defined in the LPI technical documentation.

animals, we use data from the Food and Agriculture Organization Corporate Statistical Database (FAOSTAT).

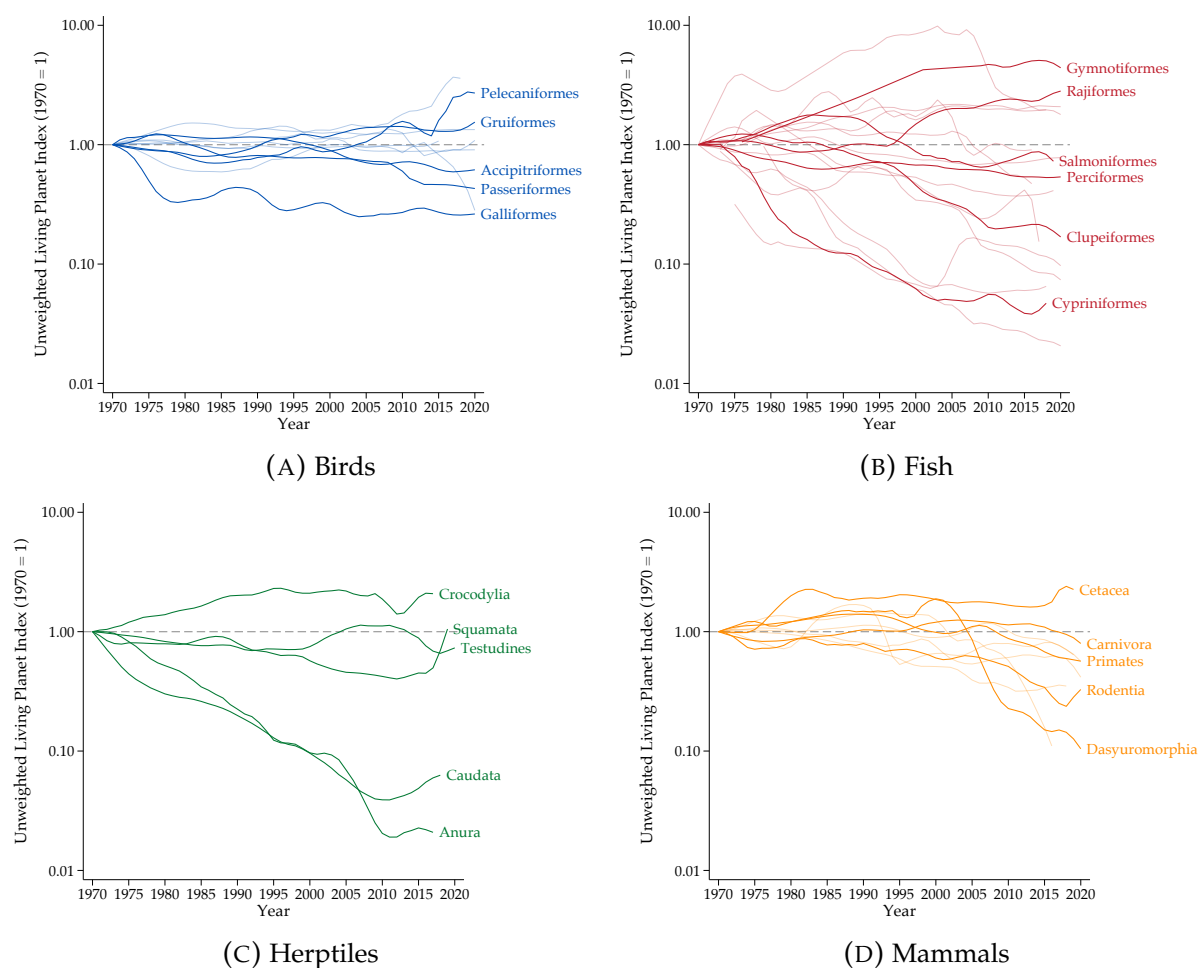
Final population dataset. We combine present population counts and changes to estimate animal population losses since 1970. We perform this calculation in two ways. Our first approach prioritizes the most granular level of disaggregation: we use species-level counts and changes when available for mammals, otherwise fall back to order-level changes, which are valid only for mammals and birds, and then fall back to changes for the major taxon when a species is not covered at the species or order level. Our second approach uses only changes and counts for the major taxa. Reassuringly, both approaches deliver similar results.¹⁷

Table 1 summarizes the data at the level of the five major taxa we consider. The most numerous group is insects, outnumbering the other groups by many orders of magnitude. Mammals are the least numerous group, on the order of 10^{10} . These substantial disparities in baseline levels are an important quantitative determinant of the results below. Therefore, we explore the robustness of our conclusions to excluding insects from our welfare calculations.

Table 1 also reports population changes at the taxon level since 1970. All taxa, from the least to the most numerous, have experienced large population declines. Figure 1 plots population declines at finer taxonomic levels. Most orders have seen substantial population declines since 1970, ranging from 5 to 10% for carnivores and primates to 90% for frogs and salamanders. Cetaceans and Crocodylia are among the few orders to have increased in population, the former likely due to the cessation of commercial

¹⁷For fish, the LPI population series mostly cover marine fish species living less than 200 meters deep and freshwater fish. There are very few data on mesopelagic fish, living deeper in the sea. To be conservative, our baseline assumes that the LPI population decline applies only to non-mesopelagic and freshwater fish, and assumes no change for mesopelagic fish. In robustness checks, we present results where we assume the LPI population change applies to all fish (see Appendix Figure F.4).

FIGURE 1: Population Declines for Vertebrates: Order Level



Notes: Global order-level population trends for birds, fish, herptiles and mammals. Data restricted to orders with more than 100 population time series.

whaling from an initially low population in 1970.

By contrast, farmed animals have seen population increases since 1970, particularly for chickens (Appendix Figure C.1).

Despite our efforts to validate the estimates of population levels and decline across taxa (see Appendix C), we want to emphasize that there remains considerable uncertainty about both of these variables. This reflects both genuine scientific disagreement and the difficulty of measuring populations for species that are not a conservation or management priority of humanity. We take the best available evidence, but as the science progresses, the exact magnitudes resulting from applying our methodology will change by necessity.

4.2 Spatial Ranges

To study the current distribution of sentience across space, we assign every wild vertebrate species its present-day population and impute a uniform density over its geographic range on an equal-area grid of 25-km cells. Ranges come from the expert polygons of the IUCN Red List ([IUCN 2025](#)) for mammals, herptiles and freshwater fish and of BirdLife International ([BirdLife International and Handbook of the Birds of the World 2025](#)) for birds, and predicted occurrence envelopes for marine fish as described below. Further details on the ranges and the sources for estimates of species-level counts for each taxon are provided in Appendix [C.8](#).

The resulting species distribution maps are shown in Figure [2](#). The concentration of both mammals and herptiles is highest in the Amazon basin and sub-Saharan Africa. Herptiles also appear to be abundant in Australia, much more so than mammals, given a more suitable climate for cold-blooded animals.¹⁸ Birds are more evenly spread across the global surface, though with notable exceptions.

Freshwater fish, unlike other taxa, are particularly abundant in the Canadian boreal zone and northern Europe. The highest concentration of animals, by an order of magnitude, occurs in the Northern and Southern Oceans, given the abundance of marine fish present there, in turn dependent on high plankton counts through the food web.

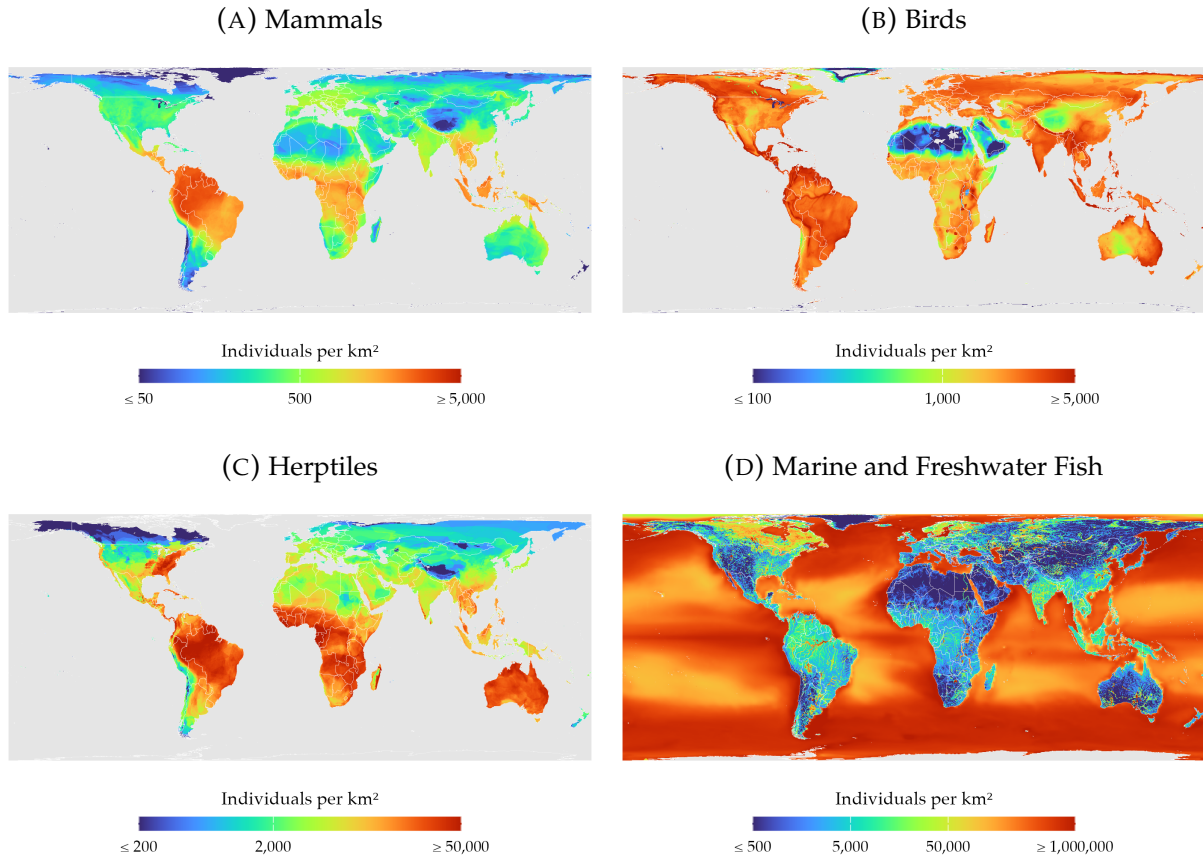
4.3 Neuron Counts

With measures of animal populations in hand, we collect data on both the total number of neurons and the number of pallial neurons across animal species. We harmonize data from a variety of recent studies ([Kverková et al. 2022](#); [Herculano-Houzel et al. 2015](#); [Sol et al. 2022](#); [Olkowicz et al. 2016](#); [Marhounová et al. 2019](#); [Ahrens et al. 2013](#); [Teles et al. 2025](#); [Barron and Mourmourakis 2023](#)) and from [Wikipedia](#) after verifying its sources. Appendix Table [D.1](#) describes these data sources in more detail.

While extensive, these data do not have complete coverage. Appendix Table [D.2](#) lists all the assumptions we use to interpolate missing data. We detail the main assumptions in this section. We rely primarily on total neurons for our analysis. We do not lose much information in doing so, as Appendix Figure [D.1](#) indicates that pallial and total neurons follow a tight log-log relationship across a wide range of species with an

¹⁸In addition, [Morton and James \(1988\)](#) hypothesize that the abundance of spinifex grass and infertile soils in Australia leads to large termite populations, which in turn favor small reptiles.

FIGURE 2: Current Density of Wild Animals by Group



Notes: Individuals per km² on an equal-area grid of 25-km cells, with each species spread uniformly over its range at its assigned population. Panel (d) combines marine fish outside the mesopelagic zone, from the BOATS model scaled to our baseline count, with freshwater fish expressed per km² of land. Each panel has its own logarithmic color scale with limits at the 5th and 95th percentiles of mapped cells. Cells beyond them take the end colors. See Appendix Figures C.2–C.3 for separate maps for marine and freshwater fish, and for reptiles and amphibians.

R-squared of 0.96. In addition, we observe neither total nor pallial neurons for some species. In this case, we appeal to cellular scaling laws. The biological literature finds that within taxonomic orders (but not across), the number of cells and neurons across species is well explained by a power function of body mass (Herculano-Houzel 2011; Herculano-Houzel et al. 2006, 2007; Sarko et al. 2009). Therefore, within each order for which we have at least three species with a total neuron count, observed or predicted from pallial neurons, we fit a log-log regression of total neurons on body mass. We fit whales and dolphins separately from other even-toed ungulates, and we exclude humans from the primate regression. We then use this regression and each species' body mass to predict the total number of neurons for species for which we observe neither total nor pallial neurons.¹⁹

¹⁹We take body masses from PanTHERIA (Jones et al. 2009), EltonTraits (Wilman et al. 2014), and

4.4 Cognitive Tests

With measures of neuron counts in hand, we collect information on cognitive performance. We collect a wide range of cognitive tests from the literature, detailed in Appendix Table D.3. Among those tests, we consider two groups.

The first group consists of tests that are broad enough to compare taxonomic groups to each other and thus provide a widely representative estimate of the elasticity of cognition to neuron counts. We consider tests that cover at least four taxonomic groups and whose test-level success rate includes at least 15% of interior probabilities. These restrictions lead us to leverage the following tests: the ‘A-not-B task’ and the ‘cylinder task’ (MacLean et al. 2014; Barron and Mourmourakis 2023), string-pulling tests, and numeration tests. These tests have been performed on species from a broad range of taxa, including great apes, lemurs, monkeys, rodents, corvids, other birds, and fish. Appendix Table D.4 lists our data sources for these tests.

The A-not-B task consists of first familiarizing an animal with finding food in one location (location A). In the test trial, the animal sees the food hidden in the same location (location A) but then moved to a new location (location B) before it is allowed to search. The cylinder task consists of familiarizing an animal with finding food hidden inside an opaque cylinder. The cylinder is then replaced by a transparent cylinder. To successfully retrieve the food, the animal needs to inhibit the impulse to reach for the food directly (bumping into the cylinder) and instead take the detour to the open ends. Both the A-not-B and cylinder tasks are cognitively demanding because they require subjects to inhibit a previously reinforced or prepotent response and select an alternative action. Performance depends strongly on the development of cortical, and especially prefrontal, systems supporting working memory and inhibitory control.

In the vertical string-pulling task, food hangs at the end of a string attached to a perch. To retrieve it, the animal must pull the string up and secure each pulled length, for example with a foot, before pulling again. The task assesses whether animals understand the causal connection between the string and the food. It has been performed on 91 species, including primates, other mammals, corvids, and many other birds (Jacobs and Osvath 2015).

Numeration tests assess whether animals can discriminate between different quantities. Typically, an animal must distinguish two sets of objects differing in number, for example choosing between 1 and 3 objects. One can increase the difficulty by making

FishBase (Froese and Pauly, eds 2026).

the ratio of the two sets of objects closer to 1 (for example 7 vs. 8). These experiments have been conducted on a range of species, ranging from honeybees to chimpanzees. We aggregate 26 different studies and detail the data we use from each one in Appendix Table D.6.

An important concern is that these tests may not accurately represent cognition because adult humans succeed trivially at them. As a result, projecting success rates on neuron counts would underestimate the gap in cognition between humans and other species, and thus, sentience.

Our second group of tests is designed to alleviate this concern. It consists of tests that we use for our piecewise analysis at the top of the cognition distribution. We restrict those tests to include young humans and primates and cover more than five species. These restrictions lead us to focus on the Primate Cognition Test Battery (PCTB) and the reversal learning test.

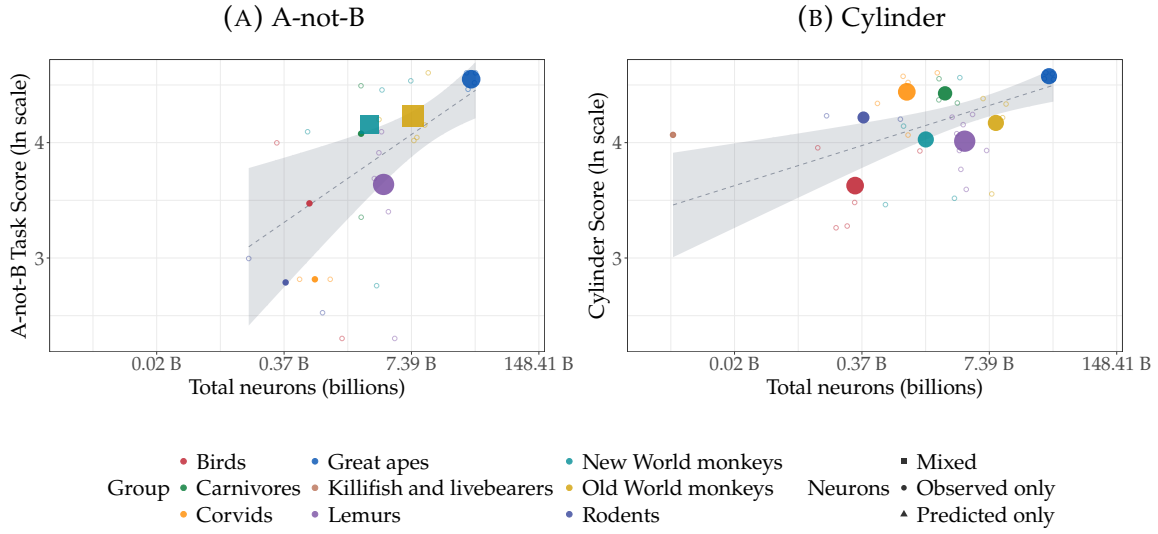
The PCTB consists of a collection of cognition tests introduced to compare primates and young humans, and recently extended to parrots (Krasheninnikova et al. 2019). The tests are designed to assess a broad range of abilities such as understanding of space, quantities, causality, social learning, and communication (detailed in Appendix Table D.5). They assess a broader range of cognitive abilities, but on a narrower range of species. In the reversal learning test, the animal first learns that one of two options is rewarded, after which the reward is moved to the other option. The test assesses how quickly the animal abandons the previously rewarded choice. It has been performed on 29 species, including humans, other primates, other mammals, and birds (Dorschner et al. 2024). In turn, society typically treats young humans as having similar sentience to adult humans. We leverage these tests to assess the robustness of our results: we design a piecewise approach that allows for more convexity in welfare ranges at the top of the distribution.

4.5 Neuronal Capacity and Cognition

Equipped with measures of test performance and neuron counts for a subset of species, we are ready to leverage equation (8). We estimate the relationship between cognition and total neuron counts separately for each test, before aggregating.

Figure 3 plots the relationship between log neuron counts and log performance on the A-not-B and cylinder tests at the species and taxonomic group level, under the identity cardinalization. These plots show that cognitive performance is increasing in

FIGURE 3: Cylinder and A-not-B Scores as a Function of Total Neurons



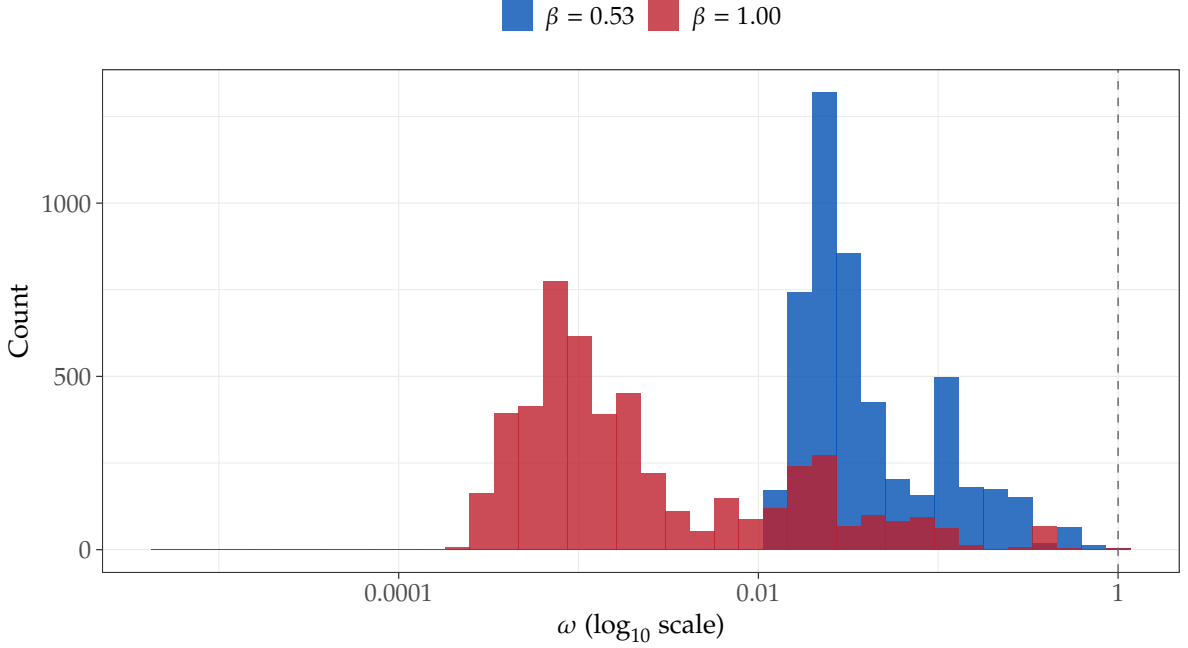
Notes: Panel a: A-not-B test score versus total neuron count (ln–ln). Panel b: cylinder test score versus total neuron count (ln–ln). Large points correspond to group-level means, small unfilled points correspond to individual species. The fit line is estimated at the species level using GLM, under the identity cardinalization $c = p$. The shaded area is its 95% confidence band. See Appendix Figures E.1 and E.2 for species-level plots and pallial neurons.

the number of total neurons. Relating log cognition at the A-not-B test to the log total neurons as per equation (8), we obtain slopes of 0.25, 0.53 and 0.39 under the identity, logistic and Poisson cardinalizations, respectively, all of which are statistically different from 0 at the 5% level. The same regression for the cylinder test leads to slopes between 0.12 and 0.23, which are also statistically significant.

We estimate these relationships at the species level. While individual species exhibit some dispersion around the fitted line, the fit is tight at the group level. Appendix Figures E.1 and E.2 display the scatter plots at the species and group levels, for both total and pallial neurons.

Appendix Figure E.3 plots the relationship between total neuron counts and performance on the numeration tests. The estimates of the slope are always smaller than for the A-not-B task. One reason is that insects, which have very few neurons, tend to perform very well on numeration tests. Estimates for the string-pulling task are also smaller than for the A-not-B task under each cardinalization. A larger slope implies a stronger decay of cognition with neuron count. Therefore, to remain conservative, we use the largest estimate across tests and cardinalizations. It corresponds to the A-not-B test under the logistic cardinalization, $\hat{\beta}^{A-B} = 0.53$, which we use as a conservative value for the relationship between cognition and neuron counts across species. In what follows, we denote it by β for simplicity.

FIGURE 4: Relative Welfare Ranges



Notes: Histograms of welfare capacity relative to humans $\omega_s = (z_{tot,s}/z_{tot,0})^\beta$ for $\beta \in \{0.53, 1\}$. These histograms use observed or predicted neurons at the species level for mammals, means at the order level for birds, and means at the group level for other groups. The dashed line corresponds to $\omega = 1$: welfare capacity equal to that of humans.

4.6 Conversion into Utility

Equipped with a baseline estimate of β , we now leverage equation (6) and construct our lower bound on sentience, or welfare ranges, for all groups relative to humans: $\omega_s \equiv \bar{u}_s/\bar{u}_0 = (z_{tot,s}/z_{tot,0})^\beta$. Figure 4 displays a histogram of ω_s . We first display this histogram for our baseline estimate $\beta = 0.53$. In that case, many animals have sentience above 0.1 times human sentience.

We also include a larger value of β to illustrate its importance. Changing β is isomorphic to rescaling cognition, and thus by varying β we explore the range of alternative cardinalizations of cognition. A smaller value of β implies that all species are similar to humans. A larger value of β implies that species with fewer neurons are substantially less sentient than humans.²⁰ Varying β equivalently spans the range of values implied by our various cognitive tests. We display the histogram using $\beta = 1$ as a natural alternative in which sentience is linear in neuron counts. When we use this larger value, the distribution of sentience shifts to the left: few species achieve

²⁰Three species have more total neurons than humans in our data: the African elephant, whose count is measured, and the African forest elephant and the killer whale, whose counts are imputed. We cap their relative welfare range at 1, the human value, so that they do not drive our results.

sentience above 0.1 times that of humans. Most have sentience less than 0.01 times that of humans.

Quality of life adjustment. Each species enters the welfare function with weight $\lambda_s \times \tilde{u}_{st} \times \bar{u}_s$, where the term $\tilde{u}_{st}$ captures realized welfare relative to potential: quality of life. We assume that in a state of nature, denoted $t = 0$, humans and all animals had equal realized welfare relative to potential, i.e., $\tilde{u}_{s0} = \tilde{u}_{00}$ for all species $s \in \mathcal{S}$. Since then, human quality of life has improved dramatically due to economic growth.

We thus proceed in two steps. First, we perform our exercise without accounting for economic growth. In this case, $\tilde{u}_{st}$ is constant for all species s and time period t , including humans. The chosen level of quality of life $\tilde{u}_{s0}$ is inconsequential, since only animal welfare relative to human welfare matters, which turns out to coincide with animal sentience relative to human sentience. In particular, $\tilde{u}_{s0}$ may be arbitrarily small if life in the state of nature is so difficult that wild lives have close to zero net welfare.²¹ Second, in Section 5.5, we account for the diverging quality of life trends between humans and animals due to economic growth.

Behavioral markers of sentience. How do our estimates of sentience compare to other measures based on behavior? For a few select species, Fischer (2023) evaluates whether they display a range of ninety behavioral sentience proxies similar to those in Section 3.2. These include cognitive proxies such as communication, performance on cognitive tests, and gratification delay, as well as hedonic proxies such as anxiety, fear, and joy-like behavior. The number of proxies exhibited by each species is divided by the number exhibited by humans, with various scaling assumptions, such as using the cube of the number of proxies, or only using high-confidence proxies. The score is interpreted as the welfare range of a species relative to that of a human.

The advantage of this approach is that it incorporates rich information that relates to sentience. At the same time, it is more difficult to scale and involves a certain amount of unavoidable judgment in converting behavior into a fraction of human sentience. In contrast, our approach is scalable, based on observed cognition, and clarifies the assumptions behind the mapping from cognition to sentience.

²¹There is an active debate over whether the average welfare of wild lives is even positive (Groff and Ng 2019; Browning and Veit 2023). We take positive but arbitrarily low realized welfare, together with symmetry between humans and other animals in the state of nature, as a natural benchmark. While it is impossible to know whether humans had positive average welfare in the state of nature, the few studies of happiness in hunter-gatherer, forager-horticulturalist and semi-nomadic pastoral societies with limited contact with the outside world find average levels above the midpoint of the scale (Frackowiak et al. 2020; Reyes-García et al. 2021; Martin and Cooper 2017) (and the two studies that include a comparison group find them higher than in samples from wealthy countries).

We compare our sentience measures to those in [Fischer \(2023\)](#) in Appendix Table C.11. The measures correlate well with each other. Our measure is uniformly lower, making it once more a conservative lower bound.

4.7 Merged Population and Sentience Data

The final merged dataset pairs each animal group’s population change from 1970 to today with a neuron count, yielding a proxy for its sentience as described above.

Each group is described at a different resolution, matched to the data available. Mammals contain one entry per species. Birds are represented by one entry for each order with eligible order-level population and decline data, together with one class-wide residual entry for the remaining birds. For both groups, the data construction falls back to order- or class-level averages when more disaggregated data are not available. Fish, herptiles, and insects contain a single entry at the class level, pairing the aggregate population loss and average neuron counts for these groups. The merged data construction is detailed in Appendix D.2.

5. THE COST OF BIODIVERSITY LOSS

5.1 Aggregate Sentience Changes

For each group g , we consider the sentience gain or loss between 1970 and t relative to a counterfactual where species s ’s population remains constant. As described in Section 4.6, the aggregate sentience gain or loss of group g is:

$$\Delta \mathcal{W}_{gt}(\mathbf{1}) = \sum_{s \in g} \bar{u}_s \tilde{u}_{st} (N_{s,t} - N_{s,1970}).$$

It corresponds to the contribution of group g to welfare if its welfare weight were equal to 1. We then express aggregate sentience losses relative to the gain in aggregate human sentience:

$$(14) \quad \frac{-\Delta \mathcal{W}_{gt}(\mathbf{1})}{\Delta \bar{\mathcal{W}}_{0t}} = \sum_{s \in g} \frac{\bar{u}_s}{\bar{u}_0} \frac{\tilde{u}_{st}}{\tilde{u}_{0t}} \frac{N_{s,1970} - N_{s,t}}{N_{0,t} - N_{0,1970}} \geq \sum_{s \in g} \omega_s \frac{N_{s,1970} - N_{s,t}}{N_{0,t} - N_{0,1970}},$$

where $\bar{\mathcal{W}}_{0t}$ denotes human welfare holding quality of life fixed at $\tilde{u}_{0t} = \tilde{u}_{00}$, and we assume fixed and equal animal and human quality of life $\tilde{u}_{st} = \tilde{u}_{00}$ for all species s and

time periods t . The inequality exploits our lower bound and applies to groups whose populations decline. We introduce the welfare effect of improvements in human quality of life due to economic growth in Section 5.5.²²

We report our results for various values of β , the elasticity of sentience to neuron counts. Varying β simultaneously spans alternative cardinalizations of cognition and different choices of cognitive tests. Our baseline is conservative: we use the largest value of β across the A-not-B, cylinder, string-pulling and numeration tests, which cover the most species, and across the three cardinalizations. This choice yields $\beta = \hat{\beta}^{A-B} = 0.53$, obtained under the logistic cardinalization. Figures 5, 6 and 8 also mark three less conservative alternatives from Section 3.7: the mean of the test-specific slopes across tests and cardinalizations (0.09), and the pooled slopes under the logistic (0.06) and Poisson (0.04) cardinalizations.

5.2 The Aggregate Sentience Cost of Biodiversity Loss

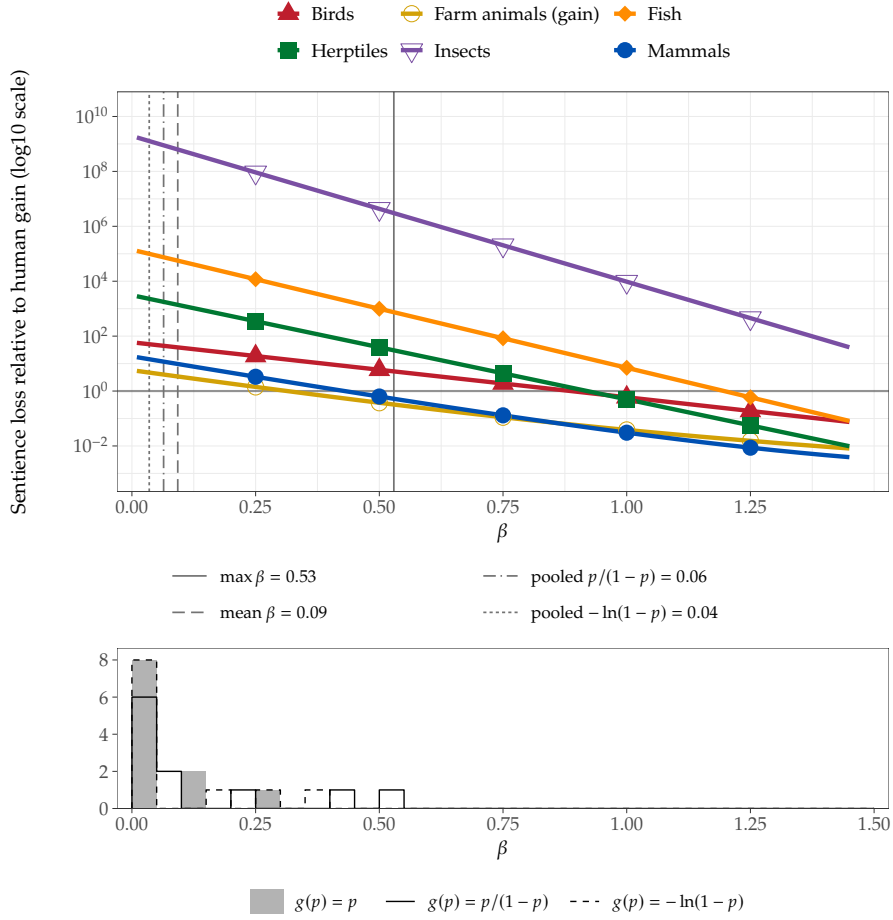
We start with aggregate sentience gains or losses by animal taxa (14), as a function of β . Figure 5 displays our results. Sentience losses for every taxon except mammals exceed human sentience gains at our baseline estimate $\beta = \hat{\beta}^{A-B}$. Mammal sentience losses are of the same order of magnitude as human sentience gains. For birds alone, the sentience losses are several times larger than human sentience gains. Herptile sentience losses are one order of magnitude larger than human sentience gains, and fish sentience losses two orders of magnitude larger. Insect sentience losses are six orders of magnitude larger. At larger values of β , these gaps fall. Yet, even when sentience scales linearly with neuron counts ($\beta = 1$), sentience losses for fish remain larger than human sentience gains, and losses for insects remain several orders of magnitude larger than human sentience gains. Herptile sentience losses fall below human sentience gains at $\beta = 1$.

To assess whether $\beta = \hat{\beta}^{A-B}$ or $\beta = 1$ are plausible values, we additionally plot the histogram of estimates of β across all the cognitive tests that cover multiple taxonomic groups and across the three cardinalizations. Our baseline value $\beta = \hat{\beta}^{A-B}$ is already conservative: no other test or cardinalization implies a stronger scaling of cognition

²²Alternatively, we can express sentience changes relative to the sentience of a single human:

$$\frac{-\Delta \mathcal{W}_{st}}{\bar{u}_0} = \frac{\bar{u}_s}{\bar{u}_0} (N_{s,1970} - N_{s,t}) \geq \omega_s (N_{s,1970} - N_{s,t}).$$

FIGURE 5: Sentence Losses or Gains by Taxa



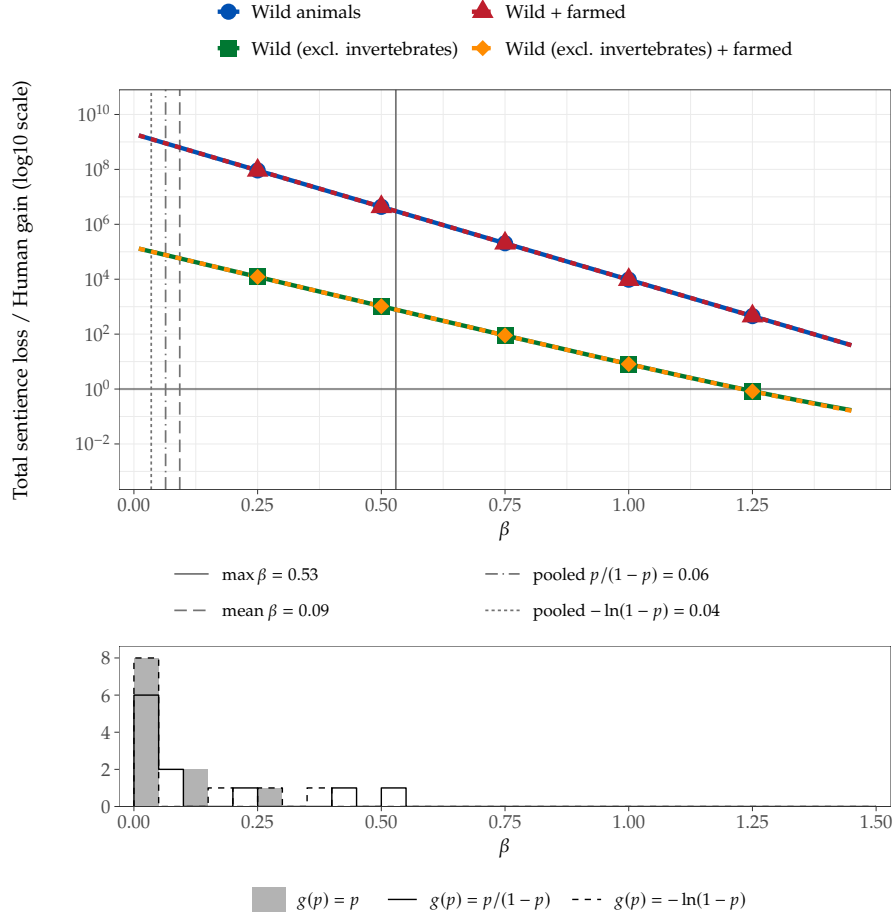
Notes: The y-axis is the sentence loss or gain for each taxon expressed relative to human sentence gains (as defined in (14)). The x-axis is the sentence-to-neurons elasticity β . The “Farm animals” category includes all farmed animals, irrespective of taxon. We use population data at the most granular level of disaggregation available. The histogram plots the sentence-to-neurons elasticity estimated using GLM for various cardinalizations of cognition $c = g(p)$. The vertical lines correspond to the maximum and mean β across cardinalizations, and the elasticity estimated from two pooled estimates. See Appendix Figure F.5 for histograms estimated using OLS and PPML.

with neuron counts than the A-not-B test. In particular, choosing $\beta = 1$ would be more conservative than what the data suggest. When considering the pooled cardinalizations $\beta = \hat{\beta}^{\text{pooled}}$ instead, mammal losses alone are 10 times larger than human gains.

So far we have estimated slopes using a Generalized Linear Model (GLM) estimator. It is a close variant of Nonlinear Least Squares and allows us to retain all observations, including zeros and ones, across all cardinalizations. Appendix Figure F.5 shows that the distribution of estimates is similar when we estimate the elasticity by OLS or PPML instead of GLM: the largest estimate is 0.62 under OLS and 0.39 under PPML, close to and on either side of our baseline of 0.53.

Figure 5 also depicts sentence gains from rising farmed animal populations. This

FIGURE 6: Total Animal Sentience Loss Relative to Human Gain



Notes: The y-axis is the sentience loss or gain for all taxa expressed relative to human sentience gain (as defined in (14)). The x-axis is the sentence-to-neurons elasticity β . The “farmed” series add all farmed animals, irrespective of taxon. We use population data prioritizing the most granular level of disaggregation. The histogram plots the sentence-to-neurons elasticity estimated using GLM for various cardinalizations of cognition $c = g(p)$. The vertical lines correspond to the maximum and mean β across cardinalizations, and the elasticity estimated from two pooled estimates. See Appendix Figure F.5 for histograms estimated using OLS and PPML.

group represents a marginal adjustment to positive contributions to total sentience. While farmed animals represent an important proportion of biomass on Earth, their sentience remains below that of humans. Therefore, they represent at most a doubling of positive contributions to sentience. By contrast, wild animals represent a smaller proportion of biomass but substantial aggregate sentience because small animals can be strongly sentient. Because the sentience costs of biodiversity loss are orders of magnitude larger than human sentience gains, farmed animals make little difference to this comparison. If we further adjusted their contribution to sentience due to their arguably lower quality of life, their positive contribution to aggregate sentience would be even smaller.

Next, we aggregate sentience across animal taxa following equation (1). Figure 6 shows total animal sentience losses relative to human sentience gains. We consider four cases. First, we include all wild animal welfare losses. Second, we consider all wild animal welfare losses of vertebrates, but exclude insects, whose sentience is less clearly established. Third, we include the combined welfare change of wild and farmed animals. Fourth, we include the combined welfare change of wild animals and farmed animals, excluding insects.

We find that animal sentience losses exceed human sentience gains by multiple orders of magnitude for all cases and across all plausible values for β . At our baseline estimate of $\beta = 0.53$, the sentience costs of biodiversity loss are more than 100 times larger than human sentience gains excluding invertebrates. When we include invertebrates, the ratio rises to more than one million. At the conservative value of $\beta = 1$ (with less empirical support), the sentience cost of biodiversity loss remains more than 8 times larger than human sentience gains after excluding insects. Including insects makes this ratio exceed 1,000.

In the Appendix, we assess how our results change as we vary assumptions. First, we use more aggregated population data for mammals and birds, defining neuron counts for each taxon as either the mean or the minimum of observed neuron counts within that taxon (Figure F.1). Second, we use relative welfare capacities estimated from pallial neuron counts, as opposed to total neuron counts (Figure F.2). Third, we apply the relative fish population decline from the LPI to all fish as opposed to excluding mesopelagic fish (Figure F.4). The sentience costs of biodiversity loss exceed human welfare gains by multiple orders of magnitude across all these specifications: at our baseline estimate of β , they remain more than 100 times larger than human sentience gains excluding insects.

5.3 Top Sentience Convexity

Our results rely on the estimated mapping between neuron counts and sentience. Figures 5, 6 and 8 already show how our results depend on the convexity β of this mapping. However, one might reasonably suspect that this mapping is not exactly log-linear. In particular, the relationship between neuron counts and cognitive abilities may steepen as we go from the smartest animals to humans.

Excess convexity at the top might be difficult to detect empirically: most cognitive tests are designed so that they might be completed by animals and humans alike. For

instance, our cognitive tests do not include inverting a 3×3 matrix by hand, at which humans might succeed but most animals would likely fail (although humans usually require extensive prior training to even take this sort of test). These considerations suggest that our estimate of β might be biased downward at the top of the neuron count distribution.

To gauge whether this concern is likely to affect our conclusions, we leverage the Primate Cognition Test Battery and the reversal learning test. These tests are more demanding than the A-not-B and cylinder tests, and are primarily run on primates. Appendix Figure E.4 shows our estimates of the scaling between neuron counts and performance on the PCTB. Across the 16 PCTB tasks and the reversal learning test, under the three cardinalizations, most slopes are below our baseline $\hat{\beta}^{A-B} = 0.53$, but two tests (social learning and tool use) deliver substantially higher slopes. We take the largest estimated slope $\beta_2 = \hat{\beta}^{PCTB} = 3.5$, obtained for social learning under the logistic cardinalization, among the primate cognition tests as our most conservative value, and use it to parametrize a piecewise relationship between cognition and neuron counts. We assume that it is governed by β_2 between humans and the top primate, the primate with the largest measured neuron count in our data (the western gorilla, *Gorilla gorilla*, with 33.4 billion neurons), and by β_1 between the top primate and other animals. This choice additionally downweights primates, and hence all animals, relative to humans using β_2 . Namely, we define a new sentience value ω_s^{conv} by

$$\omega_s^{\text{conv}} = \begin{cases} \left(\frac{z_s}{z_{\text{top primate}}} \right)^{\beta_1} \left(\frac{z_{\text{top primate}}}{z_{\text{human}}} \right)^{\beta_2}, & \text{if } z_s \leq z_{\text{top primate}} \\ \left(\frac{z_s}{z_{\text{human}}} \right)^{\beta_2}, & \text{if } z_{\text{top primate}} < z_s \leq z_{\text{human}} \end{cases}$$

Figure 7 displays how our results jointly depend on β_1, β_2 . Panel 7a shows that incorporating higher sentience convexity at the top (at $\beta_2 = \hat{\beta}^{PCTB} = 3.5$) reduces sentience losses for all groups by a factor of about 17. Mammal sentience losses, already below human sentience gains before the adjustment, fall further below them. Bird sentience losses now also fall below human sentience gains. They are 0.31 times as large. Herptile sentience losses remain about 2 times as large as human sentience gains, fish sentience losses 44 times as large, and insect sentience losses five orders of magnitude larger. As a result, Panel 7a shows that aggregate animal sentience losses remain 46 times as large as human sentience gains without insects, and five orders of magnitude larger with them.

Below the graph, we display the histogram of estimates of β across primate cognition tests, which highlights that our baseline value of $\hat{\beta}^{\text{PCTB}} = 3.5$ is strongly conservative. Appendix Figure F.6 shows that the largest slope among these tests is 2.2 under OLS and 4.3 under PPML, on either side of our GLM estimate of 3.5.

Our pooled estimates imply a smaller adjustment than our conservative baseline. The pooled slopes are 0.23 for the primate tests under the logistic cardinalization, and 0.14 under the Poisson cardinalization. At these values, the adjustment for convexity at the top divides the sentience of all animals below the top primate by only 1.2 and 1.1, instead of 17 with the most conservative scaling. With pallial instead of total neurons, both in the sentience weights and in the estimated slopes, the adjustment divides the sentience of animals below the top primate by 12 instead of 17, and animal losses excluding insects are 110 times human gains (Appendix Figure F.3).

These results indicate that our conclusions are not driven by humans being relatively close to animals with the highest cognition. Rather, they reflect the fact that cognitive, and hence sentience, differences are relatively compressed across animal species.

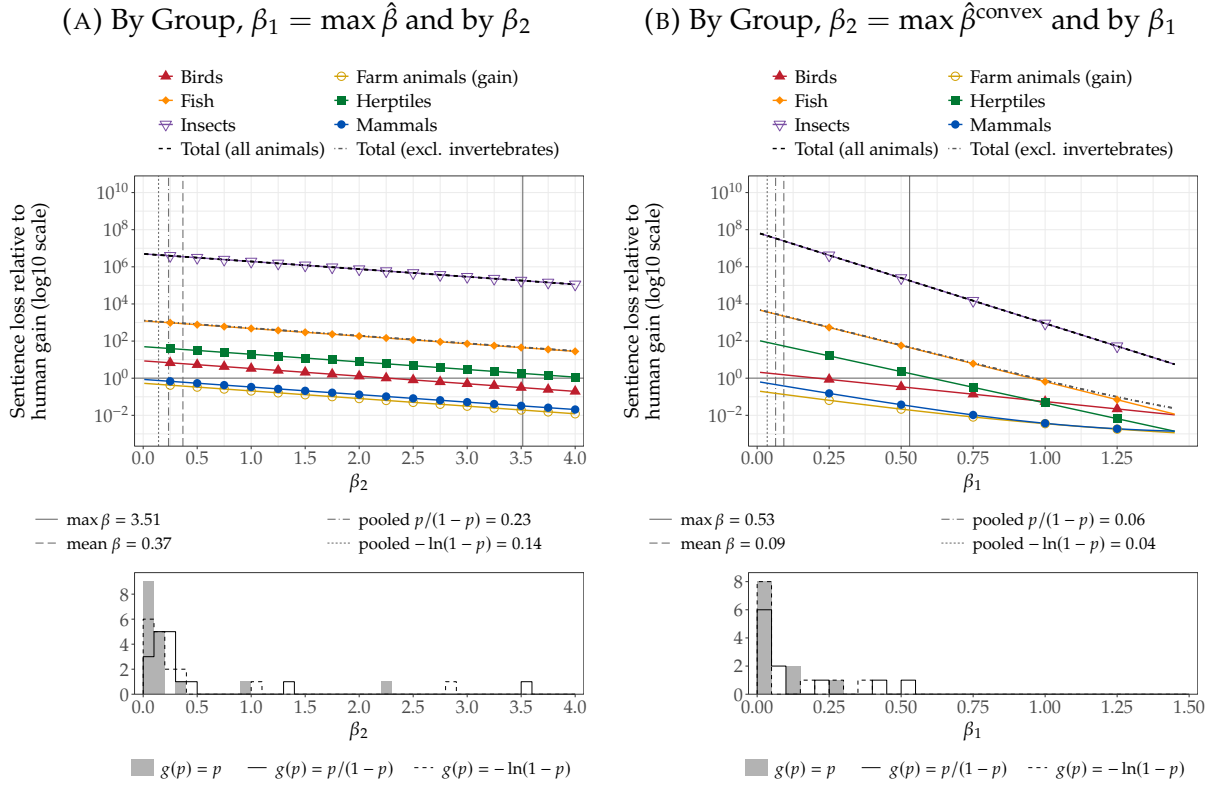
Panel 7b displays animal sentience losses by group as a function of β_1 , holding $\beta_2 = \hat{\beta}^{\text{PCTB}}$ fixed. After incorporating higher convexity at the top of the sentience distribution, the sentience loss of several animal groups becomes smaller than human sentience gains. Mammal sentience losses are smaller than human gains for every value of β_1 . Bird sentience losses become smaller for values of β_1 above 0.21, below our baseline estimate. Herptile and fish sentience losses do so only for values of β_1 above 0.61 and 0.95, above our baseline estimate of 0.53. Even then, Panel 7b indicates that aggregate animal sentience losses, driven by insects, remain above human sentience gains over the whole range of β_1 .

5.4 Animal Welfare Weight

We now convert the aggregate sentience costs from biodiversity loss into welfare. For this exercise, we consider animal welfare $\mathcal{W}_{At}(\boldsymbol{\lambda})$ as a function of the vector of animal welfare weights $\boldsymbol{\lambda}$. For simplicity, we use a common welfare weight $\lambda_s = \lambda \leq 1$ for all species other than humans, so $s > 0$. $\lambda < 1$ corresponds to a social welfare function that is more human-centric than what would be consistent with a strict interpretation of utilitarianism.

To ease the interpretation of our results, we convert our sentience metrics into a willingness to pay to preserve animal life. We consider three species groups or species

FIGURE 7: Animal Sentence Loss Relative to Human Gain under ω^{conv}



Notes: The y-axis is the sentence loss or gain for all taxa expressed relative to human sentence gain (as defined in (14)). The welfare loss is calculated using ω_s^{conv} . The x-axis is the sentence-to-neurons elasticity β_2 in Panel (a) and β_1 in Panel (b). The “Farm animals” category includes all farmed animals, irrespective of taxon. Panel (a): animal sentence loss by group under $\beta_1 = \max \hat{\beta}$ across cardinalizations as a function of β_2 . Panel (b): animal sentence loss by group under $\beta_2 = \max \hat{\beta}^{\text{convex}}$ across cardinalizations as a function of β_1 . The dashed black line is the total over all animals, and the dash-dotted gray line the total excluding invertebrates. The histograms plot the sentence-to-neurons elasticity estimated using GLM for various cardinalizations $c = g(p)$. The histograms in Panel (a) include estimates only from the “convexity at the top” tests. The histograms in Panel (b) include estimates only from the “main” tests. The vertical lines correspond to the maximum and mean β across cardinalizations, and the elasticity estimated from two pooled estimates. See Appendix Figures F.5 and F.6 for histograms estimated using OLS and PPML. Appendix Figure F.3 repeats this figure with pallial neurons.

for illustration: great apes, crows and salmon. For this calculation, we need two objects: a value of statistical life (VSL) for humans, and life expectancy for each species.²³

We start from a human VSL of USD 14 million in the United States, the central estimate in current guidance from the United States Department of Health and Human Services, in 2025 dollars. Consistent with our comparison with world GDP below, we adjust the United States VSL to world income per capita under log utility. A human life-year is then worth USD 58,000.²⁴ We scale this human value of a statistical life

²³This calculation requires us to adjust for life expectancy because we consider the value of a single life rather than a change in steady-state population.

²⁴HHS ASPE, *Standard Values for Regulatory Analysis*, March 2026 proposes a central estimate of USD 14.0 million with a range of USD 6.5 to 21.3 million, in 2025 dollars. The Department of Transportation’s current guidance is similar at USD 13.7 million.

by two factors: expected remaining life-years for each species relative to 40 years for humans, and relative sentience. Great apes are assigned 20 remaining years from adulthood, crows are assigned 6 remaining years, and salmon 3 remaining years.

With a common welfare weight $\lambda = 1$, a great ape's life is valued at USD 0.64 million, a crow's life at USD 40,000 and a salmon's life at USD 910. These values exceed stated preferences for biodiversity protection by several orders of magnitude (Loomis and White 1996). Next, we consider a welfare weight $\lambda = 1 \times 10^{-3}$ that approximately equalizes animal welfare losses and human welfare gains under conservative assumptions (when excluding insects, before accounting for economic growth). In that case, a great ape's life is valued at 640 dollars, a crow's life at 40 dollars and a salmon's life at 0.91 dollars.

Two conclusions emerge from these calculations. First, the willingness to pay for any given animal life depends crucially on the welfare weight of animals. While settling the exact value is beyond the scope of this paper, the magnitude of the potential welfare costs of biodiversity losses highlights its importance. Second, in the aggregate, even at small values of animal welfare weights, the welfare costs of biodiversity loss are at least as large as historical welfare gains to humans. Even at a smaller willingness to pay for any individual animal life, the scope of population declines is such that overall welfare costs remain large.

Importantly, even breaking even with human gains does not preclude large aggregate dollar-equivalent losses. We define human gains to include the contribution of population gains, which thus exceed annual Gross Domestic Product (GDP) at conventional VSLs. At an animal welfare weight of 1×10^{-3} that approximately equalizes human gains to animal losses excluding insects, the welfare cost from biodiversity loss represents 89% of global GDP.

5.5 Quality of Life and Economic Growth

Quality of life improvements from growth. So far, we have assumed that all species' quality of life is fixed at their state of nature. In practice, human quality of life has improved thanks to economic growth, so that $\tilde{u}_{0t} > \tilde{u}_{00}$ for humans. We now incorporate the benefits of economic growth for humans while assuming that animals remained in the state of nature. In this exercise, we first consider an animal welfare weight of $\lambda = 1$ for comparability, so that animal welfare losses equal animal sentience losses scaled by the common state-of-nature quality of life.

To parametrize the size of the welfare gain inclusive of growth, we consider the following utility function for humans, consistent with the definition of welfare ranges:

$$(15) \quad u_0(C) = \begin{cases} \bar{u}_0 + \log \frac{C}{\bar{C}} & \text{if } C \leq \bar{C} \\ \bar{u}_0 & \text{otherwise,} \end{cases}$$

where C is consumption per capita, and $\bar{C}$ denotes the value of consumption beyond which there are no utility gains anymore. The parameter $\bar{C}$ is necessary to remain consistent with the assumption that utility is bounded. In practice, we take $\bar{C}$ to be large enough so that this bound is never binding up to 2026. The functional form (15) is similar to [Hall and Jones \(2007\)](#) and [Jones and Klenow \(2016\)](#).

The utility function (15) requires us to estimate $\bar{u}_0$: it controls the marginal rate of substitution between consumption and life. We leverage that $\bar{u}_0$ is related to the VSL. In particular:²⁵

$$(16) \quad \bar{u}_0 + \log \frac{C_t}{\bar{C}} = \frac{\text{VSL}_t}{C_t}.$$

We may thus estimate a combination of $\bar{u}_0$ and $\bar{C}$ given a VSL and consumption.

The VSL in the United States is USD 14 million in 2025, as discussed in [Section 5.4](#). As our model is static, we convert the VSL to a flow measure by dividing it by the average number of years left to live for the average adult, which we take to be 40. Consumption per capita is USD 61,281 in the United States in 2025. We further adjust for the gap between consumption in the United States and the world average, using world consumption per capita at purchasing power parity of USD 13,691 in 2025.²⁶

Under these estimates, the utility of a human alive in 2020, $u_{0,2020}$, is:

$$u_{0,2020} \equiv \bar{u}_0 + \log \frac{C_{2020}^{\text{World}}}{\bar{C}} \approx \frac{\text{VSL}_{2025}^{\text{US}}}{C_{2025}^{\text{US}}} + \log \frac{C_{2025}^{\text{World}}}{C_{2025}^{\text{US}}} = 4.21.$$

Under the maintained assumption that humans and animals share the same quality of life in the state of nature, $\tilde{u}_{s0} = \tilde{u}_{00} \equiv \tilde{u}_0$. We assume that animals remain at that level throughout. Denote by C_0 human consumption per capita in the state of nature, which

²⁵Formally, we extend $u_0(C, p) = p(\bar{u}_0 + \log C/\bar{C})$, where p denotes the probability of surviving. Then: $\text{VSL}_t = \frac{\partial u(C_t)}{\partial p} \Big|_{p=1} / \frac{\partial u(C_t)}{\partial C}$. Then: $\text{VSL}_t = \left(\bar{u}_0 + \log \frac{C_t}{\bar{C}} \right) C_t$. Rearranging delivers (16).

²⁶Personal consumption expenditures per capita from the BEA for the United States, and household (and NPISH) final consumption expenditure per capita at PPP, in current international dollars, from the World Bank for the world.

we take to be observable, so that $\tilde{u}_0 = u_0(C_0)/\bar{u}_0$.

With these observations in hand, we are ready to construct welfare changes for humans and animals inclusive of economic growth that do not depend on $\bar{C}$ or $\bar{u}_0$.

Proposition 2 (Welfare changes with growth). *Suppose that $u_{0,t} = \bar{u}_0 + \log \frac{C_t}{\bar{C}}$ is given, with $C_t \leq \bar{C}$, and that animals remain at the quality of life of the state of nature $\tilde{u}_0$ for all $s \geq 1$. Human and animal welfare changes between date τ and date t satisfy:*

$$\frac{\Delta \mathcal{W}_{0t}}{u_{0,t}} = \Delta N_{0,t} + \frac{N_{0,\tau}}{u_{0,t}} \log \frac{C_t}{C_\tau} \quad \frac{\Delta \mathcal{W}_{At}(\mathbf{1})}{u_{0,t}} = \hat{u}_{0,t} \sum_{s \geq 1} \omega_s (N_{s,t} - N_{s,\tau}),$$

where $\hat{u}_{0,t} = 1 - \frac{1}{u_{0,t}} \log \frac{C_t}{C_0}$ is the ratio of human quality of life between date 0 and date t , and is observable.

Proof. For humans, (15) implies $u_0(C_\tau) = u_{0,t} - \log \frac{C_t}{C_\tau}$. Then $\Delta \mathcal{W}_{0t} = N_{0,t}u_{0,t} - N_{0,\tau}(u_{0,t} - \log \frac{C_t}{C_\tau}) = u_{0,t}\Delta N_{0,t} + N_{0,\tau} \log \frac{C_t}{C_\tau}$. Dividing by $u_{0,t}$ delivers the first equality. For animals, an individual of species s contributes flow utility $\tilde{u}_0 \bar{u}_s = \frac{u_0(C_0)}{\bar{u}_0} \bar{u}_s = u_0(C_0)\omega_s$, and $u_0(C_0) = u_{0,t} - \log \frac{C_t}{C_0} = \hat{u}_{0,t}u_{0,t}$ by (15). Summing over species and dividing by $u_{0,t}$ delivers the second equality. $\square$

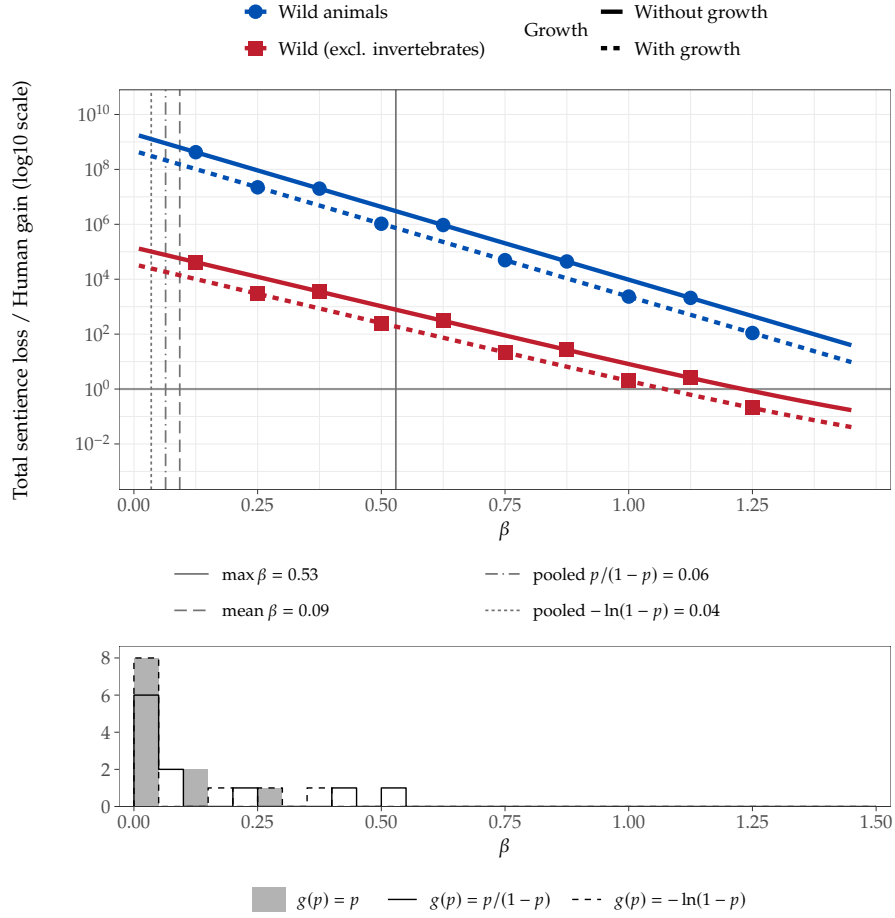
The first component of the welfare change of humans is similar to the expression without growth, with each additional person valued at the utility of a person alive today. The second component encodes the welfare gain from growth accruing to the initial population.

The welfare change of animals depends on the quality of life adjustment: because animals remain in the state of nature, their welfare is downscaled by $\hat{u}_{0,t} \leq 1$, the quality of life in the state of nature relative to life today. Proposition 2 resembles the calculations in Adhami et al. (2023), who use a total utilitarian welfare criterion to incorporate population growth into growth accounting.

We calibrate C_0 by considering two alternatives for the state of nature. Our first approach is to use estimates of consumption that go back as far as possible. The Maddison Project Database estimates world output per capita at USD 1,100 in year 1 CE in 2021 international dollars, leading to $C_{2020}/C_0 = 11.8$.²⁷ We then obtain $\hat{u}_{0,2020} = 1 - \frac{\log 11.8}{4.21} = 0.41$. Our second approach is to consider average consumption

²⁷With a pre-industrial consumption share of 90 percent against 53 percent at PPP today, consumption per capita grew by a factor of $19.9 \times 0.53/0.90 = 11.8$ since antiquity, so that $C_0 \approx$ USD 1,160 in current international dollars.

FIGURE 8: Total Animal Sentence Loss Relative to Human Welfare Gain With Growth



Notes: The y-axis is the welfare loss or gain for all taxa expressed relative to human gain (as defined in (14)), where the human welfare gain either excludes growth (solid lines) or includes growth (dashed lines). The x-axis is the sentence-to-neurons elasticity β . We use population data prioritizing the most granular level of disaggregation. The histogram plots the sentence-to-neurons elasticity estimated using GLM for various cardinalizations of cognition $c = g(p)$. The vertical lines correspond to the maximum and mean β across cardinalizations, and the elasticity estimated from two pooled estimates. See Appendix Figure F.5 for histograms estimated using OLS and PPML.

below today's extreme poverty line as an approximation to the state of nature.²⁸ We obtain $C_t/C_0 = 16.1$ and $\hat{u}_{0,2020} = 1 - \frac{\log 16.1}{4.21} = 0.34$. To remain conservative, we use $\hat{u}_{0,2020} = 0.3$ in our calculations.

We use Proposition 2 to compute the welfare changes of humans and animals inclusive of economic growth. Figure 8 depicts aggregate animal welfare losses relative to human welfare gains.

We find that adjusting for economic growth reduces animal welfare losses relative

²⁸In 2024, 10.4 percent of the world population lived below the World Bank extreme poverty line of USD 3.00 per day (2021 PPP), with a poverty gap of 3.6 percent of the line. The average person below the line thus consumed about USD 2 per day, or approximately USD 850 per year in current international dollars.

to human gains, but they remain of the same order of magnitude. To understand why, consider the two adjustments in turn. For humans, welfare gains between 1970 and 2020 comprise the population increase of 4.2 billion human lives and the gains from economic growth. The additional gains from growth contribute approximately 1 billion human-life equivalents. This adjustment is comparable to human welfare gains from expanding population, consistent with [Jones and Klenow \(2016\)](#).

For animals, sentience losses are scaled by relative qualities of life. For standard estimates of consumption in the state of nature relative to today, this adjustment is no less than 0.3 and does not drastically downweight animal losses. Since the aggregate sentience costs from biodiversity loss are at least 100 times larger than human sentience gains, quality of life improvements due to economic growth reduce these losses relative to human gains, but do not alter the relative orders of magnitude.

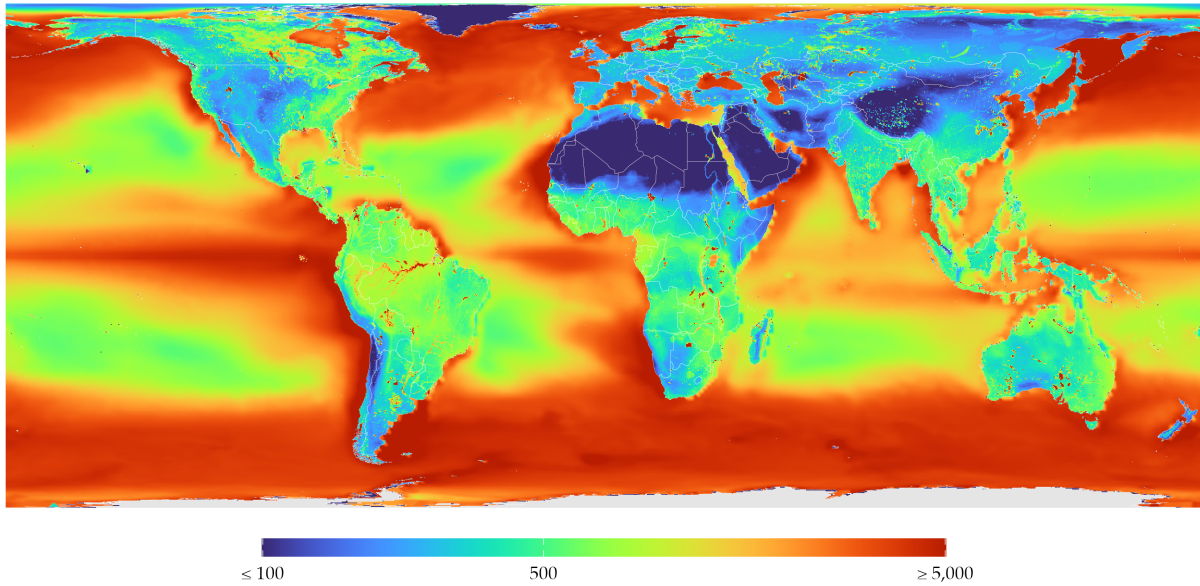
Quality of life in the state of nature. So far, we have assumed that humans and animals have equal quality of life in a state of nature, and that the only reason for divergence in quality of life is economic growth. There are two potential caveats to this approach.

First, humans and other animals may have different qualities of life in the state of nature. Which direction this might go is not obvious. Human cognition and reflection could make the state of nature more or less enjoyable than for an animal. To be conservative, we assume that animals have a lower quality of life in the state of nature. This assumption is isomorphic to assigning a smaller welfare weight $\lambda_s < 1$ to animals. Given that animal sentience losses are 100 to 10^6 times larger than human gains, one would need to assume that animals achieve only 3×10^{-7} to 1×10^{-3} of their welfare range relative to humans to break even.

We can also relax the normalization that utility is weakly positive introduced in Section 2.3 by choosing $\lambda_s < 0$. Doing so introduces the idea that animals would prefer not to live in the state of nature and assigns negative utility to them. If a sizable share of wild animals have negative utility in the state of nature, our conclusions would change meaningfully: indeed, removing net negative lives is a net positive.

Second, there may have been diverging trends in human and animal quality of life since the state of nature that go beyond the adjustment for economic growth. One candidate is declining wild animal quality of life due to habitat loss and other environmental pressures. A second candidate is farmed animals' quality of life being substantially below that experienced in the state of nature. Accounting for these

FIGURE 9: Sentience Density Across the Globe



Notes: Sentience per square kilometer relative to human sentience. Aggregation of wild animal populations multiplied by their relative sentience $\omega_s = (z_s/z_0)^\beta$ with $\beta = 0.53$. Summed over mammals, birds, reptiles, amphibians, freshwater fish and non-mesopelagic marine fish. Mesopelagic fish and invertebrates are excluded. Each species is spread uniformly over its range at its assigned population. Cells beyond range limits take the end colors.

would only increase our estimated welfare losses, so that omitting these forces is conservative.²⁹

5.6 Sentience Across the Globe

In this final subsection, we quantify the distribution of today's remaining animal sentience across the world. Together with an animal welfare weight, these measures can be used in cost-benefit analysis of conservation policies.

We combine our spatial range maps from Section 4.2 with our sentience measures. Figure 9 displays the resulting map of sentience across the globe. Two observations emerge.

First, marine fish are by far the largest contributor to the world's sentience. Even though they have relatively low sentience, they contribute substantially to sentience because they are the most numerous group we consider. They are concentrated in the temperate and polar oceans rather than in the tropical seas.

²⁹Even negative utility for farmed animals would not affect our results much because their aggregate sentience is several orders of magnitude below that of wild animals.

TABLE 2: The Intrinsic Value of Biodiversity Hotspots

REGION	AREA (10 ⁶ km ²)	ANIMALS	SENTIENCE	$\lambda = 1$		Break-even $\lambda = 5 \times 10^{-3}$	
				VALUE (USD tn)	VALUE/KM ² (USD bn)	VALUE (USD tn)	VALUE/KM ² (USD m)
Amazon basin	6.06	4.3×10^{11}	5.1×10^9	1,312	0.22	6.6	1.08
Australia	7.69	4.6×10^{11}	4.7×10^9	1,182	0.15	5.9	0.77
Canadian boreal zone	5.49	3.7×10^{11}	3.2×10^9	767	0.14	3.8	0.70
Congo basin	3.81	2.1×10^{11}	2.3×10^9	563	0.15	2.8	0.74
South-East Asia	2.55	1.1×10^{11}	1.5×10^9	402	0.16	2.0	0.79
Indonesia	1.85	6.2×10^{10}	9.4×10^8	265	0.14	1.3	0.71

Notes: This table sums density maps over each region’s land cells: mammals, birds, reptiles, amphibians and freshwater fish (marine fish are excluded). The Amazon and Congo basins are the cells drained by each river in HydroRIVERS. South-East Asia is Myanmar, Thailand, Laos, Cambodia, Vietnam, Malaysia, Singapore, Brunei, the Philippines and Timor-Leste. The Canadian boreal zone is the part of Canada with an annual mean temperature between -6 and 3°C . Sentience weights each individual by $(z_s/z_0)^\beta$ with $\beta = 0.53$. The value of animal lives follows Section 5.4: a human VSL of USD 14 million adjusted to world income (USD 2.3 million), scaled by remaining life-years relative to 40 for humans (6 for birds, 3 for fish, 4 for mammals, reptiles and amphibians), by sentience, and by the welfare weight: $\lambda = 1$ in the first two value columns, and in the last two the break-even weight that includes the gains from economic growth (Section 5.5), $\lambda = 5 \times 10^{-3}$.

Second, on land, tropical rain forests in the Amazon, the Congo basin and South-East Asia, and parts of the boreal forest, appear as large reservoirs of sentience. Australia also contains a large quantity of sentience because of its large herptile populations. Unsurprisingly, the Sahara, the Arabian Desert and the Gobi Desert hold little sentience compared to most other regions.

Table 2 presents estimates for the intrinsic dollar valuation of some of these biodiversity hotspots. As in Section 5.4, we combine sentience-weighted populations with a human VSL.

With an animal welfare weight of 1, each hotspot is worth more than global GDP. In particular, we estimate a value for the Amazon basin of USD 1,300 trillion, which is 6 times global GDP. Next, we consider the break-even animal welfare weight. We now use the break-even weight that accounts for economic growth because we compare the value of biodiversity with economic value. When we use the corresponding animal welfare weight of $\lambda = 5 \times 10^{-3}$, a square kilometer in the Amazon holds USD 1.1 million worth of biodiversity.

This implication is not simply a consequence of the Amazon’s large total area. With equal weighting, we value the Amazon basin at USD 220 million per square kilometer (km²). When we downweight animals to the break-even weight, we obtain USD 1.1

million per km².

These values can be compared to the economic value embedded in this land. Cleared cattle land in the Brazilian Amazon sells for roughly USD 2,000 per hectare, or USD 200,000 per km² ([INCRA 2025](#), see Appendix C.9 for details). Thus, even when down-weighting animals, there is a large gap between the sentience value of places like the Amazon and their value in economic production.

6. CONCLUSION

This paper explores the welfare costs of the loss of animal life in a consequentialist total utilitarian framework that incorporates the intrinsic value of living beings through their sentience. Using cognitive tests to construct a conservative lower bound on sentience under plausible neuroscientific assumptions, we estimate the relationship between cognition and neuron counts to cardinalize sentience. Combined with population and neuron data for a wide range of animal species, we find that aggregate sentience losses of animals in the past half century outstrip the sentience gains to humans from population growth by at least 100-fold. Breaking even in welfare terms requires assigning a correspondingly small welfare weight to animals.

These results reveal a large discrepancy between a sentience-based valuation of animal welfare and humans' valuation of animal welfare as revealed by their behavior and willingness to pay. Through the lens of our framework, this suggests that humans assign welfare weights to other animals that are substantially below one. We do not argue for a particular value of the animal welfare weight. Rather, our objective is to highlight the importance of this question given the substantial losses in animal sentience over the past half century. In particular, seemingly small departures from a zero animal welfare weight can lead to large aggregate welfare costs from biodiversity loss.

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Appendix

A. DEFINITIONS OF WELFARE

A.1 Utilitarianism: The Mental-State and Preference-Satisfaction Views

Utilitarianism is based on two main theories of welfare. The first theory is the mental-state theory (or, more restrictively, the hedonic theory): welfare consists of positive mental states. The second theory is the preference-satisfaction theory: welfare consists of the satisfaction of preferences.³⁰ These theories coincide on the aspects that are relevant for our purposes in this paper. While it is beyond the scope of this paper to resolve the specific distinction between the mental-state and preference-satisfaction theories, we present the key arguments for and against the mental-state view, discuss the extent to which it is compatible with the preference-satisfaction theory, and explain why we adopt it for our project.

There are two key arguments in favor of the mental-state view. The first is subject-relativity. Welfare must be for the subject: it must matter to them. Only experiential states qualify without further argument. [Crisp \(2006\)](#) writes: “how could anything else benefit me except in so far as I enjoy it?” In their most general interpretation, preference-satisfaction theories fail this criterion: preference-satisfaction can imply positive welfare from the satisfaction of a preference that is entirely other-regarding and that the subject is unaware of. Of course, the standard counter-argument restricts the relevant preferences to self-regarding ones. With this restriction, however, defining “self-regarding” requires a prior account of what is good for the self, which is what a fundamental theory of welfare is supposed to provide. Under this counter-argument, the preference-satisfaction theory becomes very close to the mental-state theory rather than a distinct theory.

The second argument in favor of the mental-state theory is self-evidence. The intrinsic goodness of pleasant experience and the intrinsic badness of suffering are plausibly self-evident from the perspective of any individual. [Sidgwick \(1907\)](#) argued that one cannot intelligibly ask of pleasure “why is that good for me?”, as the question

³⁰The main alternatives outside utilitarianism are objective list theories and capability theories of welfare (Sen, Nussbaum, perfectionism), which treat goods such as autonomy, wisdom, and friendship as incommensurable and intrinsically valuable independently of their experiential impact.

is otherwise circular. The preference-satisfaction view, by contrast, does face the further question: “Why does the satisfaction of this or that preference make me better off?” Most, if not all, philosophers agree that some preferences should not count (uninformed, manipulated, antisocial), so the preference-satisfaction view must specify an additional restriction. The most common restriction is idealization: welfare is the satisfaction of those preferences the individual would have if they were well-informed and rational. But the criterion for “well-informed and rational” cannot itself be derived from preferences without circularity. The criterion has to be supplied by a substantive theory of welfare that specifies which preferences track welfare and which do not. The criterion thus relies on theory-laden judgments that preference satisfaction is meant to dispense with. As [Rosati \(1995\)](#) writes, the idealized agent whose preferences now define welfare may bear so little resemblance to the actual person that it is no longer clear whose welfare is being tracked.

Standard economics is sometimes read as endorsing a preference-satisfaction view of welfare. We emphasize that standard economics is entirely compatible with a mental-state view of welfare, because preference-based views can be read as evidential. The evidential reading states that preference-satisfaction is evidence of welfare, but welfare is something else. This reading is compatible with the mental-state view because welfare can be valenced experience, and preference-satisfaction reveals welfare.³¹ We adopt the evidential reading, which also solves the issues with preference-satisfaction highlighted above.

While most work in economics does not explicitly distinguish the evidential reading from other interpretations, the evidential reading is consistent with economists’ actual practice. For example, when economists use revealed preference to measure utility, they are not treating preferences as identical to welfare. Rather, they are treating them as indicators of welfare. This evidential reading is also in line with the behavioral economics literature, which distinguishes between *decision utility* (the weight of an outcome in choice) and *experienced utility* (the hedonic quality of an outcome). More recently, advances in behavioral economics and neuroeconomics have provided further support for the mental-state view by providing evidence on the neural basis of choice and utility ([Caplin et al. 2010](#)).

Specific interpretations of preference-satisfaction utilitarianism hold that genuine

³¹ Another reading of preference-satisfaction views is the constitutive reading. In that interpretation, welfare is inherently preference-satisfaction. This reading is incompatible with the mental-state view, since preferences are not themselves valenced experiences.

preferences require language. Under this view, non-linguistic animals have no genuine interests and accordingly no moral status (Frey 1980). However, this criterion would also exclude human infants and severely cognitively impaired humans, which is at odds with current legal frameworks.

A.2 Alternatives to Utilitarianism

Theories consistent with sentientism. For Korsgaard’s neo-Kantianism, the obligation is to treat all subjects as ends in themselves rather than as mere means. Respecting a being as an end in itself amounts to treating as valuable the ends that the being itself values. The sense in which an animal can be said to “value” an end is precisely that it has positively valenced experiences in pursuit of it.

For Nussbaum’s capabilities approach, the imperative is to enable each being capable of “significant striving” to achieve a flourishing appropriate to its form of life. Nussbaum restricts the class of beings who can be wronged to those with “a felt orientation towards what is seen as good and a felt aversion to what is seen as bad” (Nussbaum 2023, p. 119), that is, those with valenced experience.

Theories inconsistent with sentientism. At one extreme, rationality-based moral status (orthodox Kantianism) contradicts sentientism, but it faces strong arguments against it. According to rationality-based moral status, moral status is tied to rational agency. Yet it cannot consistently include human infants and severely cognitively impaired humans without also including many sentient non-human animals.

At the other extreme, biocentrism or ecocentrism extends moral status to all living beings or to ecological wholes. However, it lacks a principled link between the entity (e.g., a plant or an ecosystem) and any subject for whom things can be non-instrumentally good or bad.

B. THEORIES OF THE NEURAL BASIS OF VALENCED EXPERIENCE

B.1 The Materialist View of Consciousness

Materialism and dualism. *Materialism* (or physicalism): conscious experience is fundamentally a physical phenomenon, ultimately realized in physical processes occurring

in the brain. To explain conscious experience, we do not need to posit any fundamentally non-physical properties, processes, forces, entities, substances, or laws.

Dualism: conscious experience, while dependent on brain function, is not itself a fundamentally physical phenomenon. To explain it, we do need to posit at least some fundamentally non-physical properties, processes, forces, entities, substances, or laws.

Materialism as the mainstream view. Materialism is the majority view among contemporary philosophers ([Bourget and Chalmers 2014](#)) and the near-universal view among neuroscientists ([Koch et al. 2016](#)).

The standard contemporary argument for materialism is the *causal-closure argument*, reconstructed most influentially by [Papineau \(2002\)](#) and [Kim \(2005\)](#). The argument has three premises. First, some conscious experiences have physical effects: experiencing pain causes one to withdraw, experiencing fear causes one to flee, and so on (mental causation). Second, all physical effects have fully physical causes, the principle of causal closure of the physical, supported by the cumulative success of physics in identifying complete physical-cause chains for physical effects. Third, the physical effects of conscious experiences are not, in general, overdetermined by distinct mental and physical causes. From these three premises it follows that at least some conscious experiences are themselves purely physical causes, and if that is so, materialism is true (at least for those conscious experiences that have physical effects).

The empirical balance has reinforced this argument. Over a century of experimental physiology and neuroscience has found zero evidence for a fundamentally non-physical force or substance operating in conscious beings ([Papineau 2002](#); [Churchland 2002](#)). Every documented mental phenomenon, from sensory perception to abstract reasoning to emotion to selfhood, has been progressively localized to specific neural substrates whose activity is necessary and, to varying degrees, sufficient for the phenomenon. The principle of causal closure has held up across the full reach of contemporary physics. No plausible candidate for a “mind force” has been identified that would interface with neural matter without leaving experimental traces. Dualism is consequently a position held primarily on philosophical rather than empirical grounds, and the empirical balance of evidence weighs heavily on the materialist side.

Other alternatives. *Panpsychism and Russellian monism.* Panpsychism proposes that all matter possesses some basic form of subjective experience. Contemporary panpsychists hold that some elements of conscious experience can be found in primitive form in some fundamental physical particles, sometimes called “protoconsciousness” or

proto-experiential properties (Chalmers 2015; Goff 2017). The view is held by a small minority of professional philosophers and is widely regarded as a fringe position for two reasons. First, panpsychism faces the *combination problem*: how vast multitudes of micro-experiences combine into the unified macro-experience of a sentient being. Second, it commits to fundamentally non-physical mental properties absent any independent empirical motivation. Physicalists view this as an ontological extravagance that does not earn its keep.

Integrated Information Theory (IIT). IIT departs sharply from the materialist mainstream: it treats consciousness as fundamental and identifies it with integrated information Φ (a measure of the cause-effect power of a system), with what we ordinarily call physical brains being the way conscious systems appear extrinsically to other conscious observers (Tononi 2004; Tononi and Koch 2015; Oizumi et al. 2014). That is, instead of running from physical objects to conscious experience, causality in IIT runs the other way: consciousness is what exists, and its physical substrate is the extrinsic shadow that a Φ -structure casts in the perceptions of other Φ -structures. IIT is not a mainstream position, for several connected reasons. First and most fundamentally, the metaphysical commitment just described is incompatible with materialism, and the physicalist supermajority documented above accordingly rejects it. The position is closer to neutral monism, panpsychism, or a modernized monadology (Tononi himself draws the Leibniz comparison) than to physicalism (Birch 2024). Second, IIT’s central axioms are not widely accepted even within the non-materialist literature. The inference from the phenomenal structure of experience to the mathematical structure of Φ is held to be under-motivated (Bayne 2018). Third, Φ is intractable to compute exactly for systems above trivial size, so the theory cannot be directly applied to real brains and must rely on proxies of contested validity. Fourth, the recent “unfolding argument” (Doerig et al. 2019) contends that IIT and other causal-structure theories cannot in principle be distinguished by empirical evidence from functionally equivalent feed-forward systems, raising a serious falsifiability concern.

B.2 The Neural Basis of Valenced Experience

We review here the leading theories of the neural basis of valenced experience. These theories differ in the neural substrate they establish as sufficient for valenced experience. However, they all link valenced experience to brain architecture, agreeing on the implication that similar brain structures support similar valenced experience.

Methodological considerations. Theories of valenced experience are claims about which physical brain mechanisms accompany felt states. The methodology is therefore organized around *markers* that are anatomically independent of the proposed substrate: (i) behavioral markers (conditioned place avoidance, motivational trade-offs, analgesic self-administration, judgment-bias paradigms), (ii) physiological markers (autonomic and hypothalamic–pituitary–adrenal axis responses), and (iii) pharmacological markers (sensitivity to analgesics, anxiolytics, and naloxone). These markers are validated as proxies by their correlation with self-report *in humans*, then carried over by analogy to non-human species. Note that the apparatus is defeasible: no third-person marker is logically sufficient for ascribing a first-person state, an instance of the hard problem of consciousness. But it is the only route available, since the strongest evidence type (verbal report) is unavailable in precisely the disputed cases.

Anatomical primer. Two broad categories of brain structure are candidates for the substrate of valenced experience. On the one hand, there are evolutionarily ancient *subcortical* structures, in particular the periaqueductal gray (PAG) and the mesolimbic dopamine system. On the other hand, there are *integrative brain regions*: the cortex in mammals, and functional analogs in other taxa.

Subcortical and cortical processes. It is useful to start with a taxonomy of layers of experience (Tulving 1985). *Anoetic* consciousness refers to raw phenomenal feel, without conceptual or self-referential content. It is sometimes called the basic “what-it-is-likeness” of experience. *Noetic* consciousness refers to experience accompanied by conceptual or propositional content and the recognition that one is experiencing, and *autonoetic* consciousness refers to self-knowing experience embedded in an autobiographical self that locates the experience in subjective time. The consensus is that anoetic affect is fully realized in subcortical machinery, and the cortex contributes to higher (noetic and autonoetic) layers of experience. The key point of disagreement is whether subcortical-based anoetic consciousness is indeed valenced experience, or whether it needs cortical re-representation to be truly experienced.

One family of theories (Panksepp 1998; Merker 2007; Damasio and Carvalho 2013) holds that anoetic consciousness is valenced experience, and hence that subcortical structures alone suffice for valenced experience.³² In this view, cortex *elaborates* affect (categorization, semantic content, self-referential framing) and *modulates* it (regulation, voluntary control, re-representation), but does not *generate* the raw affective state.

³²Anoetic differs from unconscious, e.g., a spinal withdrawal reflex, whose neural mechanism consists of local spinal circuits with no global integration.

The most basic level of valenced feeling is fully realized in subcortical machinery.³³ The competing theories hold that anoetic experience is not experience. In this view, subcortical activity is unconscious processing that needs cortical re-representation to become an experience of any kind. The family includes, among others, the Global Neuronal Workspace theory (Dehaene et al. 2017), higher-order theory (LeDoux and Brown 2017; Rosenthal 2005), and recurrent processing theory (Lamme 2006). All locate consciousness in different brain locations and mechanisms but share the requirement of cortical involvement. Holders of this view acknowledge all the behavioral and physiological evidence for subcortical affective processing, but interpret it as evidence of unconscious processing that needs cortical re-representation to become an experience of any kind.

The disagreement is hard to settle exactly because it instantiates the hard problem of consciousness. Appendix B.3 discusses the evidence in favor of both views. We do not attempt to adjudicate this dispute here.

B.3 Evidence on the Neural Basis of Valenced Experience

Evidence in favor of subcortical sufficiency. Three lines of evidence are taken to support this position. (i) *Deep brain stimulation in humans*. Stimulation of subcortical structures during clinical procedures elicits intense and vivid affective reports (panic, dread, pleasure, sexual feelings) at intensities that cortical stimulation rarely produces (Heath 1972; Panksepp 2005). (ii) *Cross-species behavioral homology*. The same stimulation in homologous midbrain regions of non-human mammals produces behaviors closely matched to the human affective reports (Panksepp 2011). (iii) *Decorticate and decerebrate individuals*. Animals whose cortex has been surgically removed, or whose cortex never developed in early developmental preparations, retain emotional behaviors: distress vocalization, fearful withdrawal, appetitive engagement, play (Panksepp 1998). The strongest empirical version of this argument concerns human hydranencephaly: Merker (2007) documents that children born without a cerebral cortex nevertheless smile, laugh, cry, recognize familiar people, prefer specific songs and toys, and engage in goal-directed behavior at the level appropriate to their motor capabilities, which Merker reads as evidence that the preserved subcortical structures suffice for some form of

³³Some researchers go further. Solms centers consciousness in the brainstem (“consciousness is affect”), which provides the felt quality, while the cortex provides the conceptual and semantic content. Damasio and Carvalho (2013) argue that subcortical mechanisms generate core consciousness (anoetic plus basic body-awareness, here-and-now) while the cortex generates extended consciousness (autobiographical).

conscious affective experience.

Evidence in favor of cortex-based theories. Three lines of evidence are taken to support the position that conscious valenced experience requires cortical (or cortex-analog) processing. (i) *Subliminal subcortical processing*. Masked fearful faces, presented too briefly for conscious perception, produce amygdala activation comparable to that produced by consciously perceived fearful faces ([Whalen et al. 1998](#)), evidence that subcortical affective processing routinely occurs without conscious accompaniment. The same pattern holds for visual perception more broadly: consciously perceived stimuli produce a late, all-or-nothing fronto-parietal “ignition” that subliminal stimuli do not, across attentional-blink and visual-masking paradigms ([Sergent et al. 2005](#); [Del Cul et al. 2007](#); [Dehaene 2014](#)). (ii) *Dissociations of conscious feeling from subcortical defensive response*. [Feinstein et al. \(2013\)](#) report that the bilateral-amygdala-lesion patient SM, who reliably fails to show autonomic or behavioral fear responses to standard threat stimuli, nevertheless reports panic and intense fear when subjected to CO₂ inhalation. [LeDoux and Brown \(2017\)](#) read this as direct evidence that the conscious feeling of fear is generated by cortical machinery the amygdala lesion leaves intact, while the amygdala itself produces a defensive survival response without phenomenal accompaniment. (iii) *Cortical disruption tracks loss of consciousness*. Across wake, deep slow-wave sleep, anesthesia, and disorders of consciousness, the perturbational complexity index of [Casali et al. \(2013\)](#), a TMS-EEG measure of cortical integration, reliably discriminates conscious from unconscious states, while subcortical activity is comparatively preserved ([Mashour et al. 2020](#)). Vegetative-state patients with preserved cortical activity perform mental-imagery tasks indistinguishably from controls ([Owen et al. 2006](#)), suggesting that conscious access tracks cortical preservation even when behavior is absent. Across these three lines, the cortex-required reading is that subcortical activity is sophisticated but unconscious, and that conscious valenced experience requires the cortical or cortex-analog processing emphasized by the Global Neuronal Workspace ([Dehaene et al. 2017](#)), higher-order theory ([LeDoux and Brown 2017](#); [Rosenthal 2005](#)), recurrent processing theory ([Lamme 2006](#)), and the attention schema ([Graziano 2013](#)).

The evolutionary roots of valenced experience. The brain-characteristics lens connects welfare range to neural similarity. A second, complementary lens grounds valenced experience in selection pressures. The same selection pressures shape similar brains and similar experiences. Both lenses point in the same direction.

The leading theories of valenced experience treat valence as an evolved adaptation

that serves to motivate fitness-improving actions and discourage fitness-reducing ones. [Cabanac \(1992\)](#) influentially proposed that pleasure functions as a “common currency” in the brain, converting competing motivational claims (hunger, thirst, fear, social bonds) into a single comparable quantity that the organism uses to choose among them. Valenced experience is thought to play three roles in support of this function: it represents fitness-relevant information, it serves as a common currency for decision-making across competing motivations, and it facilitates learning by tagging outcomes as good or bad. A sense of self evolved in conjunction with the capacity for action: once an animal moves, it becomes useful for it to distinguish stimuli arising from the external world from stimuli generated by its own actions, and the rudiments of an experiencing self become adaptive: a locus that the world impinges on, that initiates action, and that registers the consequences.

If valenced experience is an evolved adaptation, then animals facing similar challenges should have evolved similar valenced responses, supported by similar nervous systems.

B.4 Arguments on Cognition Scaling More Steeply with Neural Architecture than Affect

The conservatism claim splits into two restrictions on the Cobb–Douglas parameters: $\alpha_u \geq \alpha_c$ (the affective production function loads more on subcortical neurons than the cognitive production function does), and $\kappa_u \leq \kappa_c$ (whatever shared substrate exists is used by affect with no greater scale elasticity than cognition uses it). The arguments fall into three categories (anatomical, evolutionary, and functional), and each restriction is supported by arguments in each category. We present these in turn.

B.4.1 Affect Relies More on Subcortical Machinery, Cognition More on Cortical Machinery: $\alpha_u \geq \alpha_c$

Anatomical separation. Cognition is uncontroversially cortex-dependent: working memory, abstract reasoning, language, and flexible planning load on heteromodal cortex, and the prefrontal expansion of the primate lineage is the standard anatomical anchor for the cognitive capacities tested across species. Affect is contested: as discussed in [Appendix B.2](#), theories disagree on whether subcortical machinery is sufficient or whether cortical re-representation is required. In any case, there is no disagreement that the midbrain plays a central role in generating affect: even for cortex-based theories,

subcortical circuits provide the lower-order unconscious inputs that become conscious experiences in cortical circuits (see, e.g., [LeDoux and Brown 2017](#)). In addition, some suggest that the cortical contribution to affect, if present, is concentrated in a small *paralimbic* compartment (anterior cingulate, anterior insula, and orbitofrontal cortex) rather than spread across heteromodal cortex that supports cognition ([Craig 2009](#); [Mesulam 1998](#)). Therefore, the affective production function plausibly puts more weight on $z_{m,s}$ relative to $z_{p,s}$ than the cognitive production function does, that is, $\alpha_u \geq \alpha_c$.

Phylogenetic priority. The consensus view is that valenced experience emerged temporally prior to flexible cognition ([Panksepp 1998](#); [Damasio 1999, 2010](#); [Damasio and Carvalho 2013](#)). The logic is that the original function of any nervous system is interoceptive monitoring, registering body states and signaling them to action-coordinating circuits, and that a primitive valenced signal is what such monitoring produces. Cognition is built on top of this base, not under it: there is no fitness-relevant problem that cognitive elaboration solves prior to having a valenced signal to act on. Anatomically, the subcortical structures that produce the valenced signal (PAG, hypothalamus, amygdala, brainstem affective nuclei, mesolimbic dopamine system) are conserved across all vertebrates, while the cortical expansion that supports flexible cognition is much more recent, accelerated in the primate lineage, and most pronounced in humans. Phylogenetic priority reinforces the anatomical separation.

B.4.2 Scaling: $\kappa_u \leq \kappa_c$

Even granting any cortical contribution to affect that one wants to allow, three classes of evidence (functional, evolutionary, and anatomical) suggest the cortical scaling of affect is shallower than the cortical scaling of cognition.

Evolutionary argument. Selection on valence is for *adequacy*: once the signal is intense enough to motivate fitness-improving actions, additional cortical machinery does not improve outcomes. Emotions are evolved programs with bounded functional roles, designed to coordinate other psychological mechanisms rather than to admit indefinite refinement ([Nesse 1990](#); [Tooby and Cosmides 2008](#); [Nesse 2004](#)). Cognition, by contrast, serves competitive, maximizing functions: outsmarting conspecifics, manipulating the environment, and accumulating cultural knowledge ([Dunbar 1998](#); [Reader and Laland](#)

2002; Sol 2009; Burkart et al. 2017).³⁴ Selection truncated affective intensity at the level required to dominate competing motivations and did not reward additional capacity. Cognitive selection in social and ecological niches has no analogous diminishing-return structure. The asymmetric evolutionary pressure suggests asymmetric scaling: animals with large brains supporting advanced cognition do not need to have proportionally larger welfare ranges for sentience to achieve its evolutionary role.

Functional arguments. *Dimensionality of affect vs. cognition.* Consider a population of N neurons in some brain region. At each moment the population's state is the list of N firing rates, a point in an N -dimensional "neural state space." Over time or over a sequence of stimuli, one can characterize how many independent directions the population state actually varies along, its intrinsic dimensionality, as well as the dimension of the *readout*, i.e., the projection of the population state onto the axes along which downstream circuits decode and act. There is a notable affect-vs-cognition asymmetry in readout dimension. For affect, the readout is low-dimensional. The neuroeconomic literature finds that the brain converts heterogeneous motivational inputs (hunger, fear, social bond, prospective gain) into an essentially one-dimensional value signal read out by the action-selection circuit, the brain's "common currency" posited by Cabanac (1992).³⁵ The low-dimensional readout is not an artifact of measurement: the action-selection function *requires* a scalar comparator, so a high-dimensional affective readout would fail at the function it serves. For cognition, tasks such as discrimination, composition, and planning require many independent task-relevant variables to be simultaneously available to downstream circuits. A high-dimensional readout is required to represent distinct concepts, percepts, plans, or inferences. This is the classical Fodor-Pylyshyn argument that systematic, productive thought requires a representational system with many independently varying constituents (Fodor and Pylyshyn 1988). Cortical population codes match this high-dimensionality requirement empirically.³⁶ In turn, the high-dimensional coding capacity scales with neuron counts.

³⁴Dunbar (1998) documents the social-brain correlation across primates (neocortex ratio scales with social group size). Reader and Laland (2002) extend this to ecological cognition, finding that rates of behavioral innovation, social learning, and tool use covary with executive-brain size. Sol (2009)'s cognitive-buffer hypothesis predicts selection for greater brain capacity precisely where the environment rewards flexible problem-solving. Finally, Burkart et al. (2017) synthesize cross-species evidence for a domain-general *g*-like factor in non-human animals.

³⁵In a meta-analysis of fMRI studies, Levy and Glimcher (2012) find that the values of different rewards are coded on a common scale, mainly in a region spanning the ventromedial prefrontal and orbitofrontal cortex. Using converging evidence from neurophysiology, imaging, and lesion studies, Padoa-Schioppa (2011) localizes the same scalar representation in the same brain regions.

³⁶Rigotti et al. (2013) record from monkey prefrontal cortex during a cognitive task and find that neurons exhibit *nonlinear mixed selectivity* for combinations of task variables, generating high-dimensional

Cortical expansion therefore plausibly raises the cognitive readout’s effective dimension, and so its scaling, but cannot raise the affective readout’s dimension without breaking the function it serves: bounded for affect, open-ended for cognition.

The dimensional prediction is matched in the cross-species comparative literature: the absolute number of cortical neurons, rather than brain mass or encephalization quotient (EQ), is the strongest known neural correlate of cognitive performance across mammals. [Herculano-Houzel \(2017\)](#) synthesizes a decade of isotropic-fractionator counts in support of this conclusion, with the cognition–neuron relationship holding across rodents, primates, and afrotherians and being violated by brain-mass or EQ-based predictors (e.g., elephants have larger brains but fewer cortical neurons than humans, and rank accordingly on cognitive tests). [Dicke and Roth \(2016\)](#) converge on the same conclusion from an independent comparative review.

We can note two related points. First, the conscious channel for affect is unitary, while cognition handles parallelization. The unity-of-consciousness constraint ([Bayne 2010](#)) holds that the phenomenal field is integrated: at any moment a subject has a single overall conscious state. For affect this constraint is observed behaviorally: at high intensity, affective states do not co-occur (one cannot simultaneously feel intense fear and intense pleasure), and this corresponds to the finding of the scalar neural representation discussed above. For cognition, by contrast, there is strong evidence of parallelization.³⁷ Adding cortical capacity multiplies the parallel streams cognition can carry. It does not multiply the affective channel, which is constrained to remain unitary by the function it serves. Second, affect and cognition sit at different ends of the speed–precision trade-off.³⁸ Affective signaling must be fast, while cognition operates under a much longer time budget. Adding cortical capacity does not relax the speed constraint affect must satisfy: a brain with more cortical neurons still has to deliver

population responses whose dimensionality predicts task performance and collapses just before behavioral errors. [Fusi et al. \(2016\)](#) synthesize this evidence and argue that mixed selectivity is the substrate of higher cognition, since high dimensionality enables the linear readout of arbitrarily many task-relevant variables from a single population. [Stringer et al. \(2019\)](#) record neurons in mouse primary visual cortex during natural-image viewing and find that effective dimensionality grows with neuron population size. [Haxby et al. \(2001\)](#) establish the corresponding point in human cortex: object-category information in ventral temporal cortex is carried by distributed, overlapping multi-voxel patterns rather than localized to dedicated channels.

³⁷The dorsal and ventral visual streams operate in parallel ([Goodale and Milner 1992](#)). The seven-network cortical parcellation of [Yeo et al. \(2011\)](#) documents large-scale networks with substantial functional independence.

³⁸[Bogacz et al. \(2006\)](#) provide the canonical formalization of the trade-off. [Ratcliff and McKoon \(2008\)](#) review the empirical fit of the diffusion-decision model across two-choice tasks. [Heitz \(2014\)](#) reviews the broader empirical literature and finds that across species and tasks, faster decisions are systematically less precise, with the trade-off frontier well characterized at both behavioral and neural levels.

the affective signal before the action window closes. The trade-off thus caps the useful scaling of the affective channel. Cognition, operating on a longer time budget, is not analogously capped.

Homeostatic regulation and saturation of affect. Affective states are actively regulated back toward a baseline, while cognitive content is not. The hedonic-adaptation literature documents that sustained affective states return to a homeostatic set-point regardless of stimulus magnitude (Brickman et al. 1978; Frederick and Loewenstein 1999; Lyubomirsky 2011). This set-point regulation operates physiologically through autonomic and neuroendocrine feedback. Cognition has no equivalent: knowledge accumulates monotonically, learned skills do not regress to a baseline of ignorance, and working memory does not adapt back to a set-point after a strong cognitive episode. Cortical capacity supports a directly accumulative cognitive substrate, whereas the affective substrate is regulated, and additional capacity is partly absorbed by the regulation rather than translated into additional welfare range.

Relatedly, the neurochemical substrate of affect has hard physical limits that the cognitive substrate does not. For instance, the hedonic “liking” reaction is generated by mu-opioid binding in small, anatomically discrete hotspots in the nucleus accumbens shell and ventral pallidum. Mu-opioid receptors have finite binding capacity and downregulate under sustained activation, the standard mechanism behind opioid tolerance (Koob and Volkow 2010). The evolutionary rationale is that performance is maximized at moderate rather than extreme arousal (Yerkes and Dodson 1908; Mendl 1999). The cortical substrate of cognition is organized differently: representations are sparse and distributed across millions of pyramidal neurons, no single neurotransmitter-receptor system acts as a bottleneck, and encoding is dominated by long-term synaptic plasticity, which is essentially open-ended. Scaling cognitive capacity recruits additional neurons and cortical area (Hill et al. 2010; Van Essen et al. 2019) rather than driving a fixed substrate harder. The bounded biochemistry of the hedonic substrate and the open-ended distributed-plasticity architecture of cognition together suggest a sharper ceiling on $\bar{u}_s$ than on c_s at any given level of cortical capacity.

Anatomical argument. Finally, *within the cortex*, the cognition-relevant heteromodal compartment scales more steeply than the affect-relevant paralimbic compartment across humans, non-human primates, and mice (Hill et al. 2010; Van Essen et al. 2019).

C. DATA SOURCES: POPULATION DATA

We construct two distinct datasets:

1. For the five taxonomic groups (birds, fish, herptiles, insects, mammals), we have a current population level and a percentage decline since 1970, allowing us to back-calculate a 1970 population.
2. For birds and mammals, we also collect species-level current population counts and construct order- and species-level population changes whenever possible, as detailed below.

C.1 Population Changes for Wild Vertebrates: The Living Planet Index (LPI)

The Living Planet Database ([Zoological Society of London and World Wildlife Fund 2024](#)) contains 29,559 population data series covering 4,844 species. These population series are gathered from a variety of sources such as journals, online databases, and government reports. For each population series, the rate of change from one year to the next is calculated using generalized additive modeling. Annual rates of change in populations of the same species are then aggregated to the species level and then to higher levels (e.g., order or major taxa) using the geometric mean.

We produce population change series at two levels of aggregation. First, we aggregate species at the level of the four major vertebrate taxa (birds, fish, herptiles, mammals). As recommended by the LPI methodology for aggregation at this level, we first take a simple mean to aggregate species at the taxon $\times$ realm level (e.g., Neotropical birds) and then use the provided geographic sampling weights to aggregate from the taxon $\times$ realm level to the taxon level. Second, we aggregate species at the order level. We produce order-level estimates only for orders supported by more than 100 distinct monitored population series. This yields order-level estimates for 44 orders (18 fish, 11 birds, 10 mammals, 5 herptiles). The aggregation from the species to the order level is unweighted, as recommended by the LPI methodology for finer levels of aggregation.

Table C.1 shows the coverage of the LPI by major taxa. The coverage is best for mammals and birds, with covered shares around 15%, while fish and herptiles have covered shares below 10%.

We now discuss a number of potential issues with the LPI. First, the species covered

TABLE C.1: LPI Species Coverage

TAXA	SPECIES COVERAGE (%)	SPECIES IN LPI	SPECIES TOTAL
Birds	14.60 (12.20–14.60)	1,365	9,350 (9,350–11,185)
Fish	6.12 (5.68–7.21)	2,118	34,633 (29,371–37,288)
Herptiles	2.93 (2.68–3.11)	574	19,595 (18,466–21,420)
Mammals	15.22 (11.55–15.22)	787	5,171 (5,171–6,815)

Notes: Total species counts are taken from the Living Planet Report 2024 Technical Supplement, IUCN and Our World in Data. The “Species in LPI” column gives the number of distinct species in the public LPI analytical dataset.

in the LPI may not be representative. This would be particularly problematic if the LPI coverage is systematically biased toward species with larger population losses. The 2024 Living Planet Report Technical document shows that there are more data from regions with a longer history of monitoring and publishing data, such as North America and Europe, compared to other continents or remote or tropical areas. In the existing population series, North America and Europe have experienced the smallest population declines compared with the other continents. When aggregating to the taxon level, this sampling bias is addressed by using the realm weights. [McRae et al. \(2017\)](#) test for over- or under-representation of species classified as threatened by the IUCN in the LPI. While representation in the LPI is not perfectly balanced, the bias does not systematically go in the direction of better coverage of threatened species. For instance, the most threatened fish and amphibian species tend to be under-represented, while the most threatened reptile species are over-represented. For mammals, there is no significant over- or under-representation. Finally, the LPI may be biased toward species that are easier to track. There is, however, no straightforward way to quantify this bias.

Second, the LPI may be sensitive to outliers ([Leung et al. 2020](#); [Murali et al. 2022](#)). To alleviate this concern, we reproduce the LPI when trimming the population-level changes at different percentiles. This analysis is reported in Table C.2. For birds, fish, and herptiles, the estimated declines are virtually unchanged. For mammals, we obtain a smaller decline in the 5% tail trimming scenario. Using this smaller decline for mammals does not materially affect any of the paper’s conclusions about aggregate welfare changes.

Third, the LPI weights all population series equally when aggregating to the species, order, or major taxon level (except for the geographic reweighting). Underlying population series come in diverse units (density, absolute number, abundance index),

rendering weighting by the relative importance of species impossible.

TABLE C.2: LPI Population-Series Trimming Robustness

Panel A: ± 1 capping

TAXON	BASELINE	
Birds	45.37 [21.48, 61.18]	
Fish	77.16 [66.01, 84.90]	
Herptiles	89.95 [72.75, 96.51]	
Mammals	45.86 [9.16, 69.87]	

Panel B: 1% tail trimming

TAXON	POOLED	TAXON
Birds	40.76 [14.78, 57.29]	45.58 [20.79, 61.35]
Fish	76.95 [65.58, 84.58]	76.05 [64.48, 83.97]
Herptiles	89.46 [71.48, 96.28]	89.23 [70.35, 96.19]
Mammals	42.69 [4.44, 67.92]	44.27 [6.59, 68.56]

Panel C: 2.5% tail trimming

TAXON	POOLED	TAXON
Birds	44.36 [21.14, 59.49]	43.69 [19.78, 59.43]
Fish	75.92 [63.96, 83.64]	74.68 [62.70, 82.85]
Herptiles	88.47 [69.11, 95.94]	85.40 [63.33, 94.65]
Mammals	44.28 [2.83, 68.44]	22.52 [-14.22, 52.20]

Panel D: 5% tail trimming

TAXON	POOLED	TAXON
Birds	46.47 [24.92, 60.50]	46.56 [25.29, 60.57]
Fish	74.01 [62.68, 82.13]	71.55 [59.13, 80.33]
Herptiles	82.40 [55.18, 93.10]	84.80 [61.90, 94.02]
Mammals	20.74 [-12.32, 49.27]	21.41 [-10.81, 49.73]

Notes: This table reports percentage decline from 1970 to 2020, with 95% confidence interval bounds in brackets. Interannual $\log_{10}$ changes are capped at -1 and $+1$ in all specifications.

Fish population trend. Two aspects of the fish population trend deserve further discussion. First, the LPI population series mostly cover marine fish in the non-mesopelagic zone (living above 200 meters deep) and freshwater fish. There is virtually no evidence on population trends for the more numerous group of mesopelagic fish (living below 200 meters deep). Our baseline estimates assume that the LPI trend applies only to the non-mesopelagic and freshwater fish count.

Although no direct series can confirm this, assuming no change for mesopelagic fish appears plausible. Two pressures on these fish have moved in the unfavorable direction. First, the ocean lost more than 2% of its dissolved oxygen between 1960 and 2010 (Schmidtke et al. 2017), and mesopelagic fish respond to oxygen at the depths where they live (Koslow et al. 2011). Second, mesopelagic fish feed on zooplankton, whose

supply depends on the productivity of the surface ocean. A decline in phytoplankton has been reported in the North-East Atlantic over 1960–2019 (Holland et al. 2023) and over the past century globally (Boyce et al. 2010) (noting that this last finding is debated (McQuatters-Gollop et al. 2011)). At the same time, mesopelagic fish are barely fished, and the decline in the number of predatory fishes (Juan-Jordá et al. 2011; Christensen et al. 2014) may have been a counteracting force. Without further evidence, we take zero to be a plausible trend.

Second, one risk is that the LPI trend is driven by larger marine fish species that are commercially exploited, for which much better data exist. This would make the population drop implied by the LPI series overstate the reality of the population-weighted average change if these large commercially exploited species have also experienced larger population declines. We do not find evidence for this type of selection bias. First, among marine species, the coverage of the LPI is approximately similar (around 10% of the species listed in FishBase) for larger predatory fishes (trophic level equal to or above 3.5) and smaller prey fishes (trophic level between 2 and 3). Second, the LPI implies broadly similar trends for predatory fishes (43% decline over 1970–2010, C.I. [23%, 58%]) and prey fishes (64%, C.I. [2%, 87%]).³⁹ Even zooming within prey fishes on trends by commercial exploitation status, we do not find meaningful differences: relying on the fishery status classification of species in FishBase, we find similar trends for unexploited species (67% decline, C.I. [−91%, 92%]) and exploited species (64% decline, C.I. [4%, 87%]).⁴⁰

Validation of the LPI population changes against other sources. Taxon-level percentage declines were checked against the published and gray literature (journal articles, monitoring logs, and conservation-organization reports). We only consider large-scale independent monitoring programs, and do not report studies pertaining to a single species or site. The results of this exercise are recorded in Table C.3. Nearly every available source is narrower than a global taxon-level decline, reporting a trend for specific groups of species or regions. Overall, the orders of magnitude of taxon-level declines reported across studies line up well with the LPI estimates.⁴¹

³⁹The confidence interval is larger for prey fishes because there are fewer prey fish species, and so despite the similar coverage rate, there are about four times fewer prey fish species in the data.

⁴⁰Based on the FishBase classification, we label as exploited the species classified as highly commercial, commercial, minor commercial or subsistence, and as unexploited the species classified as of no interest or of potential interest. Including minor commercial and subsistence with exploited yields very similar results.

⁴¹We note a divergence with Christensen et al. (2014), who find an increase in the biomass of prey fish. It is worth noting that in this paper this trend is not measured directly but is based on food-web models.

TABLE C.3: Validation of LPI Taxon-Level Population Changes Against External Sources

<i>Panel A: Birds</i>					
STUDY	TIME SPAN	% CHANGE		MEASURE	NOTES
		TOTAL	ANNUAL		
LPI	1970–2020	–45.37	–1.20	Abundance	Weighted, global
LPI	1970–2020	–24.47	–0.56	Abundance	Unweighted, global
Bowler et al. (2019)	1990–2015	–13.00	–0.56	Abundance	Insectivores, Europe
Kitazawa et al. (2022)	1850–2016	–72.00	–0.76	Abundance	Birds (model-based), Ishikari lowland, Hokkaido, Japan
Lehikoinen et al. (2018)	2002–2014	–7.00	–0.61	Abundance	Birds, European mountains
Pollock et al. (2022)	1977–2020	–73.01	–3.00	Abundance	Birds, Parque Nacional Soberanía, Panama
Rosenberg et al. (2019)	1970–2017	–28.80	–0.72	Abundance	Birds, North America
SolB (2023)	2000–2022	–30.64	–1.65	Abundance	Birds, India
Wotton et al. (2017)	2010–2015	+64.60	+10.48	Abundance	Birds, Botswana
Wotton et al. (2017)	2009–2015	–15.00	–2.67	Abundance	Birds, Uganda
<i>Panel B: Fish</i>					
STUDY	TIME SPAN	% CHANGE		MEASURE	NOTES
		TOTAL	ANNUAL		
LPI	1970–2020	–77.16	–2.91	Abundance	Weighted, global
LPI	1970–2020	–64.10	–2.03	Abundance	Unweighted, global
Chevalier et al. (2023)	2003–2019	–75.20	–8.35	Biomass	Fish, Tonle Sap Lake, Lower Mekong Basin
Christensen et al. (2014)	1910–2010	–66.40	–1.08	Biomass	Marine predatory fish (model-based), global
Christensen et al. (2014)	1910–2010	+130.00	+0.85	Biomass	Marine prey fish (model-based), global
Hutchings et al. (2010)	1970–2009	–38.00	–1.22	Biomass	Marine fish (model-based), global
<i>Panel C: Herptiles</i>					
STUDY	TIME SPAN	% CHANGE		MEASURE	NOTES
		TOTAL	ANNUAL		
LPI	1970–2020	–89.95	–4.49	Abundance	Weighted, global
LPI	1970–2020	–82.41	–3.42	Abundance	Unweighted, global
Adams et al. (2013)	2002–2011	–28.77	–3.70	Occupancy	Amphibians, United States
Houlahan et al. (2000)	1966–1997	–46.50	–2.00	Abundance	Amphibians, primarily North America and Western Europe
Rovito et al. (2009)	1975–2007	–88.99	–6.66	Encounter rate	Salamanders, San Marcos, Guatemala
<i>Panel D: Mammals</i>					
STUDY	TIME SPAN	% CHANGE		MEASURE	NOTES
		TOTAL	ANNUAL		
LPI	1970–2020	–45.86	–1.22	Abundance	Weighted, global
LPI	1970–2020	–53.28	–1.51	Abundance	Unweighted, global
Gallego-Zamorano et al. (2020)	1992–2015	–4.84	–0.22	Distribution	Tropical mammals (model-based), global
Medd et al. (2025)	1987–2021	–71.25	–3.60	Abundance	Small mammals, North America
Ogutu et al. (2016)	1977–2013	–69.57	–3.25	Abundance	Wild mammals ≥ 15 kg, Kenya rangelands
Wolf and Ripple (2017)	1500–2016	–57.36	–0.17	Range	Large terrestrial carnivores (model-based), global
Woinarski et al. (2010)	1996–2009	–71.00	–9.08	Abundance	Small mammals, Kakadu National Park, Australia

Notes: This table compares LPI taxon-level population changes with estimates from external sources. Annualized changes are calculated using the compound annual growth rate over each reported time span. Model-based trends are classified as such when the underlying abundance or biomass history is reconstructed, projected, or simulated for periods or locations without direct biological observations. For the 1970–2010 predatory and prey indices discussed in the text, the marine index sample contains 738 predator species ($TL \geq 3.5$) and 190 prey species ($2.0 \leq TL \leq 3.0$), corresponding to 9.09% (738/8,116) and 10.19% (190/1,865), respectively, of the corresponding marine FishBase species pools. Pollock et al. (2022): Mean of 57 species-level trends in Fig. 1 (limbo plot), Parque Nacional Soberanía, Panama. SolB (2023): The equally weighted arithmetic mean of the six habitat-group indices shown in the figure on page 28. Rovito et al. (2009): Change in pooled encounter rates across 17 species-site comparisons representing eight salamander taxa, comparing historical surveys with 2005–2007. Gallego-Zamorano et al. (2020): Change in mean remaining species distribution inferred from reported average distribution losses of 38% in 1992 and 41% in 2015. Ogutu et al. (2016): Arithmetic mean of the 17 mammal species-level percentage changes in Fig. 2, excluding ostrich. Wolf and Ripple (2017): Unweighted arithmetic mean of historic range loss for all 25 large carnivore species in Fig. 2.

C.2 Other Sources for Population Changes for Wild Animals

Population change for insects. Beyond the LPI, we use a number of additional sources for population changes. Our estimate of the decline in the number of insects comes from [Dirzo et al. \(2014\)](#). [Dirzo et al. \(2014\)](#) construct an index of change in terrestrial or semi-terrestrial invertebrate abundance using a methodology similar to the LPI. Their estimate is based on 466 populations of 452 unique species across eight taxonomic groups: the family Aphididae and the orders Coleoptera, Diptera, Hemiptera, Hymenoptera, Lepidoptera, Odonata, and Rhabditida. All of these are insect taxa except Rhabditida, an order of nematodes, so we treat this estimate as representative of insects.

They estimate a 45% decline in insect populations from 1970 to 2008, corresponding to a 1.56% yearly decline on average. This corresponds to the lower range of the estimates in the literature: summarizing data from six different studies, [Wagner et al. \(2021\)](#) report 1% to 2% annual declines for terrestrial arthropods. Quantifying local declines across two distinct multi-year research programs in Europe, [Wildermuth et al. \(2026\)](#) get 5.1% and 0.5% yearly losses, and [Hallmann et al. \(2017\)](#) report a 6.1% annual decline in total flying insect biomass in protected areas.

Our analysis requires a population decline from 1970 to 2020. We use a conservative approach and use the same 45% decline, effectively assuming a 0% decline from 2008 to 2020.

Species-level population changes for mammals. Species-level historical and current totals are available only for a few mammal species in [Greenspoon et al. \(2025\)](#). Finally, we use [Ceballos and Ehrlich \(2002\)](#) for mammal range-contraction and extinction flags.

C.3 Sources for Current Population Levels of Wild Animals

Species-level population counts exist only for mammals and birds. For mammals, we combine two sources. [Greenspoon et al. \(2025\)](#) consolidates marine-mammal catch-reconstruction, unexploited-population estimates, and expert historical and current estimates for terrestrial mammals. The second is scraped Wikipedia population counts ([Wikipedia 2026](#)), which gather estimates from various scientific sources (with estimates

Studies that count small fish directly with standardized surveys find changes in both directions ([Heessen and Daan 1996](#); [Daan et al. 2005](#); [Benoît and Swain 2008](#); [Tulp et al. 2008](#); [Paddack et al. 2009](#); [Edgar et al. 2023](#)). As shown above, in the Living Planet Database, the largest global compilation of population series, the evidence points toward a decline for prey fishes.

systematically linked to the source). For each species, we choose the highest-quality and most representative source available, in the following order. For terrestrial mammals: hand-checked counts, then Wikipedia, then the expert estimates of [Greenspoon et al. \(2025\)](#), then their biomass-derived estimates. For marine mammals, where [Greenspoon et al. \(2025\)](#) provides the more systematic abundance estimates: hand-checked counts, then [Greenspoon et al. \(2025\)](#), then Wikipedia. Finally, we cross-check both of these sources against [Greenspoon et al. \(2023\)](#). For birds, our primary source is the scraped Wikipedia population counts ([Wikipedia 2026a](#)). We obtain order-level population counts using a bottom-up approach, i.e., we sum species-level counts at the order level.

At the level of the broad taxa, [Bar-On et al. \(2018\)](#) supplies population totals for birds and arthropods, and biomass estimates for fish, from which we build our own fish counts (Appendix C.4). This source does not provide a count for herptiles. We instead estimate their population bottom-up from mapped species ranges and predicted densities calibrated to national population estimates, as described in Appendix C.5. For mammals, it only provides a biomass estimate, not a population count, so we aggregate our species-level mammal counts to obtain the mammal-level estimate. For fish, [Bar-On et al. \(2018\)](#) does not provide a count for non-mesopelagic fish (only a biomass) or for freshwater fish (not considered). A separate Section C.4 below provides our derivation for a total fish count. We obtain a separate population count for insects from [Smithsonian Institution \(1996\)](#) at 1×10^{19} .

For mammals, the species-level population counts are exhaustive (by construction). For birds, the species-level population counts aggregate to 1.56% of the total from [Bar-On et al. \(2018\)](#).

Validation. We perform a number of validations of the population counts by taxon. For mammals, the sum of the species-level counts from [Greenspoon et al. \(2025\)](#) closely matches that from [Wikipedia \(2026\)](#) (6.75×10^{10} vs. 5.62×10^{10}). These numbers are also broadly in line with the estimate from [Tomasik \(2019\)](#), based on scaling up density estimates (1×10^{11} to 1×10^{12}). Finally, converting species-level population counts into biomass estimates and aggregating them up, we obtain a biomass of wild mammals at 0.0117 Gt C, very close to the 0.0099 Gt C estimate from [Greenspoon et al. \(2025\)](#). For birds, the group-level total from [Bar-On et al. \(2018\)](#) (3.00×10^{11}) lines up well with the estimate from [Callaghan et al. \(2021\)](#) (5×10^{10} – 4.28×10^{11}) based on 9,700 species (approximately 90% of all extant species per their estimate), as well as the estimate of [Tomasik \(2019\)](#) (1×10^{11} to 4×10^{11}). For insects, the estimate from [Smithsonian Institu-](#)

tion (1996) at 1×10^{19} is similar to the estimate of 1.16×10^{19} from [Royal Entomological Society \(2026\)](#). As expected, this is an order of magnitude smaller than the total number of arthropods estimated to be 1.422×10^{20} in [Bar-On et al. \(2018\)](#).

For mammals, we cross-check the species-level population counts across sources. This analysis is reported in Table C.4. We find that the three sources broadly agree, with a median species-level absolute difference around 25% for the three pairwise comparisons.

TABLE C.4: Pairwise Differences in Mammal Population Estimates

SOURCES	% MATCHED	ABSOLUTE % DIFFERENCE	
		MEDIAN	MEAN
Greenspoon et al. (2025) vs. Greenspoon et al. (2023)	99.34	25.50	25.93
Greenspoon et al. (2025) vs. Wikipedia (2026)	83.47	26.60	51.18
Greenspoon et al. (2023) vs. Wikipedia (2026)	85.17	24.84	43.35

Notes: This table compares the mammal population estimates across sources. % Matched is the number of matched species divided by the number of total species in the smaller source. Absolute % difference is calculated for each species as $100 \frac{|A-B|}{(A+B)/2}$. We report the mean and median of this statistic across matched species.

C.4 Estimation of Fish Count

The main contribution of [Bar-On et al. \(2018\)](#) is to provide estimates of the biomass of mesopelagic and non-mesopelagic fish. In addition, they provide a fish count by dividing the mesopelagic biomass by a typical mesopelagic mid-sized individual mass. This calculation is not a count of total fish individuals for three reasons: (i) the mesopelagic biomass includes juveniles while the divisor is a typical adult mass (as detailed below), (ii) the non-mesopelagic fish are not counted at all, and (iii) the calculation omits freshwater fish. We rely on the biomass estimates of [Bar-On et al. \(2018\)](#) but provide our own count calculation that addresses these three issues.

The objective is to estimate the number of fish alive at a time (a stock). An individual is a fish past metamorphosis (juveniles and adults, excluding eggs and larvae). For each group of fish the count is $N = \frac{B}{\bar{w}}$, where B is the group’s wet biomass and $\bar{w} \equiv \frac{B}{N}$ is the number-weighted mean mass of the individuals in the group. Throughout, we use the usual length–weight conversion $W = 0.01L^3$ (W in g, L in cm).

C.4.1 Methods for estimating the number-weighted mean mass of post-larval individuals

The size spectrum. A well-known empirical regularity of pelagic ecosystems, the Sheldon size spectrum, is that when a community is taken over many orders of mag-

nitude in body mass, biomass is roughly equal in each logarithmic size class, i.e., the number of individuals per unit of body mass falls as mass to the power of about -2 . It was first described for plankton by [Sheldon et al. \(1972\)](#) and has since been found to hold across marine life from bacteria to whales, with a slope of -2.04 for the ocean as a whole ([Hatton et al. 2021](#)). It underlies the indicators routinely used to monitor the effects of fishing and the global model of fish biomass by size of [Jennings et al. \(2008\)](#). This allows us to estimate the number-weighted mean mass of individuals between a minimum mass $w_{\min}$ and a maximum mass $w_{\max}$ as:

$$\bar{w} = w_{\min} \ln(w_{\max}/w_{\min}) / (1 - w_{\min}/w_{\max}).$$

For fish, the regularity is well measured above about 10 g, the size from which survey trawls retain fish ([Jennings and Blanchard 2004](#); [Daan et al. 2005](#)). Below 10 g it is more of an assumption, because no survey measures a whole fish community at those sizes.

Surveys. A survey gives $\bar{w}$ directly as the weight caught divided by the number caught. However, a survey is representative only over the size range the gear retains. In particular, every gear misses one end: fine nets and enclosed patches catch the small fish and lose the large mobile ones, while trawls and sonar do the reverse.

Method used for each group. We use the spectrum where the community spans many orders of magnitude and no single gear can see all of it (non-mesopelagic fish, freshwater fish) and use surveys where one net can see the whole community and the habitat is more uniform (mesopelagic fish).

C.4.2 Non-mesopelagic fish

Biomass estimate. For non-mesopelagic fish, [Bar-On et al. \(2018\)](#) use the biomass estimate of [Wilson et al. \(2009\)](#). [Wilson et al. \(2009\)](#) use 84 food-web models, one for each of the world's 66 large marine ecosystems (the coastal regions where most fishing happens) and 18 for the rest of the ocean. In each model the fish are sorted into groups by habitat and size, and the model finds the smallest biomass each group must have had in 1950 for two things to have been sustainable: the fish caught between 1950 and 2004, and the fish eaten by predators. Within each group the biomass is built up from monthly age cohorts from age one month, so it covers every post-larval fish. This yields 1.0 Gt of wet mass in 1950.

As a validation, the results in [Costello et al. \(2016\)](#) allow us to recover a wet mass

around 1.5 Gt for 2010s data, confirming the order of magnitude of the [Wilson et al. \(2009\)](#) estimate using a completely independent method and suggesting that it lies on the conservative end.

Number-weighted mean mass of post-larval individuals. Non-mesopelagic fish run from 0.05 g gobies to 500 kg tuna, six to seven orders of magnitude, where the spectrum model is likely to hold well. By contrast, no gear covers gobies to tuna and habitats differ widely. Hence our baseline comes from the spectrum model.

Most marine fish take their juvenile form at 1–2.5 cm: reef fishes at 0.8–1.5 cm ([Juncker et al. 2006](#)), herrings and cods at 2.5–5 cm. This implies the small species enter at about $w_{\min} = 0.05$ g. The largest common fish (tunas, billfish) reach several hundred kilograms, so we set $w_{\max} = 10^6$ g. A -2 spectrum run from 0.05 g to 10^6 g gives $\bar{w} = 0.84$ g.

Benchmarking the small end against survey data. As explained above, the spectrum model has best empirical support above 10 g, while our estimate assumes a power law down to 0.05 g. We therefore collect the survey evidence that exists on the small end of the range, test the model there, and correct it where it fails.

The only surveys that retain every fish are complete censuses of small enclosed patches. This is done only in shallow, clear water, in practice on coral reefs. We found four reefs for which the sizes of all the fish collected are available: Orpheus Island on the Great Barrier Reef, sampled in 1993–1997 ([Ackerman and Bellwood 2000](#); [Ackerman et al. 2004](#)), and Moorea, Lizard Island and Belize, sampled in 2016–2018 ([Brandl et al. 2019](#)). We find that the very small fishes (below 0.2 g, about 2.7 cm) are less numerous than the spectrum predicts. The reef data therefore suggest a correction to the spectrum at the small end: we start the spectrum at 0.2 g, which yields a mean weight of 3.1 g, and add the fish below 0.2 g as a measured share s taken from the surveys, at about 0.1 g each, so that $\bar{w} = (1 - s) \times 3.1 + s \times 0.1$. The three reefs in [Brandl et al. \(2019\)](#) yield $s \in [0.50, 0.67]$ so that $\bar{w} \in [1.1, 1.6]$ g, 1.3 g at the midpoint. The Orpheus Island reef has fewer small fishes: s is 0.20–0.35, which gives a mean mass of 2.3 g at the midpoint. Note that this may be an overcorrection, because reefs are very specific environments: we assume that the share of small fish measured on reefs also holds on the shelves and in the open sea that carry most of the 1 Gt. We use 1.3 g as our baseline, because it rests on three reefs in three oceans. The values 0.84 g and 2.3 g provide the two ends of the range.

Counts. Table [C.5](#) gives the resulting counts for the three mean mass estimates.

TABLE C.5: Estimated Number of Post-Larval Non-Mesopelagic Fish

Biomass (Gt wet)	Mean mass $\bar{w}$		
	$\bar{w}=2.3$ g	$\bar{w}=1.3$ g	$\bar{w}=0.84$ g
1.0	4.4×10^{14}	7.5×10^{14}	1.2×10^{15}

Notes: The biomass is the 1.0 Gt of [Wilson et al. \(2009\)](#) used by [Bar-On et al. \(2018\)](#). Each column corresponds to one of the three mean masses of this section. The central estimate is the middle cell.

Share of non-mesopelagic fish living in the deep sea (below 1,000 m). The non-mesopelagic group bundles fish living above 200 m with fish living below 1,000 m, whose populations may follow different trends. This is not a problem in practice: fish living below 1,000 m are at most a few percent of the non-mesopelagic count, so the count in Table C.5 can be read as a count of fish living above 200 m. Indeed, the deep-sea groups in the non-mesopelagic aggregate hold under 2% of its biomass ([Christensen et al. 2009](#)). Only the smallest of them, 0.7% of the biomass, is of the small-bodied kind (under 30 cm) that yields many fish per metric ton. The others are medium and large fish (30 cm and above), whose share of the count is below their share of the biomass. The deep-sea fish are therefore at most a few percent of the non-mesopelagic count.

C.4.3 Mesopelagic fish

Mesopelagic fish are the small fish of the open ocean between 200 and 1,000 m depth. Two groups dominate them everywhere: lanternfish (family Myctophidae) and bristle-mouths (genus *Cyclothone*).

Biomass estimate. We use the estimate of [Bar-On et al. \(2018\)](#): 0.5 GtC, or 3.3 Gt of wet mass at their carbon content of 15%. It is the geometric mean of two estimates obtained by different methods.

The *sonar-based estimate*, 1.8 GtC (12 Gt wet), starts from the echo that the layer of mesopelagic fish returns to ship sonar. The global echo is the geometric mean of two published totals, that of [Irigoien et al. \(2014\)](#), measured on a round-the-world research cruise in 2010–2011, and that of [Proud et al. \(2017\)](#). It is converted into biomass using the echo that a fish of a given weight returns.⁴² The *trawl-based estimate*, 0.15 GtC (1 Gt wet), is the global map of [Lam and Pauly \(2005\)](#), built from the trawl catches

⁴²Most of a fish's echo comes from its swim bladder, the gas-filled organ that controls buoyancy, and many mesopelagic fish have a reduced or fat-filled one. The biomass inferred from a given echo therefore depends on the share of fish with a gas-filled swim bladder. [Bar-On et al. \(2018\)](#) assume a share of 0.5 and get an estimate similar to [Irigoien et al. \(2014\)](#). They put the uncertainty of the sonar-based estimate at about eightfold.

compiled by [Gjøsæter and Kawaguchi \(1980\)](#). It predates 1980 and is known to be too low, because mesopelagic fish avoid nets ([Kaartvedt et al. 2012](#)). The geometric mean of the two is 0.52 Gt C, which [Bar-On et al. \(2018\)](#) report as ≈ 0.5 Gt C.

Two properties of this biomass matter for a count. First, *it includes juveniles but not larvae*. Sonar detects fish once their swim bladder has developed, which includes juveniles. The trawls retain fish from about 2 cm, while mesopelagic fish take their juvenile form at 1–2 cm. Both estimates therefore cover juveniles and adults, and exclude larvae, as intended. Second, *we assume it includes the numerous but tiny and hence hard-to-detect bristlemouths*. Indeed, most evidence points toward bristlemouths being appropriately captured by sonar-based estimates ([Irigoien et al. 2014](#); [Davison et al. 2015](#); [Sarmiento-Lezcano et al. 2023](#)).⁴³

The sonar data are from 2010–2011 and the trawl data from before 1980.

Number-weighted mean mass of post-larval individuals. Mesopelagic fish span three orders of magnitude (0.01 g to tens of grams) in a peaked community of two or three families that hold both most numbers and most biomass. Hence, the spectrum method is likely to perform poorly, while surveys based on fine-mesh midwater nets should be representative.

A survey is usable if its net keeps every juvenile and adult, does not count larvae, and is not dodged by the larger, faster fish. The two stations of [Davison et al. \(2015\)](#) meet these conditions with a mesh that keeps fish from about 1 cm (0.01 g), and excludes larvae. We summarize the features of these surveys in Table C.6. The same net gives $\bar{w} = 0.80$ g at a station in nutrient-rich coastal water (the California Current) and 0.20 g in the nutrient-poor center of the North Pacific (a subtropical gyre). The factor of four between them is therefore geography, not gear. To combine the two stations into a global mean, we take the biomass-weighted harmonic mean of the regional means $\bar{w}_{\text{meso}} = (f_{\text{gyre}} w_{\text{gyre}}^{-1} + (1 - f_{\text{gyre}}) w_{\text{Calif}}^{-1})^{-1}$, where f_{gyre} is the share of mesopelagic biomass that lives in gyre-like water. Using $f_{\text{gyre}} = 0.3$, we obtain $\bar{w}_{\text{meso}} = 0.42$ g.⁴⁴

There exist two other surveys covering mesopelagic communities, but each fails

⁴³If not, the bristlemouth biomass would have to be added at its share f_C of the biomass caught in surveys, and the count would rise by $1/(1 - f_C)$: a factor of 1.3–1.8 for the f_C of 0.25–0.45 at the two stations of [Davison et al. \(2015\)](#). We treat this as an upper bound.

⁴⁴We obtain f_{gyre} as follows. Gyres cover about 40% of the ocean’s area but hold fewer fish per square meter. [Irigoien et al. \(2014\)](#) find that the sonar echo per unit area rises with the growth of plankton at the base of the food chain (primary production, PP, measured by satellite) with an elasticity of about 1.5 ($\ln s_A = 1.52 \ln \text{PP} - 1.36$, where s_A is the echo per unit area). Primary production is 200–300 mg C m⁻² d⁻¹ in gyres and 500–700 elsewhere, so a square meter of gyre holds 2.2–6.5 times less biomass, and f_{gyre} is 0.1–0.25. We set $f_{\text{gyre}} = 0.3$, above that range, which leans toward the lighter gyre fish and hence toward a higher count.

one condition. [Sutton et al. \(2010\)](#) use a very fine mesh in the Sargasso Sea and obtain 0.125 g. But at least 5.6% of their count is larvae and their small nets are dodged by larger fish: this value is too low. [Sutton et al. \(2008\)](#) use a coarse mesh over the Mid-Atlantic Ridge and obtain 1.8–4.0 g. But that mesh lets bristlemouths and small lanternfish through: this value is too high. It is reassuring that these values bracket our baseline.

TABLE C.6: Surveys of Mesopelagic Fish Used to Estimate the Mean Mass

Study, region, date	Net	Smallest fish kept	$\bar{w}$ (g)
Davison et al. (2015) , California Current, 2008, four tows	midwater net (Matsuda–Oozeki–Hutaw), 5 m ² opening, 1.7 mm mesh	about 1 cm (0.01 g), larvae excluded	0.80 (most common weight 0.12)
Davison et al. (2015) , North Pacific Subtropical Gyre, 2009, three tows	midwater net (Matsuda–Oozeki–Hutaw), 5 m ² opening, 1.7 mm mesh	about 1 cm (0.01 g), larvae excluded	0.20 (most common weight 0.12)
Sutton et al. (2010) , Sargasso Sea, 5 stations, April 2006, 0–5,000 m	plankton nets (MOCNESS), 1 and 10 m ² openings, 0.335 mm mesh	a few mm, juveniles and adults plus at least 5.6% larvae	0.125
Sutton et al. (2008) , Mid-Atlantic Ridge, June 2004, three groups of stations (0–750 m Reykjanes Ridge to the Faraday Seamounts, 0–750 m Azores, and 750 to >2,300 m along the ridge)	large midwater net (Krill trawl), 36 m ² opening, 6 mm mesh	about 2 cm (0.08 g), juveniles and adults	2.0, 1.8, 4.0

Notes: Surveys of mesopelagic fish from which a number-weighted mean mass can be computed. Number-weighted mean mass $\bar{w}$ is calculated as the total biomass divided by the total number of individuals.

Check against other benchmarks. [Bar-On et al. \(2018\)](#) divide the same biomass by 3 g, the weight of a mid-sized lanternfish built from the length ranges in [Fock and Ehrich \(2010\)](#). The difference arises because a number-weighted mean is far below the weight of a typical individual when most individuals are small. The size spectrum of the previous section, run from the size at which these fish take their juvenile form (0.02 g) to the largest common ones (50 g), gives 0.16 g, and 0.08–0.38 g across plausible bounds. This is below the survey means, as expected for a community concentrated in a few species groups rather than spread evenly over sizes.

Counts. Table [C.7](#) gives the resulting counts as a function of the mean mass $\bar{w}$ assumption, with our central estimate in the middle cell.

TABLE C.7: Estimated Number of Post-Larval Mesopelagic Fish

Biomass (Gt wet)	Mean mass $\bar{w}$		
	$\bar{w}=0.80$ g	$\bar{w}=0.42$ g	$\bar{w}=0.125$ g
3.3	4.2×10^{15}	7.9×10^{15}	2.7×10^{16}

Notes: The biomass is the 0.5 GtC (3.3 Gt wet) of [Bar-On et al. \(2018\)](#). The middle column is the baseline (biomass-weighted harmonic mean of the two [Davison et al. \(2015\)](#) stations). The outer columns are the survey values of the California Current ([Davison et al. 2015](#)) and the Sargasso Sea ([Sutton et al. 2010](#)), which bracket the baseline. The central estimate is the middle cell.

C.4.4 Freshwater fish

Biomass estimate. We multiply habitat areas (large lakes, small lakes and ponds, rivers, reservoirs, flooded floodplains) by measured standing stocks per hectare. Table [C.8](#) lists our estimates for areas and densities for each habitat, and the resulting biomass (refer to table notes for sources). For each habitat, we give a low, central and high estimate of area and density, which yields a low, central and high estimate of biomass. The total biomass is 0.015–0.126 Gt wet, with a central estimate of 0.050 Gt wet.

TABLE C.8: Freshwater Fish Biomass from Habitat Area and Standing-Stock Density

Habitat	Area (10 ⁶ km ²)	Density (kg ha ⁻¹)	Biomass (Gt wet)
	low / central / high	low / central / high	low / central / high
Lakes above 1,000 km ² (a)	1.27 / 1.27 / 1.27	30 / 45 / 70	0.004 / 0.006 / 0.009
Other lakes of at least 10 ha (b)	1.40 / 1.40 / 1.40	20 / 50 / 100	0.003 / 0.007 / 0.014
Lakes and ponds of 0.2–10 ha (c)	0.6 / 1.0 / 1.3	15 / 40 / 80	0.001 / 0.004 / 0.010
Rivers and streams (d)	0.77 / 0.77 / 0.77	35 / 120 / 300	0.003 / 0.009 / 0.023
Reservoirs (e)	0.25 / 0.27 / 0.32	50 / 150 / 300	0.001 / 0.004 / 0.010
Floodplains and marshes, time-averaged flooded area (f)	1.0 / 2.0 / 3.0	40 / 100 / 200	0.004 / 0.020 / 0.060
Total			0.015 / 0.050 / 0.126

Notes: Densities are wet mass of all fish retained by survey gear per hectare of water. Each row's biomass is area times density. Sources by row. (a) Area: [Messenger et al. \(2016, Table 1\)](#). Density: a lake-by-lake sum of published lake-wide stocks for the twelve largest lakes with a figure, two thirds of the row's area: Caspian ([Khodorevskaya et al. 2014](#)), Superior ([Vinson et al. 2022](#)), Victoria ([Lake Victoria Fisheries Organization 2023](#)), Huron and Michigan ([O'Brien et al. 2024](#); [Warner et al. 2024](#)), Tanganyika ([Szczucka 1998](#)), Malawi ([Thompson and Allison 1997](#)), Erie ([Zhang et al. 2016](#)), Ontario ([Weidel et al. 2024](#)), Turkana ([Muška et al. 2012](#)), Kivu ([Guillard et al. 2012](#)), and Baikal, for which no lake-wide survey is published and a judgment value is used. The remaining third takes 10 / 25 / 50 kg ha⁻¹. (b) Area: [Messenger et al. \(2016, Table 1\)](#). Density: the lake censuses of [Randall et al. \(1995\)](#) grouped by climate band and weighted by where the world's lake area lies, with [Hanson and Leggett \(1982\)](#), [Samarasin et al. \(2015\)](#) and [Rypel and David \(2017\)](#). (c) Area: the size classes of [Verpoorter et al. \(2014\)](#) (central), [Messenger et al. \(2016\)](#) (low) and [Downing et al. \(2006\)](#) (high). Density: as (b), with an allowance for small waters that hold no fish ([Tammi et al. 2003](#); [Schilling et al. 2008](#)). (d) Area: [Allen and Pavelsky \(2018\)](#). Density: the river censuses of [Randall et al. \(1995\)](#) and [Welcomme \(1985\)](#), and the measured main channels of [Welcomme \(1985, Table 6.13\)](#) and [Dettmers et al. \(2001\)](#). (e) Area: [Lehner et al. \(2011\)](#), less the small reservoirs already inside the lake inventory. Density: rotenone censuses of United States reservoir coves ([Parisek et al. 2024, Dataset S1](#)), and [Gerasimov et al. \(2025\)](#), [Fernando and Holčík \(1991\)](#) and [Kolding and van Zwieten \(2012\)](#) for the rest of the world. (f) Area: the wetland classes of [Lehner et al. \(2025, Table 3\)](#), weighted by the share of the year each class is under water (own calculation). Density: [Welcomme \(1985\)](#) and [Bayley \(1988\)](#). Rows may not sum to the totals because of rounding.

The majority of the central estimate rests on censuses and yield compilations from the 1950s to the 1980s. Only the large-lake row is measured in the 2010s and 2020s.

Number-weighted mean mass of post-larval individuals. Freshwater fish run from 0.05 g to tens of kilograms, about six orders of magnitude, across lakes, rivers, reservoirs and floodplains whose communities differ, and no gear covers all sizes. Hence, we rely on the spectrum model. The evidence for the spectrum was established in lakes as early as in the sea, and lake and ocean communities fall on the same line ([Sprules and Barth 2016](#)). Most freshwater fish take their juvenile form at 1.5–2.5 cm. Carps and their relatives do so gradually, between 1.5 and 4 cm ([Copp and Kováč 1996](#)), while the many small live-bearing species are born at about 1 cm. This is the same range as in the sea, so the small species enter at about $w_{\min} = 0.05$ g. The largest common freshwater fish (carps, catfishes, pikes, sturgeons) are 10–100 kg, so $w_{\max}$ is 10^4 – 10^5 g. We take 3×10^4 g. A -2 spectrum run from 0.05 g to 30 kg gives 0.67 g.

As for marine fish, the spectrum may fail at the small end, and no census of small freshwater fish is both exhaustive and representative of lakes and rivers, so it cannot be tested there. We therefore assume that the share of fish below 0.2 g is the same as on the reefs. The spectrum from the 0.2 g mode with the three-reef share gives 1.05 g, and with the Orpheus Island share 1.76 g. The upper limit matters less than the share: across $w_{\max} = 10^4\text{--}10^5$ g the three-reef analogue runs 0.96–1.15 g. We use 1.05 g as our central estimate.

Counts. Table C.9 gives the resulting counts.

TABLE C.9: Estimated Number of Post-Larval Freshwater Fish

Biomass (Gt wet)	Mean mass $\bar{w}$		
	$\bar{w}=1.76$ g	$\bar{w}=1.05$ g	$\bar{w}=0.67$ g
0.015	8.8×10^{12}	1.5×10^{13}	2.3×10^{13}
0.050	2.8×10^{13}	4.8×10^{13}	7.5×10^{13}
0.126	7.2×10^{13}	1.2×10^{14}	1.9×10^{14}

Notes: Each line corresponds to one of the three biomass scenarios of Table C.8, each column to one of the three mean masses. The central estimate is the middle cell.

In our baseline, we use a lower and more conservative freshwater count: the bottom-up count of our density maps (Appendix C.8), 3.3 trillion fish today. Backcast to 1970 with the LPI decline for non-mesopelagic fish (77%), it gives 1.4×10^{13} fish, against 4.8×10^{13} in Table C.9.

C.4.5 Summary

Table C.10 collects the central estimates for each fish group and computes a total fish count.

TABLE C.10: Estimated Number of Post-Larval Fish, Central Estimates

Group	Biomass (Gt wet)	Mean mass $\bar{w}$ (g)	Count	Share of count
Mesopelagic	3.33	0.42	7.9×10^{15}	91.2%
Non-mesopelagic	1.00	1.3	7.5×10^{14}	8.7%
Freshwater	–	–	1.4×10^{13}	0.2%
Total	4.33	0.50	8.7×10^{15}	100%

Notes: Central estimates of Tables C.5, C.7 and C.9. The mean mass in the total row is total biomass over total count. Freshwater: the bottom-up count of our density maps backcast to 1970, which we use in the baseline. The biomass-based count of Table C.9 is 4.8×10^{13} .

These fish counts correspond to different dates. We do not attempt to correct for changes over time: these level estimates are to be understood as order-of-magnitude estimates obtained from a combination of different data sources, and we do not know how to apply a correction in a way that is consistent across the three groups. We adopt the convention of treating these estimates as 1970s values. Since our fish trend implies a decline in fish numbers between 1970 and the present, this is a conservative choice.

C.5 Estimation of Herptile Count

No global count exists for any reptile or amphibian, so we build populations from the bottom up for 11,100 reptile species (IUCN, plus 1,150 from the GARD atlas of [Roll et al. 2017](#)) and 7,768 amphibians. For the 348 reptile and 119 amphibian species with measured local densities in TetraDENSITY ([Santini et al. 2018](#)), we regress log density on log body mass. The resulting slopes, -0.63 and -0.65 , are close to the -0.75 of Damuth's law ([Damuth 1981](#)). We use this to project species density across their range maps. Because densities are measured in occupied habitat where the species is common, predicted density times range area overstates a population. We calibrate the overstatement against independent national estimates, those European Union states report under the Habitats Directive ([European Environment Agency 2020](#)), 30 reptile and 23 amphibian figures from six countries, and China's national wildlife survey. The adjustment factor is 0.0048 for reptiles and 0.0046 for amphibians (0.032 for mammals, calibrated on [Greenspoon et al. 2023](#)). The result is 8.07×10^{11} reptiles and 6.46×10^{11} amphibians, 1.45×10^{12} in all.

C.6 Combined Dataset: Population Estimates Over Time

For all five broad taxa (mammals, birds, fish, herptiles, and insects), we combine our estimates of population counts with estimates of population changes since 1970 to backcast historical taxon-level populations since 1970.

For mammals and birds, we additionally construct a more disaggregated dataset: we start from current populations and apply species-level growth rates, order-level growth rates, or growth rates for the broad taxon, depending on data availability. For mammals, the current species-level population data for 4,959 species are exhaustive by construction, and this backcasting is done with species-level growth rates for 33 species, order-level growth rates for 4,165, and taxon-level growth rates for the remaining 761

species. For birds, current species-level population data exist for 2,289 species. Birds belonging to these species account for 1.56% of the total bird population. Backcasting is done with order-level growth rates for 1,479 species, and the bird growth rate for the remaining 810 species as well as the bird population not accounted for in our species-level dataset.

C.7 Farmed Animals

We process farmed animals separately from the wild-animal datasets to produce the global, wide-format dataset `pop_domestic_detailed_clean.dta`. It is constructed directly from the 2025 normalized FAOSTAT livestock release ([Food and Agriculture Organization of the United Nations 2025](#)). Processing retains records whose element is “Stocks,” removes records flagged as missing, and drops the composite items “Cattle and Buffaloes,” “Sheep and Goats,” and “Poultry Birds” to avoid double counting. We convert values reported in FAOSTAT’s “1000 An” unit to individual animals. For bees, FAOSTAT’s “No” unit denotes beehives rather than individual insects. This unit is retained and recorded in the variable `pop_unit`. The intermediate area-year file consists of both country- and global-level population estimates.

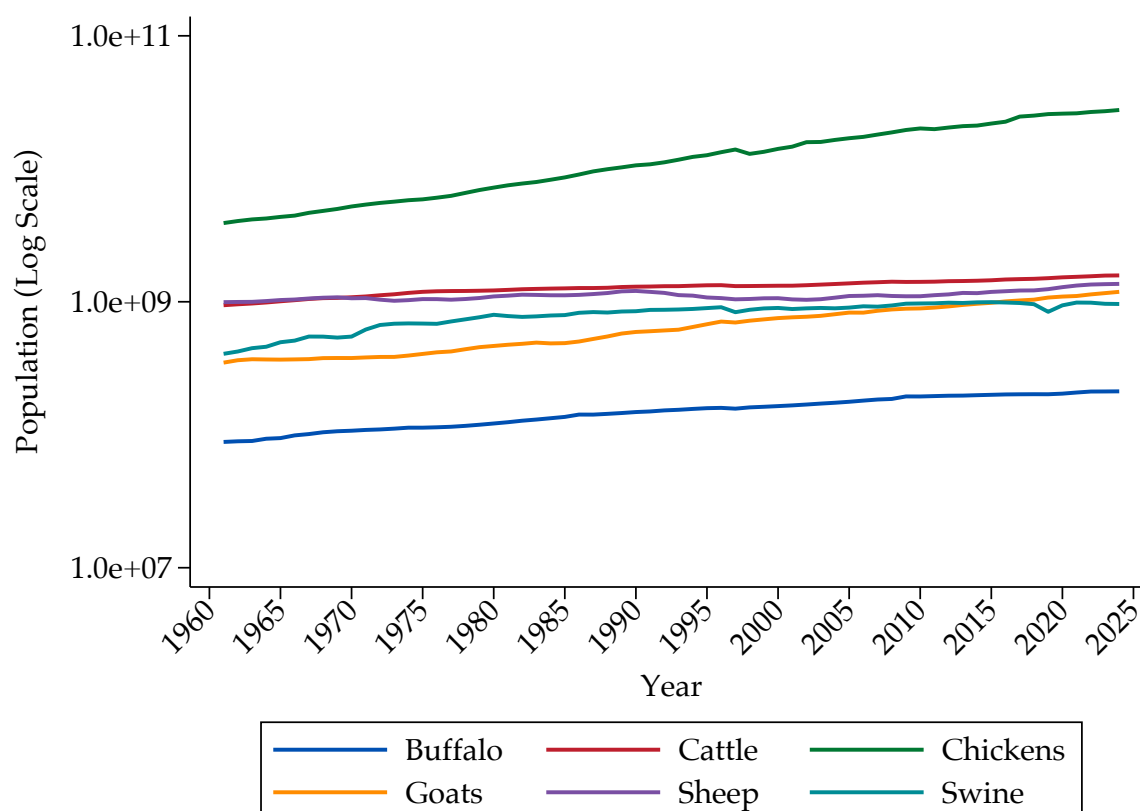
To construct the final dataset, we retain only FAOSTAT’s “World” observations and reshape the data into annual population columns from 1961 through 2024. The resulting dataset contains 18 animal categories (asses, bees, buffalo, camels, cattle, chickens, ducks, geese, goats, horses, mules and hinnies, other birds, other camelids, other rodents, rabbits and hares, sheep, swine, turkeys). Seventeen categories have observations throughout 1961–2024, while the “Other birds” series begins in 1991. Each category is accompanied by a scientific-name field, broad taxonomic group, taxonomic order, and categorical size group, which we assign based on typical weight assumptions: Small (< 10 kg), Medium (10–99 kg), and Large (≥ 100 kg).

Figure C.1 highlights selected farmed animals’ global population evolution since 1961.

C.8 Spatial Data

Mammals. Our data contain a positive count for 4,957 species (we have counts for 4,959 species, including two species with zero counts), summing to 7.84×10^{10} . Of these, 4,913 match a mapped species. The other 44 (1.8% of individuals, the house

FIGURE C.1: Farmed Animal Populations



Notes: This figure shows the population of major farmed animals at the global level since 1961. Data are from FAOSTAT.

mouse above all) cannot be placed. A further 968 mapped species lack a count. We impute their populations from the realized density (count over range area) of counted species in the same family, or order when the family has fewer than three counted species, times their range area. The imputations add 7% above the total in Table 1.

Birds. [Callaghan et al. \(2021\)](#) estimate the abundance of 9,695 species from eBird observations and range sizes. We use their means, which sum to 4.30×10^{11} (the medians sum to 5.05×10^{10}). For 14 seabird families, we use the counts of our population file instead (Wikipedia, citing BirdLife International), because estimates based on eBird observations fail for colonial and pelagic birds. The 1,438 mapped species without an estimate are imputed from their relatives as for mammals (2.3% of birds). The result of 3.25×10^{11} is scaled by a factor of 0.92, common to all species, to match the 3.00×10^{11} of [Bar-On et al. \(2018\)](#) in Table 1.⁴⁵

⁴⁵Scaling the medians instead would need a factor of six, and push well-measured species outside their own intervals. Adopting the unscaled totals does not materially change our overall conclusions on

Herptiles. For herptiles, we rely on our bottom-up construction of the herptile total, described in Appendix C.5, which already delivers species-level total counts and ranges.

Fish. No comprehensive published species counts exist for fish. For marine fish, we use the estimate in each gridded ocean cell from BOATS (Carozza et al. 2016), an ocean food-web model of the FishMIP project (Tittensor et al. 2018) that converts plankton growth and temperature from a climate model into fish biomass by size class in every 1° square, allowing for fishing. Dividing biomass by mean mass per size class gives 3.49×10^{14} fish above the mesopelagic zone. We scale this layer so that it sums to the present-day non-mesopelagic count behind Table 1, 1.71×10^{14} (the 7.5×10^{14} of Appendix C.4 in 1970, less the Living Planet decline), and the mesopelagic layer, which the maps in the text exclude, to the 7.9×10^{15} of the same census. Species then come from AquaMaps (Kaschner et al. 2019), which predicts where each species can occur from the conditions it tolerates. A cell's total fish from BOATS are split equally among the species present. This is credible on the continental shelves, which hold 12% of the fish and hundreds of species per cell. In the open ocean only a handful of wide-ranging species are predicted, so its fish are carried as species unknown at the fish-level sentence weight.

For freshwater fish, we estimate the number of fish per grid cell by computing water area, from HydroLAKES and HydroRIVERS (Messenger et al. 2016; Lehner and Grill 2013), multiplied by fish densities from electrofishing surveys of rivers (Comte et al. 2021) and hydroacoustic surveys of lakes (Emmrich et al. 2012), adjusted for depth and climate. This gives 3.3 trillion fish today, the freshwater count we use throughout the paper (Appendix C.4). The species for these estimates then come from the IUCN maps of river basins, divided equally among species present in a spatial cell.

C.9 Land Prices in the Brazilian Amazon

Section 5.6 compares the value of biodiversity in the Amazon with the price of land. Our benchmark is the median bare-land value of cattle land across the 29 regional land markets of Acre, Amapá, Amazonas, Pará, Rondônia and Roraima in INCRA (2025), BRL 11,900 per hectare at December 2024 prices, converted at the 2024 average exchange rate of 5.39 BRL per USD. The interquartile range across all 60 regional land markets in the Legal Amazon is USD 110,000–300,000 per km^2 . Native forest is valued at about USD 65,000 per km^2 , and consolidated soybean cropland in Mato Grosso at

the spatial values of biodiversity in the main text.

about USD 865,000 per km².

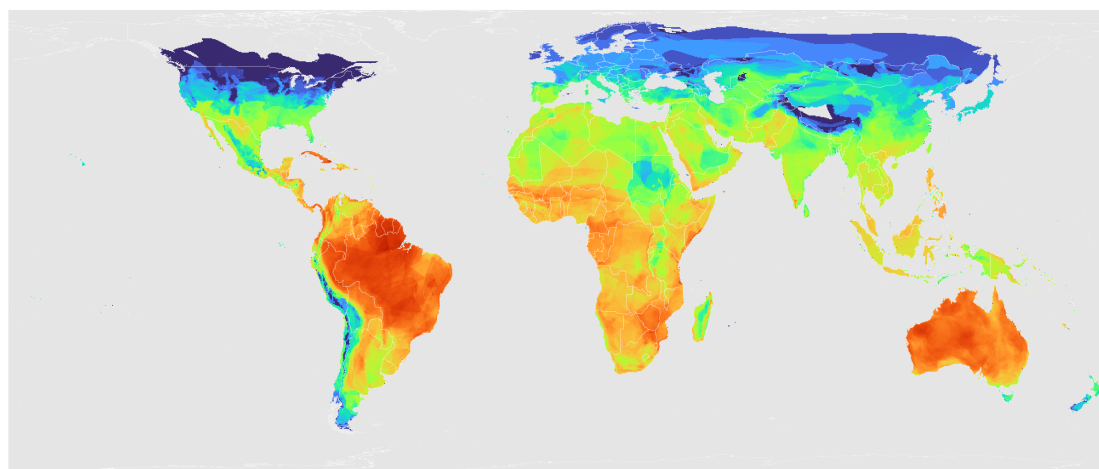
C.10 Additional Results

TABLE C.11: Welfare-Range Estimates from [Fischer \(2023\)](#)

SPECIES	MEAN	OUR WEIGHT
Pigs	0.548	0.144
Chickens	0.368	0.043
Octopuses	0.380	0.002*
Carp	0.199	0.005
Bees	0.165	0.002
Salmon	0.157	0.005
Crayfish	0.137	0.002*
Shrimp	0.210	0.002*
Crabs	0.102	0.002*
Black Soldier Flies	0.050	0.002
Silkworms	0.020	0.002

The “Mean” column is Fischer’s mean welfare range. The “Our Weight” column is $\omega_s = (z_s/z_{\text{human}})^\beta$ with $\beta = 0.53$ and $z_{\text{human}} = 86$ billion total neurons. The neuron count is the count used in the baseline welfare calculation. It is the species count for pigs and chickens, the fish average for carp and salmon, and the insect average for bees, black soldier flies, and silkworms. *Crustaceans and cephalopods are not included in our baseline welfare-loss quantification. The weight shown applies the invertebrate (arthropod) neuron average.

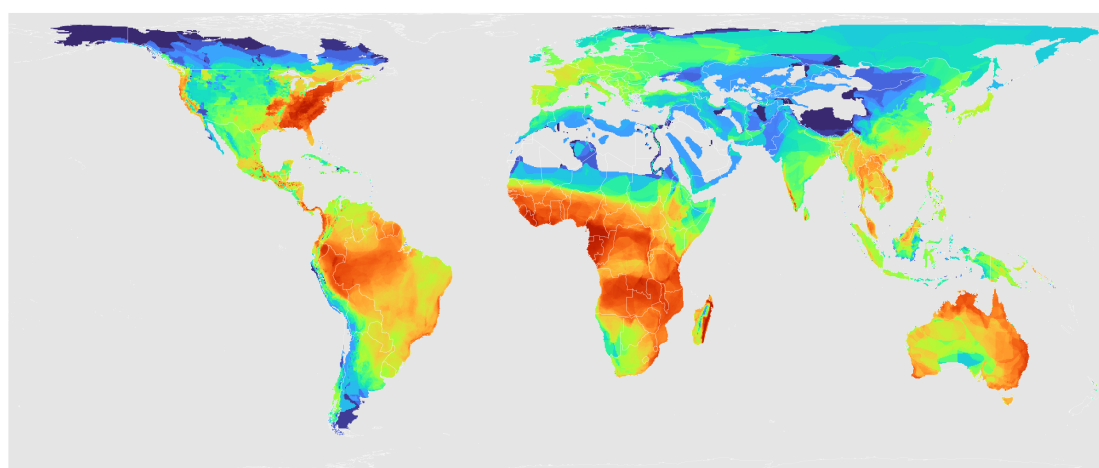
FIGURE C.2: Current Density of Wild Reptiles and Amphibians



Individuals per km²

≤ 200 1,000 5,000 ≥ 50,000

(A) Reptiles



Individuals per km²

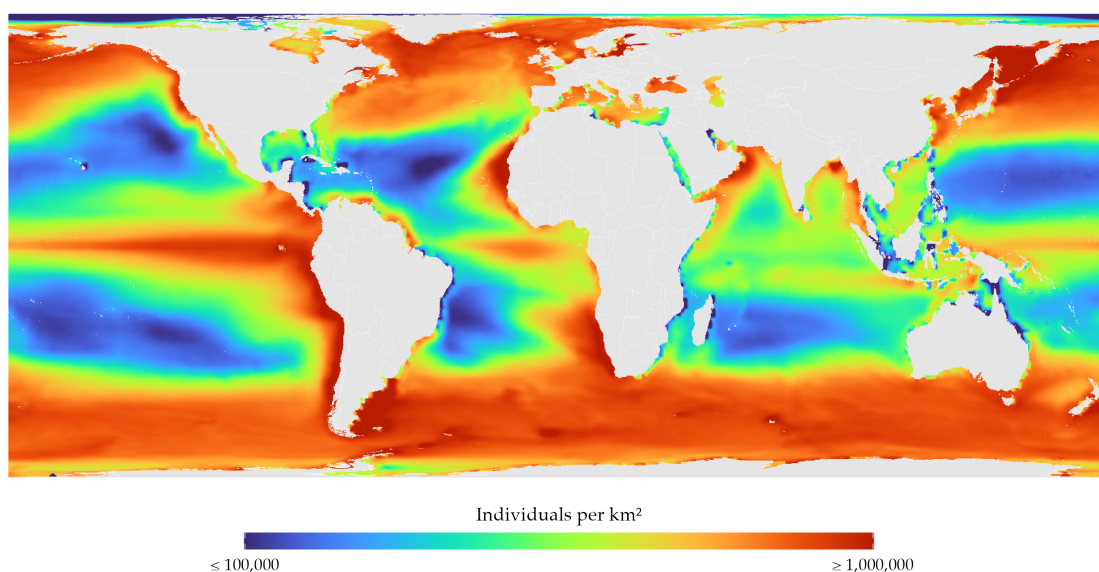
≤ 100 500 2,000 10,000 ≥ 50,000

(B) Amphibians

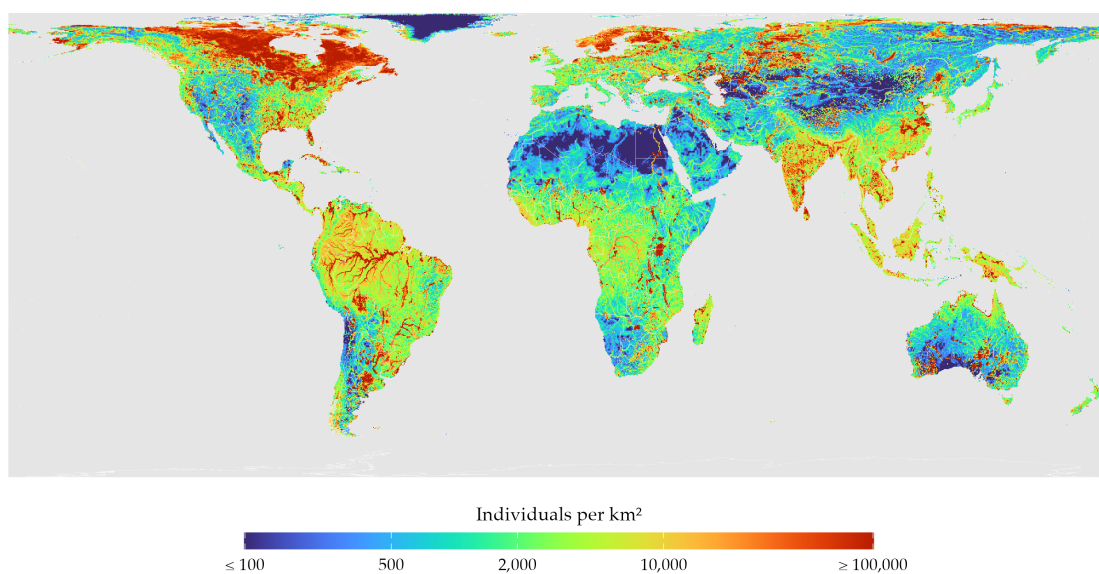
Notes: Individuals per km² for 11,100 reptile species with IUCN or GARD ranges and 7,768 amphibian species with IUCN ranges. Populations come from the density model calibrated on national estimates (Appendix C.5).

FIGURE C.3: Current Density of Marine and Freshwater Fish

(A) Marine Fish Outside the Mesopelagic Zone



(B) Freshwater Fish



Notes: Panel (a) shows fish per km² of sea surface from the BOATS model of the FishMIP project, 2005–2014 means, with scale limits at the 5th and 95th percentiles of ocean cells. Panel (b) shows lake and river area from HydroLAKES and HydroRIVERS times survey densities, expressed per km² of land rather than of water.

D. DATA SOURCES: NEURON DATA, COGNITIVE TESTS, MERGED DATASETS

D.1 Neuron Data

Table D.1 describes the data sources used for neuron counts.

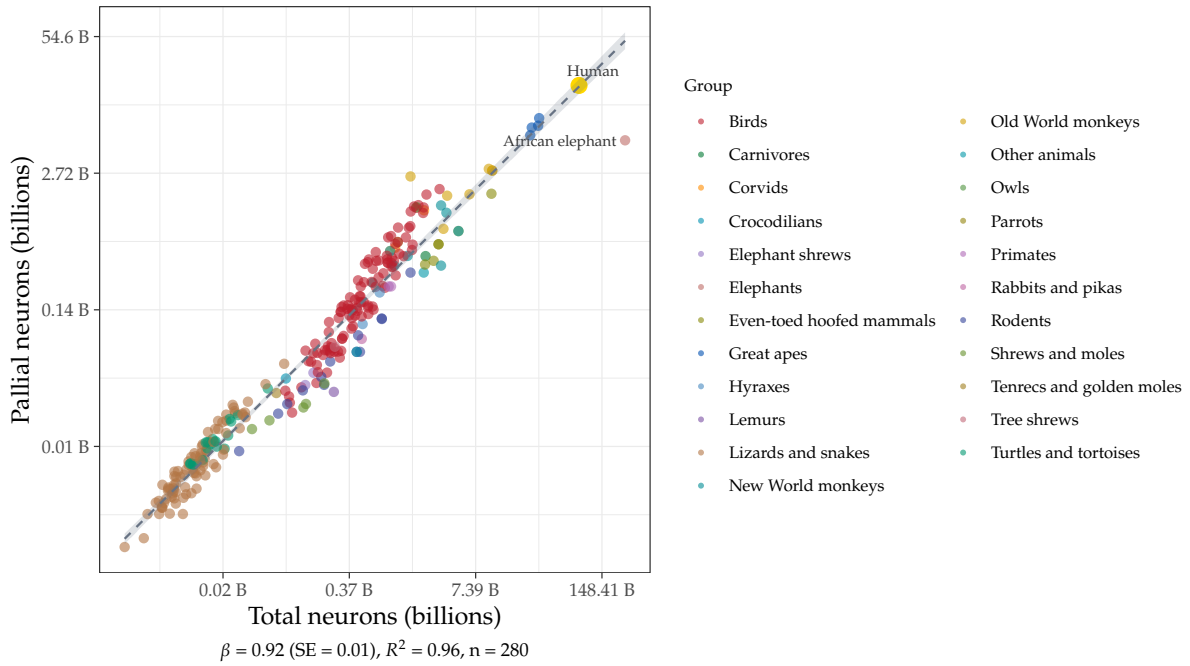
TABLE D.1: Data Sources — Neurons

Source	Description of Dataset
Kverková et al. (2022)	Dataset contains neuron counts for the whole brain, telencephalon, pallium, and other brain regions. Covers bird and reptile species. We use the whole-brain counts as total neurons. As pallial neurons, we use the pallium counts where published (37 bird species) and the telencephalon counts otherwise.
Wikipedia	Scraped from the Wikipedia list of animals by number of neurons. Contains pallial neurons only, but covers many species.
Herculano-Houzel et al. (2015)	Contains data on total neurons and cerebral cortex neurons for 39 mammalian species across six clades: Glires, Primates, Scandentia, Eulipotyphlans, Afrotherians, and Artiodactyls.
Sol et al. (2022)	Contains data on total and pallial neurons for 111 bird species.
Olkowicz et al. (2016)	Contains data on total and pallial neurons for 28 bird species.
Marhounová et al. (2019)	Contains data for whole brain and telencephalon neurons in guppies.
Ahrens et al. (2013)	Contains data on total neurons for zebrafish. Larval count, not used.
Teles et al. (2025)	Contains total and telencephalon neurons for tilapia. Not used in the fish average.
Raji and Potter (2021)	Contains whole-brain neuron counts for the fruit fly.
Menzel and Giurfa (2001)	Reports the whole-brain neuron count of the honeybee.
Barron and Mourmourakis (2023)	Contains the number of total neurons for most of the species for which we have cognition tests (A-not-B/cylinder).

TABLE D.2: Assumptions Used in Neurons/Cognition Code

Assumption short name	Description
Manual species coding	Some instances required manual species coding. For example, some Wikipedia entries listed only the genus, as in the honeybee entry, so we manually added a species name. In other cases, the datasets had mistyped names (e.g., <i>Furoscalops</i> $\leftrightarrow$ <i>Parascalops</i>).
Aggregation of dog breeds	The Wikipedia dataset contained pallial neuron counts for multiple dog breeds (the same species, <i>Canis lupus familiaris</i>). We take the average of all the dog breeds.
Within-species aggregation	More generally, if we have multiple observations for the same species across datasets, we take a within-species average.
Pallial neurons can be predicted from total neurons	We assume that pallial neurons can be predicted from a regression of log pallial neurons on log total neurons. This assumption appears to be reasonable given Figure D.1. For some species, we observe only pallial neurons. For these, we use the inverse mapping.
Total neurons can be predicted from body mass	For some species where we observe neither total nor pallial neurons, we fit a regression of log total neurons on log body mass within each order that has at least three species with observed body mass and a total neuron count, observed or predicted from pallial neurons. Whales and dolphins form their own group, and humans are excluded from the primate regression.
Group aggregation	When aggregating to the group level, we assume that the mean or minimum number of neurons that we observe represents the true group mean. This assumption is stronger for groups where we have fewer species and higher populations (e.g. insects).
PCTB human brain neurons	In the PCTB, we assume that a 2.5-year-old human has 80% of adult brain neurons.

FIGURE D.1: Relationship Between Pallial and Total Neurons



Notes: Relationship between pallial and total neurons for species in which both are observed, on a natural log scale. The dashed line is the fitted line from a regression of log pallial neurons on log total neurons. For the species of [Kverková et al. \(2022\)](#), the pallial count is the telencephalon count.

D.2 Merged Population and Neuron Data

Mammals. The dataset has one entry per species. The species-level dataset is exhaustive. Each species has:

- A current population count (observed), and a historical population count (obtained by backcasting using the species-level, order-level, or mammal-level growth rate, as described in [Appendix C.6](#))
- A neuron count: we use, in order of priority:
 - (i) the true or body-mass-predicted neuron count
 - (ii) the order mean: species whose order still cannot be fitted (too few anchors, or no body mass) inherit the mean neuron count of their order
 - (iii) the mammal mean: species in an order with no mean of its own fall back to the mammal class mean

Birds. Birds enter at the order level when the order has a neuron mean of its own. Each of them has:

- A current population count (obtained by aggregating the observed species counts at the order level), and a historical population count (obtained by backcasting

using the order-level or bird-level growth rate, as described in Appendix C.6)

- A neuron count: we use, in order of priority:
 - (i) the order mean (the average of the true or body-mass-predicted neuron count for species in the order)
 - (ii) orders with no mean of their own are not entered separately: their birds join the class-wide residual below

The order-level entries account for 0.54% of the current bird population. The remaining birds enter through a single class-wide residual carrying the average bird neuron count and the class-wide decline rate.

Fish, herptiles and insects. Fish, herptiles and insects are based on a class-wide average of species neuron counts and a class-wide population decline. These averages yield a count of 4 million neurons for fish, 14 million for herptiles, and 0.4 million for insects. The herptile average covers reptiles only, since no amphibian in our data has a neuron count. The fish average is the guppy alone: the only other fish count in our data, for the zebrafish, is for larvae, so we drop it. For the honeybee and the fruit fly, we use whole-brain neuron counts (Menzel and Giurfa 2001; Raji and Potter 2021). For the cricket, we only observe the neurons of the mushroom bodies, and we predict its total with the same regression as for pallial neurons.

D.3 Cognitive Tests

TABLE D.3: Coverage of Cognitive Tests by Taxonomic Group

Test	Primates	Non-primate mammals	Corvids	Non-corvid birds	Fish	Insects	Total	Groups	% int.	Hum.
Paper battery										
Cylinder	20	5	5	5	1	–	36	5	94	
A-not-B	18	5	2	2	–	–	27	4	85	
Composite score	15	4	2	2	–	–	23	4	100	
Quantity/numeration										
Ratio 0.25	14	10	3	3	–	1	31	5	100	
Ratio 0.33	14	11	3	2	–	1	31	5	100	
Ratio 0.50	14	12	3	3	–	1	33	5	100	
Ratio 0.60	14	10	3	2	1	–	30	5	97	
Ratio 0.67	14	12	3	3	–	2	34	5	100	
Ratio 0.75	14	11	3	2	1	2	33	6	100	
Ratio 0.80	14	5	3	2	–	–	24	4	100	
Ratio 0.90	11	2	1	1	–	–	15	4	100	
Puzzle box (carnivores)										
Ever solved	–	39	–	–	–	–	39	1	49	
Trial success rate	–	39	–	–	–	–	39	1	54	
Trials to first solve	–	23	–	–	–	–	23	1	–	
Audet (passerines)										
Solved: any of four tasks	–	–	1	14	–	–	15	2	53	
Detour reaching (trials)	–	–	1	14	–	–	15	2	–	
Learning (trials)	–	–	1	14	–	–	15	2	–	
Reversal learning (trials)	–	–	1	14	–	–	15	2	–	
ManyPrimates										
Delay 0 s	39	–	–	–	–	–	39	1	97	
Delay 15 s	39	–	–	–	–	–	39	1	100	
Delay 30 s	39	–	–	–	–	–	39	1	100	
String pulling										
Horizontal	34	12	2	9	–	–	57	4	9	
Vertical	10	5	7	69	–	–	91	4	41	
Downward	10	5	–	1	–	–	16	3	12	
Reversal learning										
Reversal index	12	10	1	6	–	–	29	4	97	✓
PCTB: space										
Spatial memory	8	–	–	4	–	–	12	2	100	✓
Object permanence	8	–	–	4	–	–	12	2	100	✓
Rotation	8	–	–	4	–	–	12	2	100	✓
Transposition	8	–	–	4	–	–	12	2	100	✓
PCTB: quantities										
Relative numbers	8	–	–	4	–	–	12	2	100	✓
Addition of numbers	8	–	–	4	–	–	12	2	100	✓
PCTB: causality										
Noise	8	–	–	4	–	–	12	2	100	✓
Shape	8	–	–	4	–	–	12	2	100	✓
Tool properties	8	–	–	4	–	–	12	2	100	✓
Tool use	8	–	–	4	–	–	12	2	33	✓
PCTB: social learning										
Social learning	8	–	–	–	–	–	8	1	50	✓
PCTB: communication										
Comprehension	8	–	–	4	–	–	12	2	100	✓
Pointing cups	8	–	–	4	–	–	12	2	100	✓
Attentional state	8	–	–	4	–	–	12	2	100	✓
PCTB: theory of mind										
Gaze following	8	–	–	4	–	–	12	2	92	✓
Intentions	8	–	–	4	–	–	12	2	100	✓
<i>Distinct species, any test above</i>	69	63	10	92	1	2	237	6	–	

Notes: Each row corresponds to a cognition test. The number in each column (Primates–Insects) corresponds to the number of species in that group in our sample. The ‘Groups’ column corresponds to the number of groups covered in the test. ‘% int.’ refers to the percentage of species with an interior test score (a test score $p \in (0, 1)$). ‘Hum.’ corresponds to whether humans are tested in that test. Shaded rows are rows that satisfy our inclusion criteria. Light gray corresponds to our “main tests”, and dark gray corresponds to our “convexity at the top” tests.

TABLE D.4: Data Sources — Cognitive Tests

Source	Description of Dataset
Barron and Mourmourakis (2023)	Aggregates cylinder task scores from MacLean et al. (2014) and Kabadayi et al. (2016).
MacLean et al. (2014)	Contains A-not-B task scores.
Lucon-Xiccato et al. (2017)	Contains cylinder task scores for guppies.
Herrmann et al. (2007)	Contains PCTB tests for humans and primates.
Fichtel et al. (2020)	— lemurs.
Schmitt et al. (2012)	— Old World monkeys.
Krasheninnikova et al. (2019)	— parrots.
Jacobs and Osvath (2015)	Contains string-pulling results for 91 species.
Dorschner et al. (2024)	Contains reversal learning results for 29 species, including humans.

Domain	Scale	Task name	Description
Physical	Space	Spatial memory	Locating a reward.
		Object permanence	Tracking of a reward after invisible displacement.
		Rotation	Tracking of a reward after a rotation manipulation.
		Transposition	Tracking of a reward after location changes.
	Quantities	Relative numbers	Discriminating quantity.
		Addition of numbers	Discriminating quantity with added quantities.
	Causality	Noise	Causal understanding of produced noise by hidden rewards.
		Shape	Causal understanding of appearance change by hidden rewards.
		Tool use	Using a stick in order to retrieve a reward which is out of reach.
		Tool properties	Understanding of functional and non-functional tool properties.
Social	Social learning	Social learning	Solving a simple but not obvious problem by observing a demonstrated solution.
	Communication	Comprehension	Understanding communicative cues indicating a reward's hidden location.
		Pointing cups	Producing communicative gestures in order to retrieve a hidden reward.
		Attentional state	Choosing communicative gestures considering the attentional state of the recipient.
	Theory of mind	Gaze following	Following an actor's gaze direction to a target.
		Intentions	Understanding what an actor intended to do (unsuccessfully).

TABLE D.5: Description of the Primate Cognition Test Battery Tests

TABLE D.6: Data Sources — Numeration

Source	Description of Dataset/Paper
<i>Fish</i>	
Lucon-Xiccato et al. (2023)	Study numerosity preferences in newly hatched zebrafish (<i>Danio rerio</i>) larvae. They report the preference for larger numerosity for ratios 0.25, 0.33, 0.50.
Bisazza and Santacà (2022)	Study zebrafish (<i>Danio rerio</i>) in tests of quantities differing by one unit (i.e. 2 vs 3, 3 vs 4, etc.). They show that zebrafish learn up to 4 vs 5 (ratio of 0.8).
Gatto et al. (2020)	Study guppies (<i>Poecilia reticulata</i>) trained to choose the larger of two arrays using a Skinner box. Two ratios tested ($0.60 = 3 \text{ vs } 5$, and $0.75 = 3 \text{ vs } 4$).
Agrillo et al. (2010)	Study mosquitofish (<i>Gambusia holbrooki</i>) trained to discriminate large numerosities at ratio 0.50, from 4 vs 8 objects to 100 vs 200. They were able to distinguish between ratios of 0.5 and 0.67, but not 0.75.
Agrillo et al. (2009)	Study mosquitofish trained to discriminate a ratio of 0.67 (2 vs 3).
Stancher et al. (2013)	Study redbtail splitfin (<i>Xenotoca eiseni</i>) tested on five comparisons at four ratios ($0.25 = 1 \text{ vs } 4$, $0.50 = 1 \text{ vs } 2$ and $2 \text{ vs } 4$, $0.67 = 2 \text{ vs } 3$, $0.75 = 3 \text{ vs } 4$) in a small-quantity discrimination task.
<i>Insects</i>	
Bortot et al. (2019)	Study honeybees at ratios 2 vs 3 and 3 vs 4. Evidence suggests that bees use absolute rather than relative numerosity.

Continued on next page

Table D.6 – continued from previous page

Source	Description of Dataset/Paper
Howard et al. (2018)	Study honeybees in an ordering setting. They train bees to distinguish “greater-than” and “less-than”.
MaBouDi et al. (2020)	Study bumblebees (<i>Bombus terrestris</i>) at ratios 0.33 and 0.50.
Bengochea et al. (2023)	Study fruit flies (<i>Drosophila melanogaster</i>) at various ratios (0.25–0.75).
<i>Birds</i>	
Rugani et al. (2014)	Study domestic chicks (<i>Gallus gallus</i>) at ratios (0.5, 0.67, 0.75).
Rugani et al. (2013)	Study domestic chicks (<i>Gallus gallus</i>) at ratio 0.5 (5 vs 10 & 10 vs 20) and 0.67 (6 vs 9).
Tan and Grace (2010)	Study pigeons (<i>Columba livia</i>) at ratio 0.33 (2 vs 6, 4 vs 12, 8 vs 24).
Scarf et al. (2011)	Study pigeons in an ordering task (ordered comparisons 1–9).
Ditz and Nieder (2016)	Study crows (<i>Corvus corone</i>) in an ordering task.
Pepperberg and Gordon (2005)	Study parrots (<i>Psittacus erithacus</i>) in a vocal-labeling task for quantities 1–6 (an ordering-like task).
<i>Reptiles & Amphibians</i>	
Stancher et al. (2014)	Study fire-bellied frogs (<i>Bombina orientalis</i>) at ratios 0.50 (1 vs 2) and 0.67 (2 vs 3) in a spontaneous prey-choice paradigm.
Khatiwada and Burmeister (2021)	Study poison dart frogs (<i>Dendrobates auratus</i>) at ratio 0.50 (1 vs 2) using live flies as prey.
Gazzola et al. (2018)	Study Hermann’s tortoises (<i>Testudo hermanni</i>) at ratios 0.25 (1 vs 4) and 0.50 (2 vs 4) in a spontaneous food-choice task.

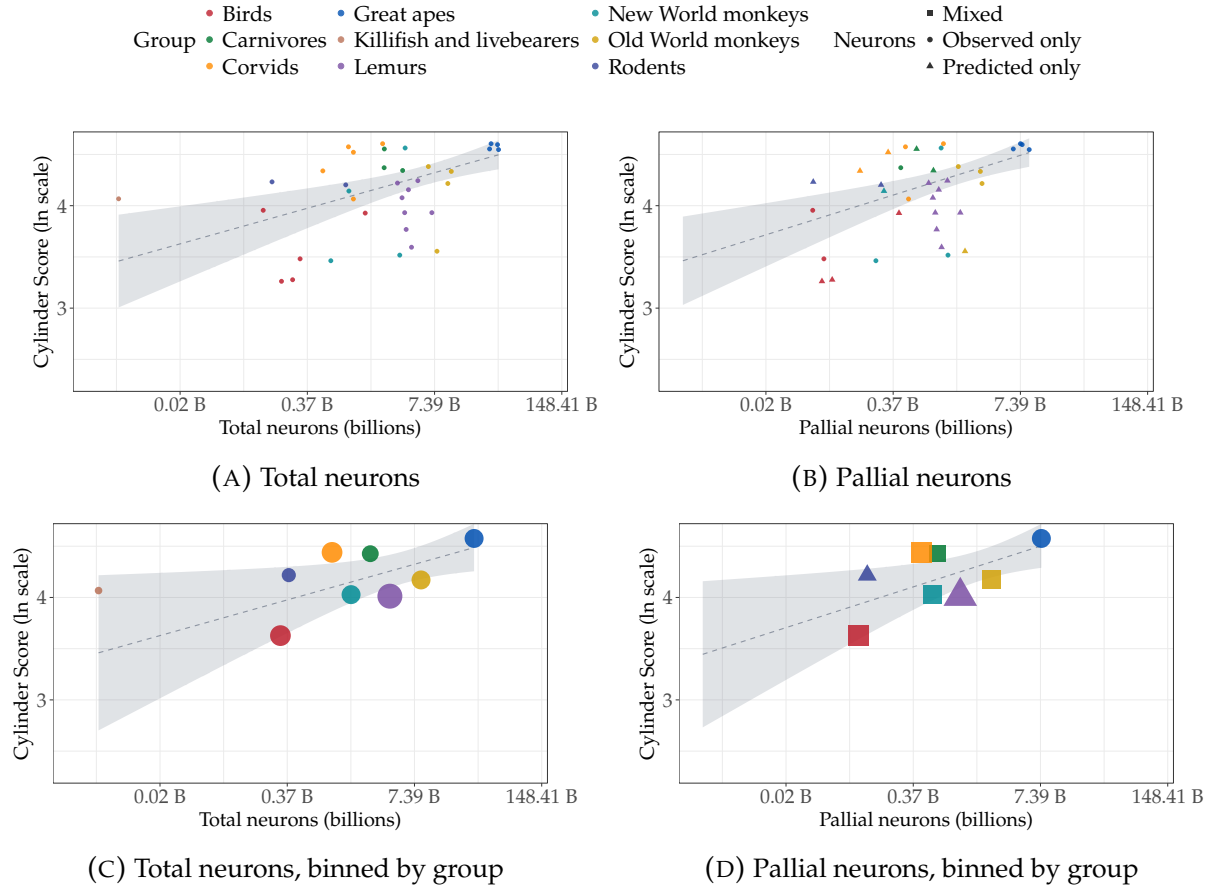
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Table D.6 – continued from previous page

Source	Description of Dataset/Paper
Lin et al. (2021)	Study Chinese pond turtles (<i>Mauremys sinensis</i>) at ratios up to 0.90 (9 vs 10), the hardest pair successfully discriminated.
<i>Mammals</i>	
Tomonaga (2007)	Study chimpanzees (<i>Pan troglodytes</i>) at numerosity pairs drawn from 1–8 in a relative numerosity task. Performance follows Weber’s Law.
Cantlon and Brannon (2006)	Study rhesus macaques (<i>Macaca mulatta</i>) in an ordering task at numerosities 1–9, tested for generalization to 10–30.
Jordan and Brannon (2006)	Study rhesus macaques (<i>Macaca mulatta</i>) at numerosities 1–9 in a delayed match-to-sample task. Performance follows Weber’s Law.
Yaman et al. (2012)	Study a bottlenose dolphin (<i>Tursiops truncatus</i>) at ratio 0.25 (1 vs 4). It fails at the harder ratio 0.75 (3 vs 4).
Miletto Petrazzini et al. (2020)	Study domestic-dog puppies (<i>Canis lupus familiaris</i>) at ratio 0.17 (1 vs 6) in a spontaneous food-quantity task.
Gabor and Gerken (2014)	Study Shetland ponies (<i>Equus caballus</i>) at ratios 0.50 (1 vs 2) to 0.80 (4 vs 5) in a matching-to-sample task.

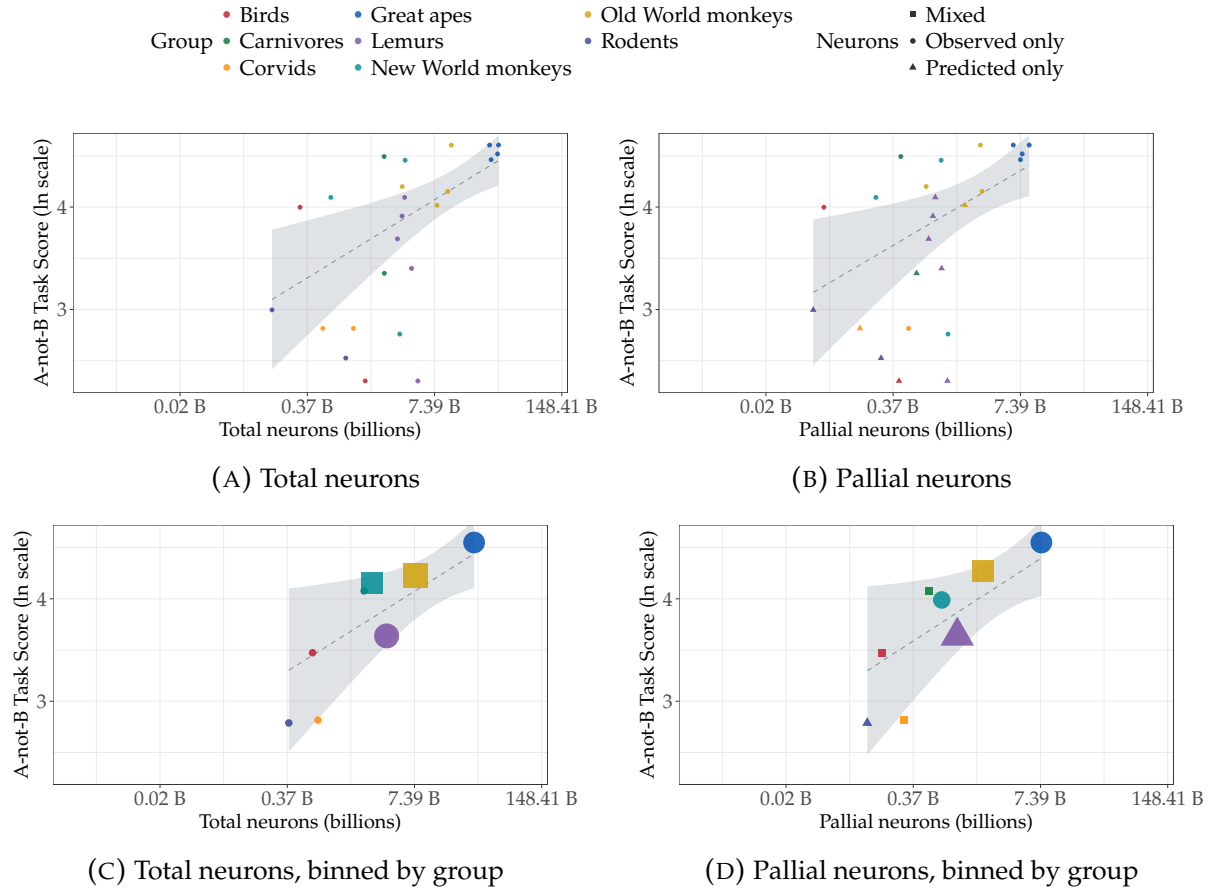
E. ADDITIONAL RESULTS: COGNITION AND NEURONS

FIGURE E.1: Cylinder Score and Neurons



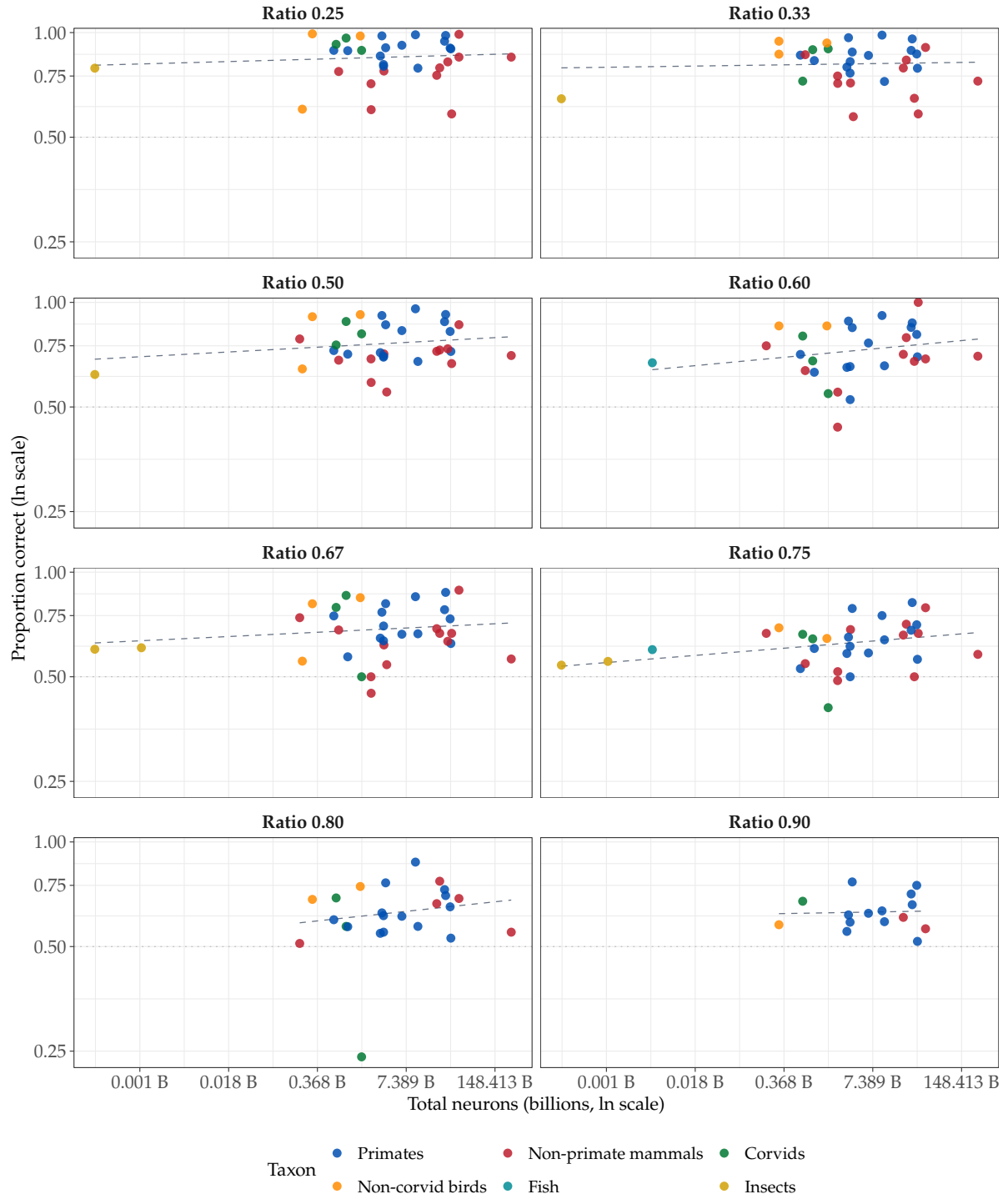
Notes: Cylinder test score versus neuron counts (ln–ln). Panels a, b show individual species. Panels c, d show bins by group, with the binned regression weighted by species count. The dashed line is the GLM fit under the identity cardinalization $c = p$, and the shaded area is its 95% confidence band.

FIGURE E.2: A-not-B Score and Neurons



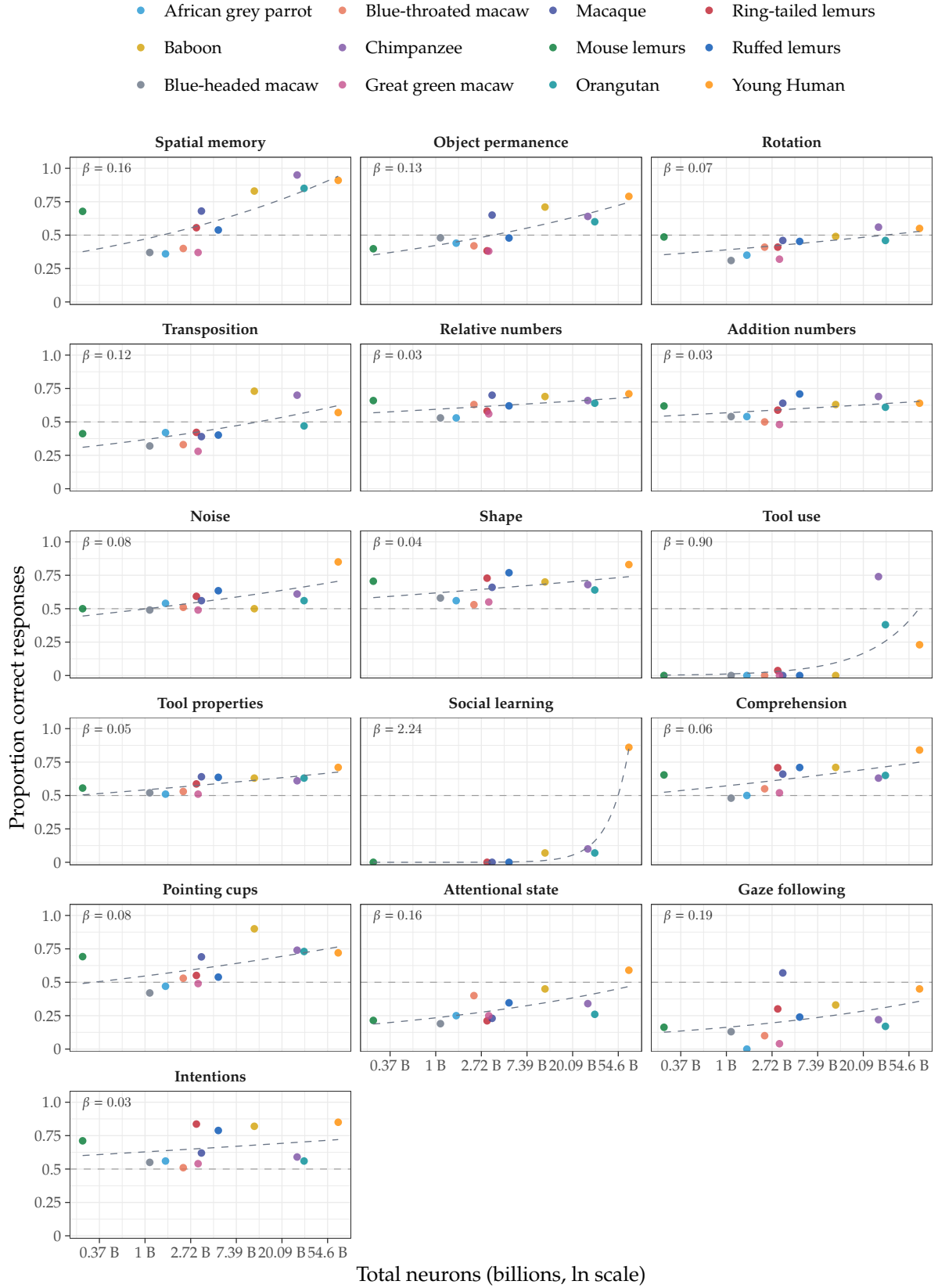
Notes: A-not-B test score versus neuron counts (ln–ln). Panels a, b show individual species. Panels c, d show bins by group, with the binned regression weighted by species count. The dashed line is the GLM fit under the identity cardinalization $c = p$, and the shaded area is its 95% confidence band.

FIGURE E.3: Numeration Tests



Notes: Relationship between total neurons and test performance on numeration tests at different ratios. The dashed line is the fitted regression line for test performance on log total neurons.

FIGURE E.4: Primate Cognition Test Battery (PCTB)

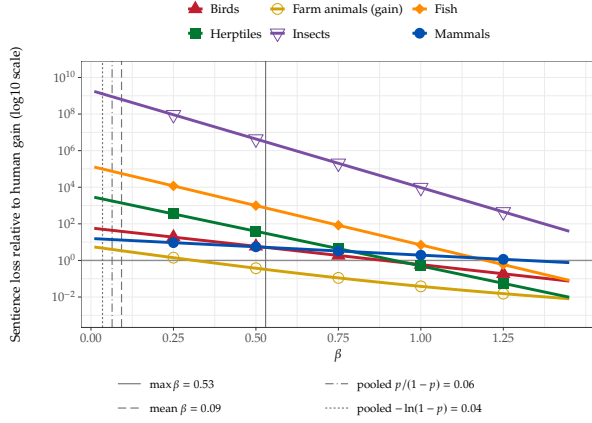


Notes: Relationship between (observed or predicted) total neurons and test performance on each of the 16 PCTB tasks. The dashed horizontal line marks a success rate of 0.5. The other dashed line is the GLM fit under the identity cardinalization $c = p$, and β is its slope.

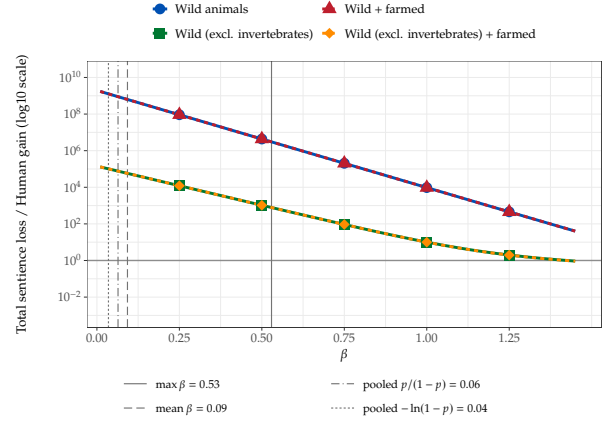
F. ADDITIONAL RESULTS: WELFARE

FIGURE F.1: Sentence Losses or Gains by Taxa: Data Aggregated by Taxon

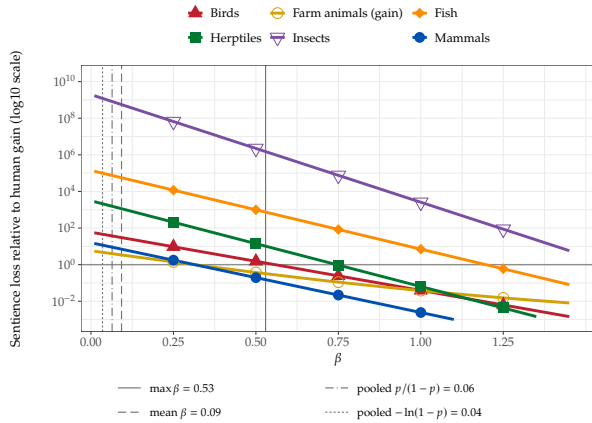
(A) By Group: Mean Neurons by Taxon



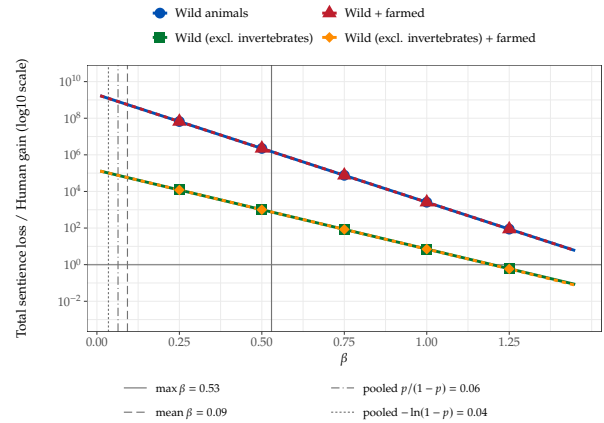
(B) Aggregate: Mean Neurons by Taxon



(C) By Group: Minimum Neurons by Taxon

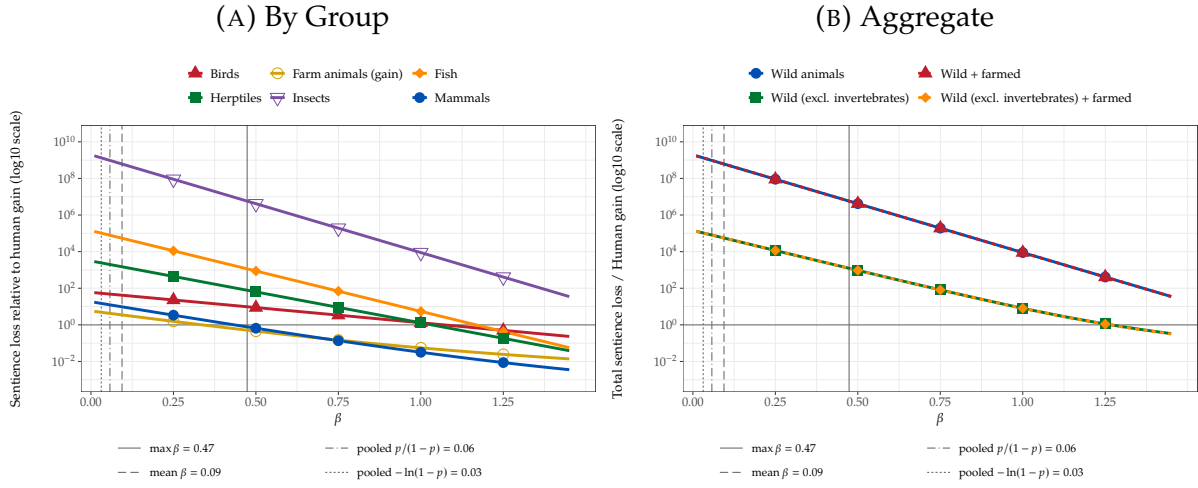


(D) Aggregate: Minimum Neurons by Taxon



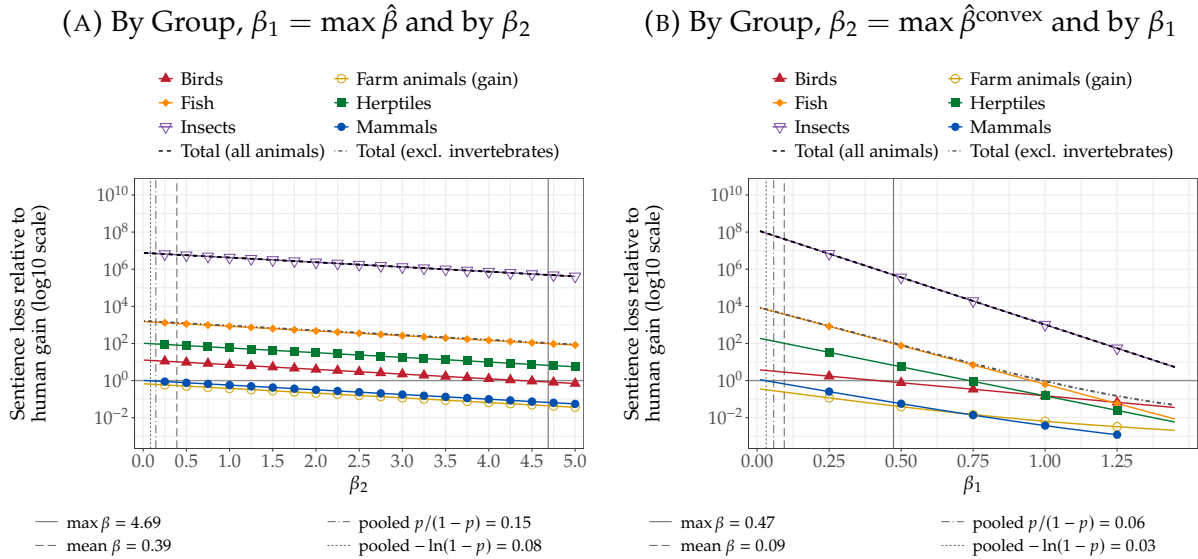
Notes: The y-axis is the cumulative sentence loss or gain for each taxon expressed in aggregate human sentence gains (as defined in (14)). The x-axis is the sentence-to-neurons elasticity β , which determines the relative welfare capacities ω_s . We use relative welfare capacities estimated from total neuron counts. The “Farm animals” category includes all farmed animals, irrespective of taxon. We use population changes by taxon for all taxa. In Panels (a) and (b), we estimate the neuron count for each taxon as the average neuron count among observed species within that taxon. In Panels (c) and (d), we estimate the neuron count for each taxon as the minimum neuron count among observed species within that taxon.

FIGURE F.2: Sentence Losses or Gains by Taxa: Using Pallial Neurons



Notes: The y-axis is the cumulative sentence loss or gain for each taxon expressed in aggregate human sentence gains (as defined in (14)). The x-axis is the sentence-to-neurons elasticity β , which determines the relative welfare capacities ω_s . We use relative welfare capacities estimated from pallial neuron counts, with the 16.3 billion cortical neurons of Azevedo et al. (2009) for humans. The vertical lines are the slopes estimated on pallial neurons (largest slope 0.47). The “Farm animals” category includes all farmed animals, irrespective of taxon. We use population data at the most granular level of disaggregation available.

FIGURE F.3: Animal Sentence Loss Relative to Human Gain under ω^{conv} : Pallial Neurons



Notes: Same as Figure 7, with sentence computed from pallial instead of total neuron counts. The human count is the 16.3 billion cortical neurons of Azevedo et al. (2009). The top primate is the western gorilla, with 9.1 billion pallial neurons, which also has the largest measured total count. Both slopes are re-estimated on pallial neurons, with the same human count: $\beta_1 = 0.47$ for the main tests and $\beta_2 = 4.7$ for the tests specific to primates and humans.

FIGURE F.4: Sentence Losses or Gains by Taxa: Including All Fish

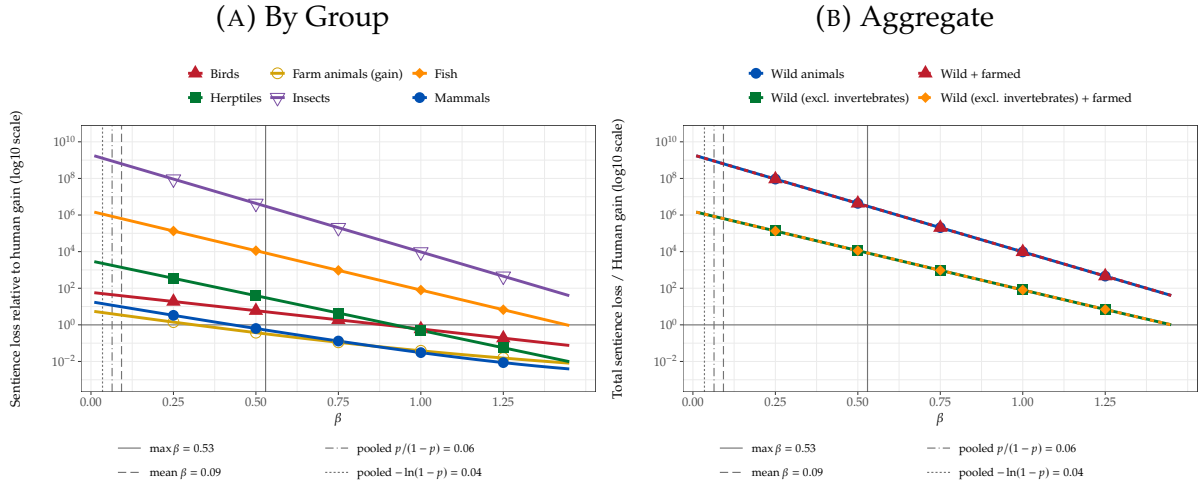


FIGURE F.5: Alternative Elasticity Estimators (Main Tests)

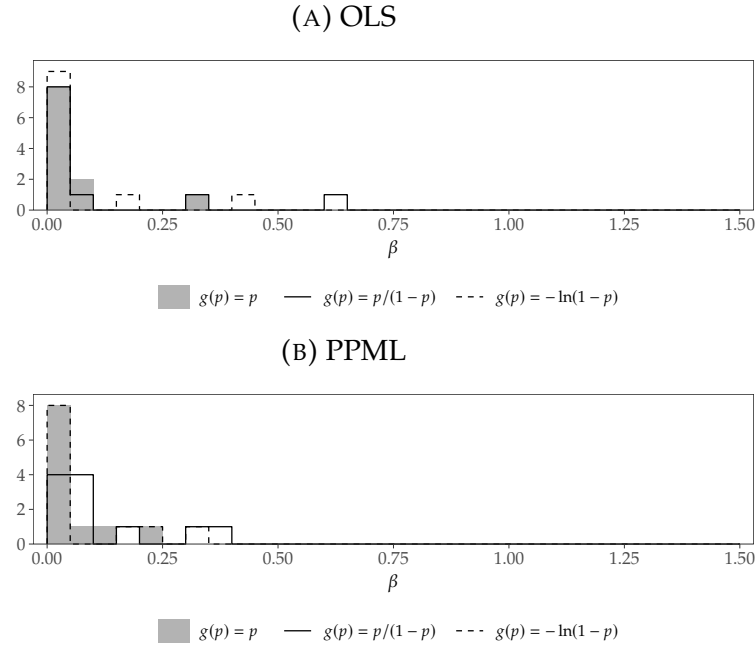
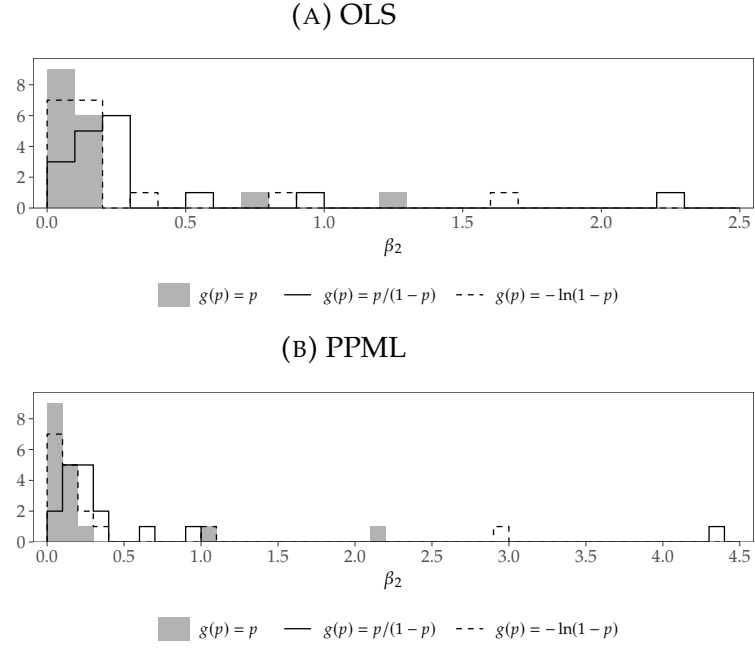


FIGURE F.6: Alternative Elasticity Estimators (“Convexity at the Top” Tests)



Notes: Histograms of the sentence-to-neurons elasticity β under various cardinalizations of cognition $c = g(p)$. The top panel estimates the elasticity using OLS, and the bottom panel estimates the elasticity using PPML. These histograms include only the “convexity at the top” tests (the 16 PCTB tasks and reversal learning), and the x-axis is the elasticity β_2 .

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